Neurons Through the Looking Glass: Constructing an Interacting Quantum Field in Two Dimensions

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Abstract

We construct an interacting quantum field from random fields called *process neurons*. We combine these building blocks and change their probability weights through a local quartic interaction. Two explicit architectures, built from random sign changes or randomly colored cells, reach the established massive, weakly coupled two-dimensional Wick-quartic field along one deterministic parameter sequence. This sequence is fixed before any measurement is chosen and works for both architectures.

The construction preserves *reflection positivity*, the positivity condition used to recover a quantum Hilbert space from Euclidean fields. At every finite stage the normalized interacting law has this property across the two central coordinate lines. We prove weak convergence of the field laws on $H^{-2}_{\mathrm{loc}}(\mathbb{R}^2)$, convergence of every power of the local seminorms, and convergence of all polynomial moments of any finite collection of smooth, compactly supported measurements chosen afterwards. The joint compact-test correlation distributions also converge at every order.

We establish the process-neuron, Gaussian-reference and fixed-volume Wick estimates needed for this construction. Attributed classical results supply the infinite-volume target and its reconstruction as a relativistic quantum field in one space and one time dimension; we verify their hypotheses at the points of use. Finite width counts whole random processes, each using countably many random variables globally. The common limiting sequence is existential, with no effective end-to-end parameter schedule or computational advantage established. The result gives a precise route from these two neural architectures to the specified interacting field.

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1 Introduction

1.1 The question

Can simple random building blocks be assembled into an interacting quantum field? We study this question by starting with random patterns on a plane, combining them, and changing the probabilities of the resulting patterns to introduce an interaction. The goal is to prove that their probability laws approach a specified quantum field as the approximations are removed.

The building blocks are *process neurons*: whole-plane random sign fields. In one architecture a neuron is formed from two independent random sign processes, one for each coordinate. In the other, a random division of the plane into rectangular cells receives random signs. A width-N field is a normalized sum of N such neurons. Before the interaction is introduced the neurons are independent; reweighting their joint probability law introduces the interaction. The width counts these random processes. Each process uses countably many random variables globally, so finite width does not mean a finite list of parameters encoding the entire plane.

Our target is the massive, weakly coupled two-dimensional Wick-quartic field, usually denoted φ_2^4. Here "quartic" means that the interaction involves the fourth power of the field, with the subtractions needed to define its continuum limit. We first work in a Euclidean plane. Classical reconstruction then identifies the limiting model with a relativistic quantum field in one space and one time dimension. The main construction result concerns this particular model and weak-coupling regime.

1.2 Why the looking glass?

Reflection gives a way to compare measurements on opposite halves of the plane. *Reflection positivity* requires certain averages formed from a measurement and its reflected counterpart to be nonnegative. In the reconstruction of a quantum theory, this supplies the positivity needed for the Hilbert-space inner product. The precise condition concerns the full probability law, including its interactions.

We preserve this condition across the two central coordinate lines at every finite stage of the construction. The limiting field has full Euclidean symmetry and reflection positivity across every line, provided by the classical theory. The looking glass in the title points to this mathematical role of reflection.

1.3 From random patterns to a field

There are several approximations to control. We increase the number of neurons, remove a stabilizing term, enlarge an auxiliary region, and remove a short-distance regulator. We then enlarge the physical region in which the quartic interaction acts. These are ordered limiting steps, assembled into one common sequence for both architectures.

The sequence is chosen once, before selecting measurements. For example, one may later choose smooth weighted averages of the field over bounded regions: their joint polynomial moments converge along that same sequence. The field laws themselves also converge in a space of distributions, which allows the limiting field to be rougher than an ordinary function. This is an existence result for the full sequence; it does not give an effective rule for how many neurons or how large a region achieves a chosen accuracy.

The target and the classical tools have a substantial history. Fröhlich and Simon constructed its

infinite-volume states and proved their regularity [8]; Albeverio, Høegh-Krohn and Zegarlinski proved a weak-coupling uniqueness theorem that identifies them [2]; Guerra, Rosen and Simon developed the free-boundary lattice approximation and the correlation inequalities used below [10]; and Simon and Griffiths proved the finite-system inequality we need [19]. Relativistic reconstruction uses the Osterwalder–Schrader theorem [16, 17] in the measure-level formulation of [8, Thm. 1.0]. These results supply the classical inputs to our architecture-specific construction. Section 14 compares that construction with earlier neural-field work.

The next subsection states the result precisely. Readers seeking an overview can then follow the three layers in Section 1.5 and the contribution summary in Section 1.6; the definitions and proofs develop the details.

1.4 The result in brief

Fix $m > 0$, a length $\ell > 0$ and $u > 0$ with $0 < u/m^2 < g_*$ (Definition 3.2). For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$ let $P^b_{N,\varepsilon,\Omega,a;\Lambda}$ be the normalized annealed neural law of Definition 2.10, a Borel probability on the Polish space $X = H^{-2}_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$. Theorem 4.1 states that there is one deterministic sequence $(a_n, R_n, \varepsilon_n, N_n)_{n\geq 1}$, common to both architectures and depending only on m, ℓ, u, a fixed localization system and fixed parameter enumerations, such that with the axis-parallel physical squares $\Lambda_n = (-n\ell, n\ell)^2$ and auxiliary squares $\Omega_n = [-R_n, R_n]^2$ the laws $P^b_n = P^b_{N_n,\varepsilon_n,\Omega_n,a_n;\Lambda_n}$ satisfy:

(A) weak convergence on X to the X-valued version ν of the classical target, with uniformly bounded and convergent powers of every local seminorm;

(B) convergence of all polynomial moments of every finite tuple of real compactly supported smooth tests chosen after the sequence;

(C) existence and weak distributional convergence of the joint correlation distributions of every order on compactly supported joint tests;

(D) identification of the limits with restrictions of the classical tempered Schwinger distributions;

(E) reflection positivity of every finite stage across the two central coordinate lines, for bounded half-plane observables and for finite polynomial forms in compact joint tests, with convergence of these forms.

Theorem 4.2 records, as attributed classical results with project-derived verification steps, the Euclidean invariance, translation mixing and ergodicity, all-line reflection positivity, exponential regularity, tempered Schwinger distributions, non-Gaussianity and $1+1$-dimensional Wightman reconstruction of the target.

1.5 What is proved, what is attributed

The paper has three layers.

(i) *Classical inputs* (Section 3). Published theorems C1–C5 are imported with attribution, in formulations whose hypotheses are matched to the application at the point of use, and C6 lists standard tools. Their proofs are not audited here, and the original Osterwalder–Schrader and Streater–Wightman texts were not available (Appendix D).

(ii) *Foundations proved here.* The process neurons and their finite-width central reflection positivity (Section 6, Appendix A); the Gaussian reference fields, including the finite-region mass tilt and the common Gaussian space with Borel distribution-valued versions (Section 7, Appendix B); the fixed-volume Wick theory (Section 8, Appendix C); the field-space facts (Section 5); and the fixed-volume chain of normalized laws (Section 9).

(iii) *Construction and target.* The reduced continuum lemmas (Section 10): the continuum limit with its Wick identification, the whole-field volume limit with all compact moments, and the target properties. The common sequence, the local moments and the joint correlations (Sections 11–12). The assembly (Section 13).

The main theorem assumes C1–C6 only. An earlier conditional version of this construction assumed, besides these, nine project premises; Appendix E records for each of them whether it is now a theorem of this paper or no longer used, and where the replacing argument is.

1.6 Contribution and organization

Our contribution is an architecture-specific realization and synthesis. In each of the two architectures the width-N field $F_N = N^{-1/2}\sum_{i=1}^{N} g_i$ is a normalized sum of N independent copies of a whole-plane process neuron with $|g| = 1/2$, $\mathbb{E}g = 0$ and the endpoint covariance $e^{-\mu|z|_1}/4$, $|z|_1 = |z_1| + |z_2|$, $\mu = 1/a$. On these fields we place normalized annealed Gibbs laws: the joint law of all neuron randomness is reweighted by $e^{-V(F_N)}$ and normalized, where for a centered axis-parallel physical square Λ inside an auxiliary square $\Omega = [-R, R]^2$,

$$V(F_N) = u\int_\Lambda (F_N^4 - 6c_a F_N^2 + 3c_a^2)\,dz + \varepsilon\int_{\Omega\setminus\Lambda} F_N^4\,dz - \frac{\kappa_a^2}{2}\int_\Omega F_N^2\,dz, \qquad \kappa_a^2 = a^{-2} - m^2.$$

The fixed-volume limits, the growing-volume target and the joint correlation statements above are proved for both architectures along one sequence. The infinite-volume target, its Osterwalder–Schrader properties and its reconstruction are classical results used with attribution. The reflection-positivity mechanism, the density-deformation mechanism and the endpoint covariance kernel all have prior instances (Section 14). We make no historical-priority claim.

Section 2 fixes notation and the order of the limiting stages. Section 3 states the classical inputs and fixes the admissible pair. Section 4 states the results. Sections 5–8 state the foundations, with proofs in Appendices A–C. Sections 9–12 prove the construction and Section 13 assembles it. Sections 14 and 15 discuss prior work and limitations.

2 Setting, notation and conventions

This section fixes the objects that appear in the main theorem. It introduces the model parameters and the three covariances, lists the order in which the approximations are removed (Section 2.2), and defines the field spaces in which convergence is measured, the two neuron architectures, the interaction potential and the normalized laws. A first reading can concentrate on Definitions 2.1, 2.6, 2.8 and 2.10 together with the list of stages; the remaining conventions can be consulted where they are cited.

2.1 Parameters, geometry and covariances

Points of the plane are $z = (z_1, z_2)$; $|z|$ is the Euclidean norm, $|z|_1 = |z_1| + |z_2|$ and $|z|_\infty = \max(|z_1|, |z_2|)$. In the reconstruction (Theorem 4.2) Euclidean time is z_2, following [8, p. 495]. $|A|$ is Lebesgue measure, $\mathcal{L}(\mathbb{R}^2)$ the Lebesgue σ-algebra and $\mathcal{B}(\cdot)$ a Borel σ-algebra. $\mathbb{N} = \{1, 2, 3, \dots\}$. The symbol N alone always denotes a width.

Definition 2.1 (Parameters and physical geometry)**.** Fix a mass $m > 0$, a length $\ell > 0$ and a quartic coefficient $u > 0$. A *regulator* is a real a with $0 < a < 1/m$; put $\mu = 1/a$, $\kappa_a^2 = a^{-2} - m^2 > 0$ and $r_a = 1 - a^2 m^2 \in (0, 1)$; the value $a = 0$ denotes the unregulated case. A *stabilizer* is $\varepsilon \geq 0$.

(i) An *auxiliary square* is $\Omega = [-R, R]^2$, $R > 0$.

(ii) A *physical square* inside Ω is a centered **axis-parallel** square $\Lambda = (-l, l)^2$ or $\Lambda = [-l, l]^2$ with $0 < l \leq R$. Then $\bar{\Lambda} = [-l, l]^2 \subseteq \Omega$, and both Λ and Ω are invariant under the two central coordinate reflections $\theta_1(z_1, z_2) = (-z_1, z_2)$ and $\theta_2(z_1, z_2) = (z_1, -z_2)$.

(iii) The physical squares of the main theorem are $\Lambda_n = (-n\ell, n\ell)^2$, $n \geq 1$, with closures $\bar{\Lambda}_n = [-n\ell, n\ell]^2$.

Open and closed physical squares differ by the Lebesgue-null boundary $\partial\Lambda$; every integrated object below is unchanged when Λ is replaced by $\bar{\Lambda}$ (Lemma 9.1).

The axis-parallel requirement in (ii) is a hypothesis, not a convention: centering alone does not give the two reflection invariances used for finite-stage positivity (Remark 6.8). Rotated physical squares are not part of this definition; the general positivity theorem of Section 6 applies to every region and potential that are invariant under both central reflections, whatever their shape.

The Fourier convention is $\hat{f}(p) = \int e^{-ip\cdot z} f(z)\, dz$, extended to L^2 by Plancherel, with inversion $C(z) = (2\pi)^{-2} \int e^{ip\cdot z} \hat{C}(p)\, dp$ and $\int |f|^2 = (2\pi)^{-2} \int |\hat{f}|^2$; for $g \in L^1(\mathbb{R}^2)$ we write $\mathcal{I}g(x) = (2\pi)^{-2} \int e^{ip\cdot x} g(p)\, dp$. With $p = (w, q)$,

$$\hat{C_{\text{free}}}(p) = \hat{C}_0(p) = (m^2 + |p|^2)^{-1}, \qquad \hat{C}_a(w, q) = (m^2 + w^2 + q^2 + a^2 w^2 q^2)^{-1}, \tag{1}$$
$$C_\mu^{\text{end}}(z) = \tfrac{1}{4} e^{-\mu|z|_1}, \qquad \hat{C}_\mu^{\text{end}}(w, q) = \frac{\mu^2}{(\mu^2 + w^2)(\mu^2 + q^2)}.$$

$C_{\text{free}} = (-\Delta + m^2)^{-1}$ is the free covariance, C_a the regulated physical covariance and C_μ^{end} the endpoint covariance. The Wick variance c_a is the variance $C_a(0)$ of the regulated Gaussian physical reference field, the centered Gaussian field with covariance C_a (Theorem 7.9):

$$c_a = (2\pi)^{-2} \int_{\mathbb{R}^2} \hat{C}_a(p)\, dp = (2\pi)^{-1} \int_0^\infty \left[(m^2 + q^2)(1 + a^2 q^2)\right]^{-1/2} dq \in (0, \infty). \tag{2}$$

It is strictly decreasing in a, satisfies $c_a > 1/4$ for every regulator (compare the integrand in (2) with its value at $a = 1/m$, where the right-hand side is $1/4$; see Lemma C.2(K4)), and tends to $+\infty$ as $a \downarrow 0$ by monotone convergence. The endpoint variance is $C_\mu^{\text{end}}(0) = 1/4$; the two variances are never interchanged.

2.2 The mass-shift identity and the order of stages

The neurons produce the endpoint covariance C_μ^{end}, whereas the target is built on the massive free covariance C_{free}. The identity below shows how this gap is bridged for the Gaussian references:

subtracting κ_a^2 from the reciprocal of the endpoint multiplier gives the reciprocal of the multiplier of C_a, and $1/\hat{C}_a$ differs from $1/\hat{C_{\text{free}}}$ only by the term $a^2w^2q^2$ in (1). The negative quadratic term $-T_{\Omega,a}$ of the potential (Definition 2.8) implements this shift (stages (3) and (5) below). The numbered list records the order in which the approximations are removed and the result controlling each stage.

Since $1/\hat{C}_\mu^{\text{end}} = \mu^2 + w^2 + q^2 + a^2w^2q^2$ and $a\mu = 1$,

$$\frac{1}{\hat{C}_\mu^{\text{end}}} - \kappa_a^2 = \frac{1}{\hat{C}_a}, \qquad \hat{C}_a = \frac{\hat{C}_\mu^{\text{end}}}{1 - \kappa_a^2 \hat{C}_\mu^{\text{end}}} = \sum_{k\geq 0} \kappa_a^{2k} (\hat{C}_\mu^{\text{end}})^{k+1}. \tag{3}$$

This is an identity between Fourier multipliers of *Gaussian reference covariances*. It is used in the following precise order of stages; it is not a formula for the covariance of a finite-width law or of a quartically tilted law.

(1) *Finite width.* The neural law $P^b_{N,\varepsilon,\Omega,a;\Lambda}$ reweights the bounded, non-Gaussian field F_N by the full potential V, whose negative quadratic term acts on all of Ω.

(2) *Width limit.* As $N \to \infty$ these laws converge to the full Gibbs tilt $\nu^v_{\varepsilon,\Omega,a;\Lambda}$ of the Gaussian endpoint reference G_μ (covariance C_μ^{end}) by the same potential (Proposition 9.2); we assert neither that this tilt is Gaussian nor that it is not.

(3) *Pure quadratic tilt on finite* Ω. The tilt of G_μ by $e^{T_{\Omega,a}}$ alone is the centered Gaussian law $G'_{\Omega,a}$ on the whole plane with covariance $C'_{\Omega,a}$ (Theorem 7.4), and $C_\mu^{\text{end}} \leq C'_{\Omega,a} \leq C_a$ pointwise. The full tilt is the residual quartic tilt of $G'_{\Omega,a}$ (Proposition 9.3).

(4) *Stabilizer removal* $\varepsilon \downarrow 0$ at fixed a, Ω, Λ (Proposition 9.4).

(5) *Auxiliary-volume removal* $\Omega \uparrow \mathbb{R}^2$ at fixed a, Λ: $C'_{\Omega,a} \to C_a$ locally uniformly (Theorem 7.6); this is where the multiplier (3) becomes the covariance of the Gaussian reference, and the residual laws converge to the regulated physical law $\nu_{a,\Lambda}$ (Proposition 9.5).

(6) *Regulator removal* $a \downarrow 0$ at fixed Λ (Proposition 9.6); the anisotropy $a^2w^2q^2$ of the Gaussian reference disappears.

(7) *Physical volume* $\Lambda_n \uparrow \mathbb{R}^2$, through classical continuum input (Section 10).

The interacting laws at every stage are non-Gaussian in general, and their covariances are not given by (3).

2.3 Field spaces and localization

Two spaces are needed. At every finite stage the neural field is a bounded function, so the quartic potential is an ordinary integral of a function in E. The limiting law lives on distributions (Theorem 10.10(iii)), so convergence is stated in the space X of local H^{-2} distributions. Its seminorm p_j cuts the field off with χ_j, regards the result as a periodic distribution on the torus $\mathbb{T}_j$, and measures it by a weighted Fourier sum; every compact smooth smearing $v(f)$ is controlled by one such seminorm through (16).

Definition 2.2 (Field spaces)**.** $E = L^4_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$ consists of real locally fourth-power integrable functions modulo null functions, with the metric $d_E(v,w) = \sum_{j\ge1} 2^{-j}\min(1, \|v-w\|_{L^4((-j,j)^2)})$. $X = H^{-2}_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$ consists of the real distributions v on $\mathbb{R}^2$ with $\chi v \in H^{-2}(\mathbb{R}^2)$ for every $\chi \in C^\infty_c(\mathbb{R}^2)$. The map $\iota : E \to X$ sends a function class v to $f \mapsto \int vf$. The *localization system* consists of:

- real cutoffs $\chi_j \in C^\infty_c(\mathbb{R}^2)$, $j \ge 1$, with $0 \le \chi_j \le 1$, $\chi_j = 1$ on $[-j,j]^2$, $\operatorname{supp}\chi_j \subset (-j-\frac12, j+\frac12)^2$ and $\chi_{j+1} = 1$ on $\operatorname{supp}\chi_j$;
- the square tori $\mathbb{T}_j$ of side $2(j+1)$, $|\mathbb{T}_j| = 4(j+1)^2$, with fundamental cell $\mathcal{C}_j = (-(j+1), j+1)^2$; a distribution supported in $\operatorname{supp}\chi_j$ is regarded as a periodic distribution on $\mathbb{T}_j$;
- the orthonormal basis $e_{j,k}(z) = |\mathbb{T}_j|^{-1/2}\exp(i\pi k\cdot z/(j+1))$, $k \in \mathbb{Z}^2$, of $L^2(\mathbb{T}_j)$, and the weights $\lambda_{j,k} = 1 + \pi^2|k|^2/(j+1)^2$, so that $(1-\Delta)\overline{e_{j,k}} = \lambda_{j,k}\overline{e_{j,k}}$ and $\lambda_{j,-k} = \lambda_{j,k}$.

For $r \in \mathbb{R}$ and a periodic distribution h on $\mathbb{T}_j$ put $\|h\|^2_{H^r(\mathbb{T}_j)} = \sum_k \lambda^r_{j,k}|h(\overline{e_{j,k}})|^2 \in [0,\infty]$. The local seminorms, the metric and the bounded-Lipschitz distance are

$$\begin{aligned} p_j(v) = \|\chi_j v\|_{H^{-2}(\mathbb{T}_j)}, \qquad d(v,w) = \sum_{j\ge1} 2^{-j}\min(1, p_j(v-w)), \\ \beta(\rho,\eta) = \sup_{|H|\le1,\ \mathrm{Lip}_d(H)\le1} |\rho(H) - \eta(H)|, \end{aligned} \tag{4}$$

where $(\chi_j v)(g) = v(\chi_j g)$ for smooth periodic g, so that $p_j(v)^2 = \sum_k \lambda^{-2}_{j,k}|v(\chi_j\overline{e_{j,k}})|^2$, with v extended complex-linearly. X carries the topology of d and its Borel σ-algebra.

Proposition 5.1 proves that E and X are Polish, that d is a complete metric, that compact smooth evaluations $v \mapsto v(f)$ are continuous and generate the Borel σ-algebra of X, that ι is a continuous injection, and that β metrizes weak convergence. A *weak law statement on* X always means convergence of $\int H\,d\rho$ for every real bounded continuous $H : X \to \mathbb{R}$.

2.4 The two process-neuron architectures

Both architectures produce a random sign field with values $\pm1/2$, mean zero and the same endpoint covariance $C^{\mathrm{end}}_\mu(z-z') = \frac14 e^{-\mu|z-z'|_1}$ (Theorems 6.1(b) and 6.4(c)). The product neuron multiplies two independent telegraph signals, one along each coordinate, each flipping at the points of a Poisson process. The Mondrian neuron colors the cells of a random rectangular partition with independent signs; two points lie in a common cell with probability $e^{-\mu|z-z'|_1}$ (Theorem 6.2(e) with $s = \mu$). Because the covariances agree, both architectures have the same Gaussian reference in the width limit (Proposition 9.2).

The neurons are defined exactly; Section 6 states their properties, proved in Appendix A. Random objects live on a probability space $(\Gamma, \mathcal{A}, \mathbb{P})$; $\mathrm{Exp}(c)$ has density ce^{-ct} on $(0,\infty)$, $U(0,1)$ is uniform on $(0,1)$, and a *uniform sign* is uniform on $\{-1,1\}$.

Definition 2.3 (Product (telegraph) neuron)**.** Let $\mathcal{N}$ be a Poisson process on $\mathbb{R}$ of intensity $\mu/2$ and ϵ an independent uniform sign. The stationary symmetric telegraph process of flip rate $\mu/2$ is $\sigma(t) = \epsilon(-1)^{\mathcal{N}((0,t])}$ for $t \ge 0$ and $\sigma(t) = \epsilon(-1)^{\mathcal{N}((t,0])}$ for $t < 0$ (set $\sigma = \epsilon$ on the null event where $\mathcal{N}$ is not locally finite or charges 0). With τ an independent copy, the product neuron is $g(z) = \frac12\sigma(z_1)\tau(z_2)$.

Definition 2.4 (Boxes, colored partitions and the canonical Mondrian)**.** A *box* is $C = [a_1, b_1] \times [c_1, d_1]$ with $a_1 \le b_1$, $c_1 \le d_1$, and half-perimeter $p(C) = (b_1 - a_1) + (d_1 - c_1)$. For a box C with nonempty interior, $\mathrm{Part}^{\mathrm{col}}(C)$ is the set of finite sets of pairs (R, ξ), R a closed box with nonempty interior and $\xi \in \{-1, 1\}$, such that the boxes R have pairwise disjoint interiors and union C; it is encoded by finite counting measures on $\mathbb{R}^4 \times \{-1, 1\}$ (Lemma A.1). For boxes $B \subseteq C$ with nonempty interiors the restriction is $\mathrm{res}_B(P) = \{(R \cap B, \xi) : (R, \xi) \in P,\ \mathrm{int}(R \cap B) \neq \emptyset\}$; images under translations and under θ_1, θ_2 act on closed boxes. The *lower-left representative* of $R = [a_1, b_1] \times [c_1, d_1]$ is $R^{\mathrm{ll}} = [a_1, b_1) \times [c_1, d_1)$.

Let Tr be the set of finite words over $\{0, 1\}$ (root $\emptyset$) and let $(E_w, D_w, X_w, \xi_w)_{w \in \mathrm{Tr}}$ be independent, $E_w \sim \mathrm{Exp}(1)$, $D_w, X_w \sim U(0, 1)$, ξ_w a uniform sign. For a box C and budget $s \ge 0$ put $R_\emptyset = C$, $b_\emptyset = 0$; a born node w with box $R_w = [a_1, b_1] \times [c_1, d_1]$ dies at $d_w = b_w + E_w / p(R_w)$ ($d_w = \infty$ if $p(R_w) = 0$). If $d_w < s$, w splits: if $D_w < (b_1 - a_1)/p(R_w)$ at the vertical line $x = a_1 + X_w(b_1 - a_1)$, otherwise at the horizontal line $y = c_1 + X_w(d_1 - c_1)$; its children $w0, w1$ receive the two closed pieces (left/lower, right/upper) and are born at d_w. Born nodes with $d_w \ge s$ are leaves. The output $\Pi^{\mathrm{can}}(C, s) = \{(R_w, \xi_w) : w \text{ a leaf}\}$ has law $M^{\mathrm{col}}(C, s)$. Thus each cell with side lengths w', h' splits at rate $w' + h'$, orthogonally to the first axis with probability $w'/(w' + h')$, at a uniform position, and the daughters evolve independently until the budget is exhausted.

Definition 2.5 (Whole-plane Mondrian neuron; closed cells and half-open representatives)**.** Let $B_n = [-n, n]^2$. A *consistent colored Mondrian tessellation* of budget μ is a sequence $P = (P_n)_{n \ge 1}$ of random elements $P_n \in \mathrm{Part}^{\mathrm{col}}(B_n)$ on one probability space with $P_n \sim M^{\mathrm{col}}(B_n, \mu)$ for every n and, almost surely, $\mathrm{res}_{B_n}(P_{n+1}) = P_n$ for all n (the *consistency event*). For $z \in \mathbb{R}^2$ let $n_z = \min\{n \ge 1 : z \in [-n, n)^2\}$, and for $n \ge n_z$ let $(R_n(z), \xi_n(z))$ be the cell of P_n with $z \in R_n(z)^{\mathrm{ll}}$. The *closed geometric cell* is $R(z) = \bigcup_{n \ge n_z} R_n(z)$ and the *half-open representative* is $H(z) = \bigcup_{n \ge n_z} R_n(z)^{\mathrm{ll}}$. The *Mondrian neuron* is the field $g(z) = \frac{1}{2}\xi_{n_z}(z)$, defined for every z and every outcome (lower-left convention). An *admissible boundary convention* is any $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable g' such that on the consistency event $g'(z) = \xi/2$ for some cell (R, ξ) of P_{n_z+1} with $z \in R$. Maps act on tessellations through closed cells; half-open representatives of an image are obtained by applying the lower-left convention to its closed cells.

Definition 2.6 (Neural fields)**.** For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \ge 1$ and independent copies $g^b_{1,a}, \dots, g^b_{N,a}$ of the neuron of Definition 2.3 or 2.5 with $\mu = 1/a$, on the product $(\Gamma_b, \mathbb{P}_b)$ of N copies of its probability space, the neural field of width N is

$$F^b_{N,a} = N^{-1/2} \sum_{i=1}^{N} g^b_{i,a}, \qquad |F^b_{N,a}| \le \sqrt{N}/2. \tag{5}$$

Remark 2.7 (What the width counts). The width N counts process neurons. Each neuron uses countably many random variables; on every bounded region it has almost surely finitely many telegraph jump lines or Mondrian cell edges, but that number is random, unbounded and grows with the region. A finite-width stage is therefore not a random field encoded by finitely many global parameters for the whole plane. For background on the Mondrian process see [18]; for stationary tessellations stable under iteration see [15]. These are background attributions; every property used below is proved in Appendix A.

2.5 Potentials, Wick and coupling conventions

The potential has three parts with separate jobs. The Wick-ordered quartic $W_{a,\Lambda}$ on the physical square is the interaction that survives in the limit; its subtraction $-6c_a x^2 + 3c_a^2$ lets the interaction

converge as $a \downarrow 0$, although $c_a \to \infty$ (Theorem 8.2(a)). The negative quadratic $-T_{\Omega,a}$ implements the mass shift of Section 2.2. The stabilizing quartic $\varepsilon J_{\Omega,\Lambda}$ on $\Omega \setminus \Lambda$ keeps the weight e^{-V} bounded, see (18), which the width limit uses; it is removed at fixed a, Ω, Λ (Proposition 9.4).

Definition 2.8 (Integrated potentials)**.** For $v \in E$, a physical square Λ, an auxiliary square $\Omega \supseteq \bar{\Lambda}$, a regulator a and $\varepsilon \geq 0$, put

$$W_{a,\Lambda}(v) = u \int_\Lambda (v^4 - 6c_a v^2 + 3c_a^2)\, dz, \qquad J_{\Omega,\Lambda}(v) = \int_{\Omega\setminus\Lambda} v^4\, dz,$$
$$T_{\Omega,a}(v) = \frac{\kappa_a^2}{2} \int_\Omega v^2\, dz, \qquad V_{\varepsilon,\Omega,a;\Lambda} = W_{a,\Lambda} + \varepsilon J_{\Omega,\Lambda} - T_{\Omega,a}. \tag{6}$$

Equivalently $V_{\varepsilon,\Omega,a;\Lambda}(v) = \int_\Omega v_V(z, v(z))\, dz$ with the Borel integrand

$$v_V(z,x) = \mathbf{1}_\Lambda(z)\, u(x^4 - 6c_a x^2 + 3c_a^2) + \varepsilon \mathbf{1}_{\Omega\setminus\Lambda}(z)\, x^4 - \tfrac{1}{2}\kappa_a^2 x^2. \tag{7}$$

The negative quadratic term acts on all of Ω, including Λ. These are integrated potentials of ordinary functions; they are never evaluated as pointwise polynomials of an arbitrary distribution.

Remark 2.9 (Wick and coupling conventions). At a positive regulator, $x^4 - 6c_a x^2 + 3c_a^2$ is the fourth Wick polynomial at the variance $c_a = C_a(0)$ of the regulated Gaussian physical reference field (the centered Gaussian field with covariance C_a): own-mass, own-regulator Wick ordering of that Gaussian reference. The endpoint variance $1/4$ is never a Wick constant, and no finite-width or interacting law is asserted to have variance c_a. At $a = 0$ the interaction is the common-space fourth-chaos L^2 Wick limit $U_{0,\Lambda}$ constructed in Section 8; its precise construction statement is Theorem 8.2(b). The coefficient u multiplies the Wick quartic with no factorial divisor, as in [8] (polynomial λX^4) and [2] ($\lambda \int :P(\phi):_0$, $P(X) = X^4$); the finite-volume neural construction of [7] uses $\lambda/4!$, so its $\lambda/24$ plays the role of u. In natural units the field is dimensionless in two dimensions; m, μ, the telegraph flip rate $\mu/2$ and the Mondrian budget μ have units of inverse length; u and κ_a^2 of inverse length squared; u/m^2 is the dimensionless coupling, and the dilation of Section 10 preserves it.

2.6 The normalized laws

Definition 2.10 introduces the laws met along the stages of Section 2.2. The neural laws (a) are the objects of the main theorem, and the full Gaussian tilt (b) is their width limit. For $\varepsilon > 0$, the residual laws (c) rewrite that tilt as a quartic tilt of the Gaussian law $G'_{\Omega,a}$; they also define its stabilizer-free version at $\varepsilon = 0$. The physical laws (d) arise when the auxiliary square is removed; at $a = 0$ they are the continuum laws on a fixed square. Item (e) names the growing-volume laws and the target. Remark 9.7 lists the four fixed-volume arrows joining (a)–(d).

The following objects are well defined by results of Sections 6–8 (finiteness and positivity of normalizers, existence of Gaussian versions and of the L^2 Wick limit); the proofs of those results, in Appendices A–C, do not use this definition.

Definition 2.10 (Laws)**.** Fix a regulator a, a physical square Λ and an auxiliary square $\Omega \supseteq \bar{\Lambda}$.

(a) *Neural laws* ($\varepsilon > 0$). With $Z_N^b = \mathbb{E}_b \exp[-V_{\varepsilon,\Omega,a;\Lambda}(F_{N,a}^b)]$,

$$P_{N,\varepsilon,\Omega,a;\Lambda}^b(B) = \frac{\mathbb{E}_b\big[\mathbf{1}_B(\iota F_{N,a}^b)\, e^{-V_{\varepsilon,\Omega,a;\Lambda}(F_{N,a}^b)}\big]}{Z_N^b}, \qquad B \in \mathcal{B}(X). \tag{8}$$

These are *annealed* laws: the weight reweights the joint law of all neuron randomness. (The same formula is meaningful at $\varepsilon = 0$ for finite N, Theorem 6.5(a); that finite-width extension is not used below.)

(b) *Full Gaussian tilt* ($\varepsilon > 0$). Let G_μ be the law on $C(\mathbb{R}^2)$ of the continuous centered Gaussian field γ_μ with covariance C_μ^{end} (Theorem 7.1). With $Z_{\mathrm{full}} = \mathbb{E}_{G_\mu} e^{-V(\gamma_\mu)}$, $\nu^v_{\varepsilon,\Omega,a;\Lambda}(B) = \mathbb{E}_{G_\mu}[\mathbf{1}_B(\iota\gamma_\mu)e^{-V(\gamma_\mu)}]/Z_{\mathrm{full}}$.

(c) *Mass tilt and residual laws* ($\varepsilon \geq 0$). Let $\mathrm{Det}_{\Omega,a} = \mathbb{E}_{G_\mu} e^{T_{\Omega,a}(\gamma_\mu)}$ and $G'_{\Omega,a} = \mathrm{Det}^{-1}_{\Omega,a} e^{T_{\Omega,a}} G_\mu$, the centered Gaussian law on $C(\mathbb{R}^2)$ with covariance $C'_{\Omega,a}$ (Theorem 7.4). Put $z_{\varepsilon,\Omega,a;\Lambda} = \mathbb{E}_{G'_{\Omega,a}} e^{-W_{a,\Lambda}-\varepsilon J_{\Omega,\Lambda}}$, $d\mathrm{Res}_{\varepsilon,\Omega,a;\Lambda} = z^{-1}_{\varepsilon,\Omega,a;\Lambda} e^{-W_{a,\Lambda}-\varepsilon J_{\Omega,\Lambda}}\, dG'_{\Omega,a}$ and $\nu^{(2)}_{\Omega,a;\Lambda} = \iota_* \mathrm{Res}_{0,\Omega,a;\Lambda}$.

(d) *Physical laws* ($0 \leq a < 1/m$). On the common Gaussian space $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0)$ of Theorem 7.7, let $\Phi_a : \Gamma_0 \to X$ be the Borel version of Theorem 7.10 and, for $a > 0$, $\tilde{\phi}_a$ the continuous modification of Theorem 7.9. Put $U_{a,\Lambda} = W_{a,\Lambda}(\tilde{\phi}_a)$ for $a > 0$, $U_{0,\Lambda} = L^2(\mathbb{P}_0)\text{-}\lim_{a\downarrow 0} U_{a,\Lambda}$ (Theorem 8.2(a)), $Z_{a,\Lambda} = \mathbb{E}_0 e^{-U_{a,\Lambda}}$ and

$$\nu_{a,\Lambda}(B) = \mathbb{E}_0\big[\mathbf{1}_B(\Phi_a)\, e^{-U_{a,\Lambda}}\big]/Z_{a,\Lambda}. \tag{9}$$

(e) *Continuum finite-volume laws and the target.* $\nu_n = \nu_{0,\Lambda_n}$. The classical finite-volume laws $\nu_{m,u,\Lambda}$ and the classical target ν^S are probabilities on the real tempered distributions $\mathcal{S}'(\mathbb{R}^2)$ with the cylinder σ-algebra $\mathcal{C}$ (Definition 10.3, Theorem 10.8); $\nu = \nu^X_\infty$ is the X-valued version of ν^S (Theorem 10.10).

Symbol	Meaning	Bounds and traps
Z^b_N	neural normalizer	$e^{-\lvert\Omega\rvert M_N} \leq Z^b_N \leq e^{\lvert\Omega\rvert M_N}$, $M_N = \sup\{\lvert v_V(z,x)\rvert : z \in \Omega, \lvert x\rvert \leq \sqrt{N}/2\}$; not ≥ 1 in general
Z_{full}	full Gaussian normalizer	$Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a}\, z_{\varepsilon,\Omega,a;\Lambda}$ for $\varepsilon > 0$; no monotonicity in Ω is asserted (Remark 9.7)
$\mathrm{Det}_{\Omega,a}$	quadratic-tilt normalizer	$1 \leq \mathrm{Det}_{\Omega,a} \leq \exp(\kappa_a^2\lvert\Omega\rvert/(8a^2m^2))$; nondecreasing in Ω
$z_{\varepsilon,\Omega,a;\Lambda}$	residual normalizer	$e^{-3u\lvert\Lambda\rvert c_a^2 - 3\varepsilon\lvert\Omega\setminus\Lambda\rvert c_a^2} \leq z_\varepsilon \leq e^{6u\lvert\Lambda\rvert c_a^2}$
$Z_{a,\Lambda}$	physical normalizer	$1 \leq Z_{a,\Lambda} \leq \Xi_1(\Lambda)$, $0 \leq a < 1/m$; physical laws only; no uniformity in Λ

Table 1: Normalizers. The bound $Z \geq 1$ holds only for the physical laws on the common Gaussian space.

2.7 Tests, distributions and pairings

Definition 2.11 (Tests and complex-linear pairings)**.** For an open set U of a Euclidean space, $\mathcal{D}(U) = C^\infty_c(U;\mathbb{C})$ carries its standard inductive-limit (LF) topology over the Fréchet spaces $\mathcal{D}_K(U)$, $K \subset U$ compact, with seminorms $\|F\|_{C^M} = \max_{\lvert\alpha\rvert\leq M} \sup \lvert\partial^\alpha F\rvert$. The dual $\mathcal{D}'(U)$ consists of continuous *complex-linear* functionals; pairings are linear in the test. A real distribution v extends by $v(f+ig) = v(f) + iv(g)$. For $f_1, \ldots, f_k \in \mathcal{D}(\mathbb{R}^2)$ put $(f_1 \otimes \cdots \otimes f_k)(z_1, \ldots, z_k) = \prod_r f_r(z_r)$. For

$F \in \mathcal{D}(((-j,j)^2)^k)$, regarded as a smooth periodic function on $\mathbb{T}_j^k$ after extension by zero,

$$A_{j,k}(F) = \Big\| \prod_{r=1}^{k} (1 - \Delta_{z_r}) F \Big\|_{L^2(\mathbb{T}_j^k)}. \tag{10}$$

2.8 Reflections and half-plane observables

The positivity statements compare an observable with its mirror image. An observable is Σ_+^θ-measurable when it depends only on smearings of the field by tests supported in the open half-plane H_+^θ; then $F(\theta v)$ is the same observable applied to the reflected field, and so depends on the opposite half-plane. Inequality (11) requires the average of $\overline{F(\theta v)}F(v)$ to be nonnegative. The form $\Pi_\rho(f)$ is the analogous expression for polynomial observables built from compact joint tests, used in clause (E2) of Theorem 4.1.

Definition 2.12 (Central reflections, half-planes and positivity)**.** For $\theta \in \{\theta_1, \theta_2\}$ let H_+^θ be the open positive half-plane ($\{z_1 > 0\}$ for θ_1, $\{z_2 > 0\}$ for θ_2), define $(\theta v)(f) = v(f \circ \theta)$ for $v \in X$, and let Σ_+^θ be the σ-algebra on X generated by the evaluations $v \mapsto v(f)$, real $f \in C_c^\infty(H_+^\theta)$. A Borel probability ρ on X is *reflection positive across θ for bounded observables* if

$$\int_X \overline{F(\theta v)}\, F(v)\, \rho(dv) \geq 0 \tag{11}$$

for every bounded complex Σ_+^θ-measurable F. For $J \geq 0$, $f_0 \in \mathbb{C}$ and $f_r \in \mathcal{D}((H_+^\theta)^r)$, $1 \leq r \leq J$, put $(\theta_r \overline{f_r})(z_1, \dots, z_r) = \overline{f_r(\theta z_1, \dots, \theta z_r)}$ (value $\overline{f_0}$ at $r = 0$) and

$$\mathcal{O}_f(v) = \sum_{r=0}^{J} v^{\otimes r}(f_r), \qquad \Pi_\rho(f) = \sum_{r,s=0}^{J} \mathcal{S}_{r+s}^\rho((\theta_r \overline{f_r}) \otimes f_s), \tag{12}$$

where $v^{\otimes r}$ and $\mathcal{S}_k^\rho$ are the tensor evaluations and correlation distributions of Section 12, $v^{\otimes 0}(c) = c$ and $\mathcal{S}_0^\rho(c) = c$.

3 Classical inputs

The inputs answer distinct questions. C1 supplies the infinite-volume states, their Osterwalder–Schrader properties and local regularity, and convergence of free-boundary finite-volume laws once the pressure is differentiable. C2 gives uniqueness at weak coupling, which makes the two states coincide; Definition 3.2 turns its existential constants into the regime $0 < u/m^2 < g_*$. C3 provides reconstruction and a kernel theorem; C4 provides lattice approximation and correlation inequalities, and C5 a third-cumulant inequality, both used for the volume-uniform bounds of Theorem 10.10. C6 lists standard tools.

The following published results are used with attribution. Their proofs are not audited here. Each formulation records exactly the content used; its application to the present objects is verified at the point of use. Page references are to the printed pages; Appendix D records for each page whether the classical-input review read it as a page image or as supplied text.

Conventions shared by C1–C4. For a mass $M > 0$, P_M is the centered Gaussian probability on $(\mathcal{S}'(\mathbb{R}^2), \mathcal{C})$ with covariance $\langle f, (-\Delta + M^2)^{-1} g\rangle$ (the free measure μ_0 of [8, p. 496]); for open U, $\mathcal{C}_U$ is generated by $\psi \mapsto \psi(f)$ with real $f \in \mathcal{S}$, $\operatorname{supp} f \subset U$. The interaction is $\lambda \int_\Lambda :\phi^4:$ with whole-plane,

same-mass Wick ordering and no factorial; "free boundary" means the whole-plane Gaussian with the interaction restricted to the bounded region and zero external field [8, pp. 496–497], [2, (4), p. 685], [10, (IX.5a), pp. 311–312]. In these sources $:\phi^4:$ denotes Gaussian Wick normal ordering at the whole-plane reference covariance; for degree four this is the fourth-chaos variable realized as the L^2 limit of own-variance Wick quartics of mollified fields. Lemma 10.1 proves that this classical variable exists and has positive finite exponential normalizers using only the free Gaussian covariance and C6. The source finite-volume convention is applied only after these provisos are proved. The separate identification with the common-space regulator limit is Proposition 10.4; it is not an input to the restricted classical route.

Classical input C1 (Fröhlich–Simon)**.** Fix $M > 0$ and $P(X) = \lambda X^4$, $\lambda > 0$ [8].

(i) [8, Thm. 4.1, p. 509]: the Section-4 limits ν_+ and ν_- at zero external field are probabilities on $(\mathcal{S}', \mathcal{C})$ satisfying all Osterwalder–Schrader axioms except a priori clustering; used are proper Euclidean invariance and Osterwalder–Schrader positivity, in the measure form of [8, p. 495, (a)–(b)] or for moments of positive-time (possibly time-ordered) tests; both readings are treated in Section 10.

(ii) [8, §6 definition, pp. 513–514, Remarks 1–2 and Thm. 6.1; Thm. 7.2, p. 517; (1.2b), p. 498]: ν_+ and ν_- are ultraregular: for every bounded open B, the restriction of $\nu_\pm$ to $\mathcal{C}_B$ has a density f_B with respect to P_M lying in every $L^p(P_M)$, $p < \infty$, and $f_B = e^{-U_B} e^{-\alpha_\infty |B|} g_{\partial B}$ with $g_{\partial B} > 0$ almost everywhere, in every L^p, and measurable for the boundary σ-algebra, which by [8, p. 514, Remark 2] is either the intersection of $\mathcal{C}_R$ over open $R \supseteq \partial B$, or generated by $\phi(f)$ with $f \in H_{-1}$ of compact support in ∂B. Perimeter estimates are stated for rectangles.

(iii) [8, Cor. 4.3(ii), p. 510]: if $\nu_+ = \nu_-$, the pressure $\alpha_\infty(P - hX)$ is differentiable at $h = 0$ (only this implication is used).

(iv) [8, Thm. 5.2, p. 512]: if the pressure is differentiable at $h = 0$, the classical boundary-condition finite-volume measures, including the free ones, converge as Λ increases to $\mathbb{R}^2$ (ordinary exhaustion, not the iterated convention of [8, p. 503]) to $\nu_+ = \nu_-$. It is used only along centered squares, and only for the characteristic functions of finitely many compact smooth tests; the conclusion is the same under the moment or Laplace-functional readings of convergence (Lemma 10.6).

The two boundary algebras in C1(ii) are denoted BC1 (the neighborhood-intersection algebra) and BC2 (the boundary-supported H_{-1} algebra, with the completion specified below). The next remark specifies the two positivity readings.

Remark 3.1 (Literal alternative readings). The positivity alternatives above are retained separately. In the measure reading (b_M), the inequality is for every complex $F \in L^2(\mathcal{S}', \Sigma_+, \nu_\pm)$. In the weaker reading (b_P), it is for every polynomial $F = \sum_r c_r \prod_{i=1}^{n_r} \psi(f_{r,i})$, $c_r \in \mathbb{C}$, $n_r \geq 0$, real Schwartz $f_{r,i}$, whose individual product supports lie in $0 < (z_1)_2 < \cdots < (z_{n_r})_2$; empty products are constants. For C1(iv), the alternatives are convergence of the characteristic value for every real compact g, convergence of every moment of $\psi(g)$, or convergence of $\mathbb{E}e^{t\psi(g)}$ for every real t. The first supplies all frequencies by replacing g with tg. In BC2 the boundary algebra is completed by P_M-null sets; its variables for boundary-supported real H^{-1} tests are defined by their Gaussian L^2 limits of Schwartz approximants. Both BC1 and BC2, both positivity readings and all three convergence readings are treated below.

Classical input C2 (Albeverio–Høegh-Krohn–Zegarlinski)**.** [2] work on a set $\Omega_A \subset \mathcal{S}'(\mathbb{R}^2)$ with the traces of the weak-Borel sets; for closed D, $\mathcal{B}_D = \bigcap_{\Lambda' \supseteq D \text{ open}} \mathcal{B}_{\Lambda'}$ and $\Sigma_{\Lambda^c} = \mathcal{B}_{\Lambda^c} \cap \Omega_A$ [2, p. 684]; the specification consists of kernels $E^\eta_{\Lambda^c}$ with (a) each $E^\eta_{\Lambda^c}$ a probability with $E^\eta_{\Lambda^c}(F) = F(\eta)$ for Σ_{Λ^c}-measurable F, (b) $\eta \mapsto E^\eta_{\Lambda^c}(F)$ Σ_{Λ^c}-measurable, (c) compatibility; a Gibbs measure satisfies $\mu(E_{\Lambda^c}F) = \mu(F)$ for all $F \in L^1(\mu)$ [2, (7)–(8), p. 686]; $\mathcal{G}_{\mathrm{ur}}$ is the ultraregular Gibbs class [2, (10)–(12), p. 686].

(i) *Membership* [2, pp. 686–687], citing [8, Thm. 7.2]: both states ν_+ and ν_- of C1 at mass m_0 belong to $\mathcal{G}_{\mathrm{ur}}$ ([2] prints the maximal measures without sign suffix; [8, Thm. 7.2] names both signs).

(ii) *Regime and uniqueness* [2, Theorem, p. 687; after (47), p. 691]: there are $M_0, \lambda_0 > 0$ such that for every $m_0 > M_0$ and $0 < \lambda < \lambda_0$ the class $\mathcal{G}_{\mathrm{ur}}$ for λX^4 at mass m_0 is a singleton.

(iii) *Boundary-condition independence* [2, (13)–(14), p. 687; completed p. 691]: in that regime, for $\mu, \tilde{\mu} \in \mathcal{G}_{\mathrm{ur}}$ and every polynomial function F measurable for Σ_{Λ_0}, Λ_0 bounded,

$$\lim_{\Lambda \in F_0} (\mu \otimes \tilde{\mu}) \big| E^\eta_{\Lambda^c} F - E^{\tilde{\eta}}_{\Lambda^c} F \big| = 0,$$

where F_0 is the increasing sequence of centered squares. "Polynomial function" is read as $Q(\phi(f_1), \dots, \phi(f_k))$ with Q a real polynomial and real $f_i \in \mathcal{S}$ supported in a bounded open Λ_0; this reading is supported by [2, p. 689, after (31)], which localizes F in a finite union of unit cubes (Appendix D).

Definition 3.2 (Fixed admissible pair, threshold and regime)**.** Fix once and for all one pair (M_0, λ_0) of C2(ii), a reference mass $M_{\mathrm{ref}} > M_0$ and $0 < \lambda_* \le \lambda_0$, and put $g_* = \lambda_* / M^2_{\mathrm{ref}} > 0$. The *regime* consists of the pairs (m, u) with $m > 0$, $u > 0$ and $0 < u/m^2 < g_*$. This instantiates an existential statement: no numerical value of M_{ref}, λ_* or g_* is supplied and no larger coupling range is inferred. For (m, u) in the regime put $s = m/M_{\mathrm{ref}}$ and $\lambda = u/s^2 = uM^2_{\mathrm{ref}}/m^2$; then $0 < \lambda < \lambda_*$.

Classical input C3 (Reconstruction and tempered kernels)**.** (i) [8, Thm. 1.0, first assertion, and conditions (a)–(d), p. 495]: let μ be a probability on $(\mathcal{S}'(\mathbb{R}^2), \mathcal{C})$ with (a) invariance under proper Euclidean motions; (b) $\int \overline{\Theta F}\, F \, d\mu \ge 0$ for Σ_+-measurable F, where Θ reflects the last coordinate z_2 and Σ_+ is generated by $\phi(f)$, $f \in \mathcal{S}$, $\operatorname{supp} f \subset \{z_2 > 0\}$, and the inequality is required for every $F \in L^2(\Sigma_+, \mu)$; (c) a norm $|||\cdot|||$ continuous on $\mathcal{S}$ such that $f \mapsto \int e^{\phi(f)} d\mu$ is uniformly bounded and $|||\cdot|||$-continuous on $\{|||f||| \le 1\}$; (d) ergodicity of the translation group of $\mathbb{R}^2$. Then the moments of μ are the Schwinger functions of a unique relativistic quantum field theory satisfying all the Wightman axioms, listed on [8, p. 494] as temperedness, Poincaré invariance, positivity, spectrum condition, locality and clustering; by [8, p. 495, Remark 2], (d) implies uniqueness of the physical vacuum. [8] states this as a consequence of [16, 17], whose bodies were not available.

(ii) [8, p. 494]: the Wightman functions are boundary values of functions holomorphic in the permuted extended tube, which contains the Euclidean points with pairwise distinct arguments; the Schwinger functions are their values there and coincide with the moments. Uniqueness of a function holomorphic in a tube with vanishing boundary values is used as a standard attributed fact [20] (not available).

(iii) [14, Lecture 15, p. 2, Thm. 1]: $K \in \mathcal{S}'(\mathbb{R}^N \times \mathbb{R}^n)$ corresponds bijectively to linear maps $L : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^N)$ with $\psi \mapsto (L\psi)(\varphi)$ continuous for each φ, through $(L\psi)(\varphi) = K(\varphi \otimes \psi)$; the pairing is bilinear.

Not used: [8, Thm. 4.4], an identification of an Osterwalder–Schrader semigroup with a Wightman energy, the spectrum condition as an input, and tube uniqueness as an input to (d). Clustering, ergodicity and the vacuum statements of the target are obtained in the order of Theorem 10.16: Euclidean mixing first, reconstruction afterwards.

Below, C3(ii') is shorthand for the tube-uniqueness assertion contained in C3(ii).

Classical input C4 (Guerra–Rosen–Simon, free boundary condition)**.** Fix $m, u > 0$ and a centered square Λ of side L; take meshes $\delta = L/(2k+1)$, so that $\partial\Lambda$ lies midway between lattice sites [10, §IX.1, pp. 309–310]. Let $I_\delta = \{i \in \mathbb{Z}^2 : \delta i \in \Lambda\}$, $A_\delta = (m^2\delta^2 + 4)I -$ adjacency on $\mathbb{Z}^2$ [10, (IX.2)], $K_\delta = (A_\delta^{-1})|_{I_\delta \times I_\delta}$ and $B_\delta = K_\delta^{-1}$ (the free precision, [10, (IX.4c), p. 311], there written A_Λ^F), whose off-diagonal entries are nonpositive; the diagonal c_δ of K_δ is the whole-lattice variance. The normalized lattice law has density proportional to $\exp[-\frac{1}{2}q^T B_\delta q - u\delta^2 \sum_{i\in I_\delta}(q_i^4 - 6c_\delta q_i^2 + 3c_\delta^2)]$ [10, pp. 311–312]. For real $h \in C_c^\infty(\Lambda)$ put $Y_\delta(h) = \delta^2 \sum_i h(\delta i) q_i$ [10, (IX.19), p. 316].

(i) *Fixed-volume convergence* [10, (IX.19)–(IX.21), p. 316; Lemmas IX.3 (with the remark covering the sharp indicator of Λ) and IX.4, p. 317; review to p. 319]: on a common Gaussian realization of mass m, the lattice smearings converge to the continuum smearings and the lattice interaction densities to the continuum density $e^{-u\int_\Lambda :\phi^4:}$ in every finite Gaussian L^p, with positive normalizer limits; the smeared Schwinger functions converge.

(ii) *Correlation inequalities* [10, §IX.2(i),(iii), p. 324], boundary condition F, zero external field: for nonnegative real $h_1, \dots, h_r \in C_c^\infty(\Lambda)$, $\mathbb{E}_{\nu_\Lambda} \prod_i \phi(h_i) \geq 0$, and for nonnegative $g \in C_c^\infty(\Lambda)$ the truncated four-point function smeared with g is nonpositive.

Not used: [10, Thm. IX.5, p. 320] (periodic and Neumann) and Thm. IX.7, and any free–Neumann comparison. *Attribution limit.* [10, pp. 316–319] review the free case using its references [29], [62] and [28], and sketch a uniform exponential estimate (p. 317); these antecedents were not available, and the complete classical proof was not audited.

Classical input C5 (Simon–Griffiths)**.** [19, Thm. 1, p. 148; (4) and Thm. 2, (5), p. 151]: for finitely many real variables with density proportional to $\prod_i F_i(s_i) \exp(\sum_{i\neq j} a_{ij} s_i s_j + \sum_i h_i s_i)$, where $F_i(s) = e^{-a_i s^4 - b_i s^2}$ with $a_i > 0$ and $b_i \in \mathbb{R}$, the ordered-pair coefficients satisfy $a_{ij} > 0$, and $h_i \geq 0$, every mixed third cumulant $\langle s_i s_j s_k\rangle + 2\langle s_i\rangle\langle s_j\rangle\langle s_k\rangle - \langle s_i\rangle\langle s_j s_k\rangle - \langle s_j\rangle\langle s_i s_k\rangle - \langle s_k\rangle\langle s_i s_j\rangle$ (repeated indices allowed) is nonpositive. This is a finite-system statement.

Classical input C6 (Standard tools)**.** Standard analysis and probability, non-exhaustively and always with hypotheses checked where used, including: Fubini–Tonelli, dominated and monotone convergence, Fatou, Hölder, Minkowski, Jensen (also conditional), Markov, Chernoff, Paley–Zygmund, the portmanteau and continuous-mapping theorems, Lévy's continuity theorem, the method of moments, Prokhorov's theorem on Polish spaces, Dynkin's π–λ and the monotone-class theorems, Doob–Dynkin factorization and countable dependence of functions measurable for generated σ-algebras, the Lusin–Souslin Borel-image theorem, conditional expectations, martingale convergence for increasing filtrations, polynomial density in L^2 of a finite-dimensional law with finite exponential moments, Fourier series and Parseval on tori and Fejér kernels, compact Sobolev embeddings on a fixed torus, Banach–Steinhaus on Fréchet spaces, density of finite sums of product tests, Gaussian likelihoods and the Cameron–Martin shift for finite-dimensional Gaussian vectors, the Hermite expansion, real and complex L^2 duality on finite measure spaces, Bochner integration in L^2, approximate identities, the layer-cake formula, uniqueness of Fourier transforms of finite measures, and the Schur product theorem. The following standard formulations are stated explicitly:

(a) *Hypercontractivity* (Nelson): for Y in the k-th homogeneous Wiener chaos of a Gaussian Hilbert space and real $r \geq 2$, $\|Y\|_r \leq (r-1)^{k/2}\|Y\|_2$.

(b) *Wick products and chaos*: for centered jointly Gaussian ξ_i, η_i, $\mathbb{E}[:\xi_1 \cdots \xi_k: :\eta_1 \cdots \eta_l:] = \delta_{kl} \sum_{\pi \in S_k} \prod_i \mathbb{E}[\xi_i \eta_{\pi(i)}]$; the k-th chaos is the closed span of $:\xi_1 \cdots \xi_k:$, chaoses of different order are orthogonal, chaos of order ≥ 1 has mean zero, and finite products of jointly Gaussian variables with dense span generate a dense subspace of each chaos.

(c) *Hausdorff–Young and Plancherel* on $\mathbb{R}^2$ with the stated Fourier convention.

(d) *Continuity criterion* (Kolmogorov–Chentsov–Loève): a real process on $\mathbb{R}^d$ with $\mathbb{E}|Y(z) - Y(z')|^\alpha \leq c|z - z'|^{d+\beta}$ has a version with continuous paths for every outcome.

(e) *Kolmogorov extension* for consistent finite-dimensional laws, in particular centered Gaussian processes with a given positive semidefinite kernel, and for sequences of standard Borel spaces.

(f) *Isonormal processes*: every separable real Hilbert space carries an isonormal process.

(g) *Operators*: Neumann series; Hilbert–Schmidt operators and trace class ($\operatorname{tr} T^*T = \|T\|_{\mathrm{HS}}^2$); the spectral theorem for compact self-adjoint operators.

(h) *Point processes*: restrictions of a homogeneous Poisson process on $\mathbb{R}$ to disjoint Borel sets are independent; $t \mapsto -t$ preserves its law; a fixed point is almost surely not an atom; $\mathbb{E}(-1)^{\mathrm{Poi}(c)} = e^{-2c}$. Finite counting measures on a Polish space form a standard Borel space.

4 Main results

Two devices keep the main theorem compact. The enumerations (13) fix in advance the countable sets from which the regulator, the auxiliary half-side, the stabilizer and the width are chosen. The discrepancy dist_n of (14) adds to the bounded-Lipschitz distance β the differences in expected values of the first n powers of the first n local seminorms, so one number controls weak closeness and moment closeness at stage n. In Theorem 4.1, clause (S) asserts the sequence, (0) that its laws are well defined, (A)–(D) the convergence and identification statements, and (E) finite-stage positivity.

Fix once and for all the parameter enumerations

$$a(k) = \frac{1}{2mk}, \qquad R(k) = k, \qquad \varepsilon(k) = \frac{1}{k}, \qquad N(k) = k, \qquad k \in \mathbb{N}. \tag{13}$$

For $n \geq 1$ and probabilities ρ, η on X with finite moments of p_j^r for $j, r \leq n$, put

$$\mathrm{dist}_n(\rho, \eta) = \beta(\rho, \eta) + \sum_{j=1}^{n} \sum_{r=1}^{n} \big|\rho(p_j^r) - \eta(p_j^r)\big|. \tag{14}$$

Theorem 4.1 (Main theorem)**.** *Fix the admissible pair of Definition 3.2, and $m > 0$, $\ell > 0$, $u > 0$ with $0 < u/m^2 < g_*$. Assume the classical inputs C1–C5 in the formulations of Section 3 and the standard tools C6. No other hypothesis is made. Fix the localization system of Definition 2.2 and the enumerations* (13)*.*

(S) One sequence. *There is one deterministic sequence $(a_n, R_n, \varepsilon_n, N_n)_{n \geq 1}$, determined by m, ℓ, u, the localization system and the enumerations alone, and the same for both architectures, with*

$$0 < a_n < \min(1/n, 1/(2m)), \quad a_n < a_{n-1}/2; \qquad R_n \in \mathbb{N}, \quad R_n \geq \max(n, n\ell), \quad R_n > R_{n-1};$$

$$0 < \varepsilon_n < 1/n, \quad \varepsilon_n < \varepsilon_{n-1}/2; \qquad N_n \in \mathbb{N}, \quad N_n > n, \quad N_n > N_{n-1}$$

(conditions with index $n-1$ only for $n > 1$), such that with $\Omega_n = [-R_n, R_n]^2$ and $P_n^b = P^b_{N_n,\varepsilon_n,\Omega_n,a_n;\Lambda_n}$,

$$\operatorname{dist}_n(P_n^b, \nu_n) < 2^{-n} \qquad \text{for both } b \text{ and every } n \geq 1. \tag{15}$$

Apart from the discrepancies dist_n, which are determined by the localization system, no observable, test function, polynomial or functional enters this choice; in particular none of the objects quantified in (A)–(E) does. The physical index n is never passed to a subsequence.

(0) Laws. *Every P_n^b is the Borel probability (8) on X with positive finite normalizer, and every power of every p_j is integrable under it. The limit ν is the X-valued version of the classical target ν^S.*

(A) Whole field and local moments. *For both b, $P_n^b \Rightarrow \nu$ weakly on X; for all $j, r \geq 1$, $\sup_{n\geq 1,b} P_n^b(p_j^r) < \infty$ and $P_n^b(p_j^r) \to \nu(p_j^r)$.*

(B) Every later compact polynomial tuple. *For every $k \geq 1$, all real $f_1, \dots, f_k \in C_c^\infty(\mathbb{R}^2)$ (repetitions allowed) and every real or complex polynomial Q on $\mathbb{R}^k$ of any degree, all expectations below are absolutely finite,*

$$\mathbb{E}_{P_n^b} Q(v(f_1), \dots, v(f_k)) \to \mathbb{E}_\nu Q(v(f_1), \dots, v(f_k)), \qquad \sup_{n,b} \mathbb{E}_{P_n^b} |Q(v(f_1), \dots, v(f_k))|^q < \infty \ (q > 0),$$

the same powers are ν-integrable, and the laws of $(v(f_1), \dots, v(f_k))$ converge weakly.

(C) Joint compact-test correlations. *Put $\mathcal{L} = \{\nu\} \cup \{P_n^b : n \geq 1,\ b \in \{\mathrm{prod}, \mathrm{Mon}\}\}$ and $M_{j,r} = \sup_{\rho\in\mathcal{L}} \int p_j^r \, d\rho$, $M_{j,0} = 1$. For every $k \geq 1$:*

- *(C1) for every $v \in X$ there is a unique $v^{\otimes k} \in \mathcal{D}'((\mathbb{R}^2)^k)$ with $v^{\otimes k}(f_1 \otimes \cdots \otimes f_k) = \prod_r v(f_r)$ for $f_r \in \mathcal{D}(\mathbb{R}^2)$; it is symmetric, real in the sense $\overline{v^{\otimes k}(F)} = v^{\otimes k}(\overline{F})$, $v \mapsto v^{\otimes k}(F)$ is continuous on X, and for $\operatorname{supp} F \subset ((-j,j)^2)^k$, $|v^{\otimes k}(F)| \leq p_j(v)^k A_{j,k}(F) \leq 3^k |\mathbb{T}_j|^{k/2} p_j(v)^k \|F\|_{C^{2k}}$;*
- *(C2) for $\rho \in \mathcal{L}$, $\mathcal{S}_k^\rho(F) = \int_X v^{\otimes k}(F)\, \rho(dv)$ is absolutely defined, lies in $\mathcal{D}'((\mathbb{R}^2)^k)$, is the unique distribution extending $(f_1, \dots, f_k) \mapsto \int \prod_r v(f_r)\, \rho(dv)$, and $\sup_{\rho\in\mathcal{L}} |\mathcal{S}_k^\rho(F)| \leq M_{j,k} A_{j,k}(F)$ for $\operatorname{supp} F \subset ((-j,j)^2)^k$, with $M_{j,k} < \infty$;*
- *(C3) with $\mathcal{S}_{n,k}^b = \mathcal{S}_k^{P_n^b}$ and $\mathcal{S}_{\infty,k} = \mathcal{S}_k^\nu$, $\mathcal{S}_{n,k}^b(F) \to \mathcal{S}_{\infty,k}(F)$ for every $F \in \mathcal{D}((\mathbb{R}^2)^k)$, that is, in $\sigma(\mathcal{D}'((\mathbb{R}^2)^k), \mathcal{D}((\mathbb{R}^2)^k))$.*

(D) Target identification. *$\mathcal{S}_{\infty,k}$ is the restriction to $\mathcal{D}((\mathbb{R}^2)^k)$ of the tempered Schwinger distribution Schw_k of ν^S (Theorem 10.16(v)).*

(E) Finite-stage central reflection positivity. *Fix $\theta \in \{\theta_1, \theta_2\}$.*

- *(E1) Every P_n^b satisfies (11) for every bounded complex Σ_+^θ-measurable F.*
- *(E2) For every $J \geq 0$, $f_0 \in \mathbb{C}$ and $f_r \in \mathcal{D}((H_+^\theta)^r)$, and every $\rho \in \mathcal{L}$: $\mathcal{O}_f$ and $\mathcal{O}_f \circ \theta$ lie in $L^2(\rho)$ with separate bounds, every term of $\Pi_\rho(f)$ is absolutely finite and $\Pi_\rho(f) = \int \overline{\mathcal{O}_f(\theta v)} \mathcal{O}_f(v)\, \rho(dv)$; for $\rho = P_n^b$ it is a nonnegative real number, and $\Pi_{P_n^b}(f) \to \Pi_\nu(f) \geq 0$.*

Theorem 4.2 (Target properties and reconstruction; attributed classical results with verified hypotheses)**.** *Fix the admissible pair and (m, u) in the regime. Assume C1–C3 and C6. The target ν^S of Theorem 10.8 has the properties (i)–(vii) of Theorem 10.16: Euclidean and field-sign invariance; reflection positivity across every line for square-integrable positive-half-plane observables; strong mixing under translations, hence clustering in this Euclidean sense and translation ergodicity; the exponential regularity of C3(i)(c) for an explicit Schwartz-continuous norm; tempered symmetric Schwinger distributions; non-Gaussianity with local $L^2(B^n)$ kernels and a nonzero even truncated Schwinger distribution of order at least four on tests in disjoint Euclidean-time slabs; and, by [8, Thm. 1.0] applied only after ergodicity has been established, the scalar relativistic Wightman theory in $1+1$ dimensions that it determines, with the Wightman axioms listed in C3(i), a unique physical vacuum, and a field that fails Gaussian factorization. The classical existence and reconstruction theorems are attributed; the verification of their hypotheses for this target is project-derived and appears in Section 10.*

Every property of Theorem 4.2 is proved along the restricted classical route in Lemma 10.1, the source portion of Theorem 10.8, Lemmas 10.13–10.15 and Theorem 10.16. This route uses only C1–C3 and C6; the separate common-space identification for the neural construction uses the Gaussian and Wick foundations. Sign symmetry, including its use in the even-cumulant witness, has the same restricted source route.

Remark 4.3 (Quantifier order; one unchanged sequence). The order of quantifiers is: fix the admissible pair, hence g_* and the regime; fix (m, ℓ, u) in the regime; fix the localization system and the enumerations; then the sequence of (S) is determined, once, for both architectures. Only afterwards are the following quantified, each independently and without reselection of parameters: every $H \in C_b(X)$; every j, r; every k, every finite real compact smooth tuple and every real or complex polynomial of any degree; every $q > 0$; every order k and every joint test $F \in \mathcal{D}((\mathbb{R}^2)^k)$; every reflection θ, every J and every list f. Clauses (A)–(E) use exactly the sequence of (S), with no subsequence. No countability restriction is placed on the later objects. The constants $M_{j,k}$ are suprema over all stages and over ν; they are finite by Proposition 11.9 and do not depend on n, b or F.

Remark 4.4 (What is not claimed). No effective end-to-end rate or schedule for the full chain of limits and no parameter cost: the $\sqrt{a}$ bounds of Theorem 8.2(d),(f) and Proposition 9.6 are upper bounds at fixed physical volume, with volume-dependent constants, and they give no rate for the width, stabilizer, auxiliary-volume or physical-volume limits. No numerical value of g_*, and no algorithm deciding the selection predicates. No convergence on Schwartz or other noncompact tests at the neural stages, no strong-dual $\mathcal{S}'$ convergence or tightness, no compact-L^2 or bounded Borel local tests, no local total variation or uniform density bound, and no interchange of the ordered limits. No finite-stage Euclidean invariance or clustering statement and no finite-stage reflection positivity across lines other than the two central coordinate lines; nothing is asserted about those other lines either way. No bound uniform over all neural parameter choices: uniformity holds along the selected sequence only. No independent neural-estimate construction of the thermodynamic limit (the volume limit uses classical continuum input), no reconstruction of the neural approximants, no new interaction, dimension or architecture class, no scattering statement, no quantitative mass gap, no computational advantage, no training or sampling procedure, and no priority. "Interacting" means the non-Gaussian quantum field of Theorem 4.2.

5 Field spaces

Sections 5–8 contain the foundations; none of them uses an interacting law as an input (Remark 13.2). This section establishes that X supports the probabilistic arguments used later. It is Polish, so Prokhorov's theorem applies and β metrizes weak convergence; compact smooth smearings are continuous and controlled by one seminorm through (16); the potentials are continuous on E, the reflections on X, and the map $L : C(\mathbb{R}^2) \to X$ for uniform convergence on compacts. Readers mainly interested in the construction can treat Proposition 5.1 as a toolbox and return to it where it is cited.

Proposition 5.1 (Field-space facts)**.** *(i) (E, d_E) is Polish. If $F : \mathbb{R}^2 \times \Gamma \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable and every path is locally fourth-power integrable (for instance $|F| \le r$), then $\omega \mapsto [F(\cdot, \omega)]$ is a Borel E-valued random element.*

(ii) For $v \in X$ and $j \ge 1$, $p_j(v) < \infty$; d is a complete separable metric on X, so X is Polish.

(iii) Duality. *For $v \in X$, $j \ge 1$ and $f \in C_c^\infty((-j, j)^2; \mathbb{C})$, with $\|f\|_{H^2(\mathbb{T}_j)}$ computed for the zero extension,*

$$|v(f)| \le p_j(v)\, \|f\|_{H^2(\mathbb{T}_j)}. \tag{16}$$

Hence every compact smooth evaluation is d-continuous. The Borel σ-algebra of X equals the σ-algebra generated by the compact smooth evaluations, and equals the one generated by the countably many real and imaginary parts of $v \mapsto v(\chi_j \overline{e_{j,k}})$, $j \ge 1$, $k \in \mathbb{Z}^2$.

(iv) $\iota : E \to X$ is a continuous injection.

(v) β metrizes weak convergence of Borel probabilities on X.

(vi) For bounded measurable $D \subset \mathbb{R}^2$ and $k \in \{2, 4\}$, $v \mapsto \int_D v^k$ is continuous on E; hence $W_{a,\Lambda}$, $J_{\Omega,\Lambda}$, $T_{\Omega,a}$, $V_{\varepsilon,\Omega,a;\Lambda}$ and their exponentials are continuous on E, and also on $C(\mathbb{R}^2)$ with the topology of uniform convergence on compacts.

(vii) Reflections. *For $\theta \in \{\theta_1, \theta_2\}$ and $v \in X$, $\theta v \in X$, $\theta\theta v = v$, $(\theta v)(f) = v(f \circ \theta)$, and $p_j(\theta v) \le C_j p_{j+1}(v)$ with $C_j = \|\chi_j\|_\infty + 2\sum_{a=1,2} \|\partial_a \chi_j\|_\infty + \|\Delta \chi_j\|_\infty$; in particular θ is continuous on X. For $w \in E$, $\theta\iota w = \iota(w \circ \theta)$.*

(viii) The map L. *$(LG)(f) = \int G\,(1 - \Delta) f$ defines a linear map $L : C(\mathbb{R}^2) \to X$ (the distribution $(1 - \Delta)G$) with $p_j(LG) \le (2j + 1) C_j \sup_{[-j-1/2, j+1/2]^2} |G|$; L is continuous for uniform convergence on compacts.*

(ix) A countable separating family. *Let $\mathcal{D}_0$ be the set of real and imaginary parts of $\chi_n P$, $n \ge 1$, P a trigonometric polynomial on $\mathbb{T}_n$ with coefficients in $\mathbb{Q} + i\mathbb{Q}$. Two real distributions that agree on $\mathcal{D}_0$ are equal.*

(x) Countable factorization. *If F is bounded, complex and Σ_+^θ-measurable, there are real $f_1, f_2, \dots \in C_c^\infty(H_+^\theta)$ and a Borel $g : \mathbb{R}^{\mathbb{N}} \to \mathbb{C}$ with $|g| \le \sup |F|$ such that $F(v) = g((v(f_j))_j)$ and $F(\theta v) = g((v(f_j \circ \theta))_j)$ for every $v \in X$; in particular $v \mapsto \overline{F(\theta v)} F(v)$ is bounded and Borel, and $\Sigma_+^\theta \subset \mathcal{B}(X)$.*

Proof. (i) Restriction embeds E isometrically as the closed subspace of compatible families in $\prod_j L^4((-j, j)^2)$ with the product metric; each factor is a separable Banach space, so E is complete and separable. For a fixed Borel representative s_0 of an element of E, Tonelli makes $\omega \mapsto \int_{(-j,j)^2} |F(z, \omega) -$

$s_0(z)|^4 dz$ measurable, so $\omega \mapsto d_E([F(\cdot,\omega)], s_0)$ is measurable; in a separable metric space every open set is a countable union of open balls.

(iii), duality. Let $f \neq 0$ be supported in $(-j,j)^2$ and let $\tilde{f}$ be the $2(j+1)$-periodic extension of $f|_{\mathcal{C}_j}$; it is smooth on $\mathbb{T}_j$, and $f_k = \int_{\mathcal{C}_j} \overline{e_{j,k}} f$ satisfies $\sum_k \lambda_{j,k}^2 |f_k|^2 = \|(1-\Delta)\tilde{f}\|^2_{L^2(\mathbb{T}_j)} < \infty$. The Fourier series of $\tilde{f}$ converges in $C^\infty(\mathbb{T}_j)$, so $\chi_j \sum_{|k|\le K} f_k e_{j,k} \to \chi_j \tilde{f} = f$ in C^∞ with supports in $\operatorname{supp}\chi_j$ (because $\chi_j = 1$ on $[-j,j]^2 \supset \operatorname{supp} f$). Since v is a distribution, $v(f) = \lim_K \sum_{|k|\le K} f_k v(\chi_j e_{j,k})$, and $v(\chi_j e_{j,k}) = v(\chi_j \overline{e_{j,-k}})$. Cauchy–Schwarz with the weights $\lambda_{j,k}^{\pm 1}$ and $\lambda_{j,-k} = \lambda_{j,k}$ give (16), provided $p_j(v) < \infty$, which is proved next.

(ii), finiteness. Let $u = \chi_j v \in H^{-2}(\mathbb{R}^2)$ and choose a real $\zeta \in C_c^\infty((-j-\frac{1}{2}, j+\frac{1}{2})^2)$ with $\zeta = 1$ near $\operatorname{supp}\chi_j$. For a trigonometric polynomial $P = \sum_{|k|\le K} c_k e_{j,k}$, ζP is a compactly supported smooth function and $\sum_k c_k u(\zeta e_{j,k}) = u(\zeta P)$, with $u(\zeta e_{j,k}) = v(\chi_j e_{j,k})$. By the product rule $(1-\Delta)(\zeta P) = \zeta(1-\Delta)P - 2\nabla\zeta\cdot\nabla P - P\Delta\zeta$, and Parseval on $\mathbb{T}_j$ gives $\|\zeta P\|_{H^2(\mathbb{R}^2)} \le C(\zeta)(\sum_k \lambda_{j,k}^2 |c_k|^2)^{1/2}$ with $C(\zeta) = \|\zeta\|_\infty + 2\sup|\nabla\zeta| + \|\Delta\zeta\|_\infty$. Taking $c_k = \lambda_{j,k}^{-2}\overline{u(\zeta e_{j,k})}$ in $|u(\zeta P)| \le \|u\|_{H^{-2}}\|\zeta P\|_{H^2}$ gives $\sum_{|k|\le K} \lambda_{j,k}^{-2}|v(\chi_j e_{j,k})|^2 \le C(\zeta)^2\|u\|^2_{H^{-2}}$ for all K; reindexing $k \mapsto -k$ yields $p_j(v) \le C(\zeta)\|\chi_j v\|_{H^{-2}(\mathbb{R}^2)}$.

(ii), metric and completeness. Each p_j is a seminorm, so d is a pseudometric; if $d(v,w) = 0$ then $p_j(v-w) = 0$ for all j and (16) gives $v = w$. Let (v_n) be d-Cauchy. Then $(\chi_j v_n)_n$ is Cauchy in the Hilbert space of periodic distributions with finite $H^{-2}(\mathbb{T}_j)$ norm. For $f \in C_c^\infty((-j,j)^2)$, $v_n(f)$ is Cauchy by (16); define $v(f) = \lim_n v_n(f)$, which does not depend on j. On each $\mathcal{D}_K$, $|v(f)| \le \sup_n p_j(v_n)\,\|f\|_{H^2(\mathbb{T}_j)} \le 3|\mathbb{T}_j|^{1/2}\sup_n p_j(v_n)\|f\|_{C^2}$, so v is a distribution. Fatou for series gives $p_j(v_n - v)^2 \le \liminf_l p_j(v_n - v_l)^2 \to 0$. Finally $v \in X$: for $\chi \in C_c^\infty((-j,j)^2)$ and $g \in \mathcal{S}(\mathbb{R}^2)$, $(\chi v)(g) = v(\chi g)$ and (16) with the product rule give $|v(\chi g)| \le p_j(v) C(\chi)\|g\|_{H^2(\mathbb{R}^2)}$, so $\chi v \in H^{-2}(\mathbb{R}^2)$. Hence $d(v_n, v) \to 0$.

(ii), separability, and (iii), Borel generation. The map $J(v) = \big((\lambda_{j,k}^{-1} v(\chi_j \overline{e_{j,k}}))_k\big)_j$ takes values in $\prod_j \ell^2(\mathbb{Z}^2;\mathbb{C})$, and $p_j(v-w) = \|J_j(v) - J_j(w)\|_{\ell^2}$; thus $d(v,w) = \sum_j 2^{-j}\min(1, \|J_j(v) - J_j(w)\|)$ is the restriction of a product metric on a separable space, and X is separable. Let $\mathcal{E}$ be the σ-algebra generated by the real and imaginary parts of the localized Fourier evaluations. For fixed w, $v \mapsto d(v,w)$ is a countable expression in $\mathcal{E}$-measurable functions, so open balls, and by separability all open sets, lie in $\mathcal{E}$. The Fourier tests $\chi_j\overline{e_{j,k}}$ have compact support, so $\mathcal{E}$ is contained in the σ-algebra of compact smooth evaluations, which is contained in $\mathcal{B}(X)$ because each evaluation is continuous: if $d(v,w) < 2^{-j}$ then $p_j(v-w) \le 2^j d(v,w)$, and (16) applies for $\operatorname{supp} f \subset (-j,j)^2$.

(iv) For $v \in E$, $(\iota v)(\chi_j\overline{e_{j,k}}) = \langle e_{j,k}, \chi_j v\rangle_{L^2(\mathbb{T}_j)}$, so Parseval and $\lambda_{j,k} \ge 1$ give $p_j(\iota v) \le \|\chi_j v\|_{L^2} \le (2j+2)^{1/2}\|v\|_{L^4((-j-1,j+1)^2)}$ by Hölder; $\chi\iota v = \chi v \in L^2 \subset H^{-2}$. A locally integrable function with zero integral against every compact smooth test vanishes almost everywhere.

(v) Every Borel probability on the Polish space X is tight. If $\beta(\rho_n, \rho) \to 0$, the bounded Lipschitz functions $(1 - k\,\mathrm{dist}_d(\cdot, F))_+ / \max(1,k)$ give $\limsup \rho_n(F) \le \rho(F)$ for closed F, hence weak convergence. Conversely, let $\rho_n \Rightarrow \rho$ and fix $\delta > 0$; choose a compact K with $\rho(K) > 1 - \delta$; its open η-neighbourhood K^η has $\rho_n(K^\eta) > 1 - 2\delta$ for large n. Choose a finite η-net of K and quantize the values of functions H with $|H| \le 1$, $\mathrm{Lip}_d(H) \le 1$ at the net points to accuracy η; this yields finitely many classes, with one representative each. A function and its representative differ by at most 3η on K and 5η on K^η, so $\beta(\rho_n,\rho) \le 10\eta + 6\delta + \max|\rho_n(H_i) - \rho(H_i)|$ over the finitely many representatives; let $n \to \infty$ and then $\eta, \delta \to 0$.

(vi) $|x^k - y^k| \le k|x-y|(|x|+|y|)^{k-1}$ and Hölder on D; the inclusion $C(\mathbb{R}^2) \to E$ is continuous.

(vii) For a smooth periodic f on $\mathbb{T}_j$ put $\zeta = (\chi_j \circ \theta)(f \circ \theta)$, supported in $(-j-\frac{1}{2}, j+\frac{1}{2})^2$

where $\chi_{j+1} = 1$. Then $(\chi_j \theta v)(f) = v(\zeta) = (\chi_{j+1} v)(\zeta)$, and the product rule with $\|\partial^\alpha (f \circ \theta)\|_{L^2(\mathcal{C}_j)} = \|\partial^\alpha f\|_{L^2(\mathbb{T}_j)}$ and $\|f\|_2, \|\partial_a f\|_2 \le \|f\|_{H^2(\mathbb{T}_j)}$ give $|(\chi_j \theta v)(f)| \le C_j p_{j+1}(v) \|f\|_{H^2(\mathbb{T}_j)}$; taking the supremum over trigonometric polynomials f with $\|f\|_{H^2(\mathbb{T}_j)} \le 1$ gives the bound. For $\chi \in C_c^\infty$, $\chi \theta v = \theta((\chi \circ \theta) v)$ and the Fourier transform of θw is $\hat{w} \circ \theta$ with $|\theta p| = |p|$, so $\theta v \in X$. The identity for $w \in E$ is the change of variables $z \mapsto \theta z$.

(viii) For real $\chi \in C_c^\infty$, testing against f gives $\chi \cdot LG = (1 - \Delta)(\chi G) + 2 \sum_i \partial_i (G \partial_i \chi) - G \Delta \chi$, so $LG \in X$. With $\chi = \chi_j$, $(LG)(\chi_j \overline{e_{j,k}}) = \lambda_{j,k} \langle e_{j,k}, \chi_j G \rangle + 2i \frac{\pi}{j+1} \sum_i k_i \langle e_{j,k}, G \partial_i \chi_j \rangle - \langle e_{j,k}, G \Delta \chi_j \rangle$, and $\lambda_{j,k}^{-2} (\pi |k_i| / (j+1))^2 \le 1$, Minkowski in ℓ^2 and Parseval give the bound, since all functions involved are supported in $[-j - \frac{1}{2}, j + \frac{1}{2}]^2$ of area $(2j+1)^2$. Continuity follows from dominated convergence for the series defining d, with summable majorant 2^{-j}.

(ix) For real $f \in C_c^\infty$ choose n with $\operatorname{supp} f \subset [-n, n]^2$; the periodic extension $\tilde{f}$ of $f|_{\mathcal{C}_n}$ is smooth, its Fourier series converges in $C^\infty(\mathbb{T}_n)$, and truncation and rational rounding give trigonometric polynomials P_l with $\chi_n P_l \to \chi_n \tilde{f} = f$ in C^∞ with supports in $\operatorname{supp} \chi_n$; hence $\operatorname{Re}(\chi_n P_l) \to f$ in that sense and $v(\operatorname{Re}(\chi_n P_l)) \to v(f)$.

(x) The sets in Σ_+^θ that are measurable with respect to the evaluations at some countable subfamily of $C_c^\infty(H_+^\theta; \mathbb{R})$ form a σ-algebra containing the generators, hence all of Σ_+^θ. A bounded F is a uniform limit of simple functions; collecting countably many tests gives $F = \tilde{g} \circ (v \mapsto (v(f_j))_j)$ by Doob–Dynkin, and $g = \tilde{g} \mathbf{1}_{\{|\tilde{g}| \le \sup |F|\}}$ still represents F. By (vii), $F(\theta v) = g(((\theta v)(f_j))_j) = g((v(f_j \circ \theta))_j)$; both factors are Borel by (iii). □

Remark 5.2 (Use and scope of these facts). In (viii), LG is the distribution $(1 - \Delta)G$, not ιG. It is used only for the Gaussian versions $\Phi_a = L\tilde{\psi}_a$, where $\tilde{\psi}_a$ carries the extra spectral multiplier $(1 + |p|^2)^{-1}$, so that $\Phi_a(f) = \phi_a(f)$ (Theorem 7.10). Items (ii) (separability and completeness) and (iii) (Borel generation) are proved here directly; they re-express arguments of the earlier width record and of a Gaussian-foundations lemma whose Borel-law part lies outside that record's accepted scope, and they require current-version review in this text (Appendix D). Completeness is used for Prokhorov compactness and for (v).

6 Process neurons and finite-width central reflection positivity

Conditional factorization gives positivity for the interacting law at finite width. For product neurons, given the complete tuple of boundary data of all N neurons, the two half-plane sides are independent with equal conditional laws (Theorem 6.1(d)); for Mondrian neurons, the trace decomposition of Theorem 6.3, followed by approximation from finite boxes (Proposition A.15), plays this role. The reflection-invariant additive potential splits into half-plane contributions and is absorbed into the side observables (Lemma A.14). Theorem 6.5 states the result, and Corollary 6.6 applies it to the tilt of the main theorem.

This section states the properties of the neurons and the finite-width positivity theorem; all proofs are in Appendix A. For open $U \subseteq \mathbb{R}^2$, $L^1_{\mathrm{loc}}(U)$ is the Polish Fréchet space of classes of locally integrable functions on U, and $\mathrm{Ball}_r(U) = \{s \in L^1_{\mathrm{loc}}(U) : |s| \le r \text{ a.e.}\}$, a closed subset. For $\theta \in \{\theta_1, \theta_2\}$ write $H_+ = H_+^\theta$. The *one-neuron sides* of a neuron g across θ are $s_+^\theta = g|_{H_+}$ and $s_-^\theta = (g \circ \theta)|_{H_+}$; the *$N$-neuron sides* are $S_+^\theta = F_N|_{H_+}$ and $S_-^\theta = (F_N \circ \theta)|_{H_+}$, random elements of $\mathrm{Ball}_{\sqrt{N}/2}(H_+)$. For a product neuron the *boundary data* are $b^{\theta_1} = (\epsilon_\sigma, \tau)$ and $b^{\theta_2} = (\sigma, \epsilon_\tau)$, where ϵ_σ, ϵ_τ are the initial signs of σ, τ.

Theorem 6.1 (Product neuron). *(a) σ is càdlàg with $|\sigma| = 1$, $\mathbb{E}\sigma(t) = 0$, $\mathbb{E}\sigma(t)\sigma(t') = e^{-\mu|t-t'|}$,*

and its finite-dimensional laws are those of the stationary symmetric two-state Markov chain with flip rate $\mu/2$.

(b) g is $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{A}$-measurable, $|g|=1/2$ everywhere, $\mathbb{E}g(z)=0$, $\mathbb{E}g(z)g(z')=C_\mu^{\mathrm{end}}(z-z')$, and almost surely g is constant on each cell of the locally finite grid formed by the jump points of σ and of τ.

(c) Given b^{θ_1}, the one-neuron sides across θ_1 are conditionally independent with equal conditional laws; likewise across θ_2 given b^{θ_2}.

(d) For independent product neurons $g_1,\dots,g_N$, the N-neuron sides $S^\theta_\pm$ are conditionally independent with equal conditional laws given the complete *tuple $(b^\theta_1,\dots,b^\theta_N)$. Conditioning on the data of a single neuron does not suffice when $N\geq 2$.*

Theorem 6.2 (Box Mondrian)**.** *Let C be a box with nonempty interior and $s\geq 0$.*

(a) Almost surely finitely many nodes are born; the mean number of born nodes is at most $2e^{sp(C)}-1$; $\Pi^{\mathrm{can}}(C,s)\in\mathrm{Part}^{\mathrm{col}}(C)$ is a Borel function of (C,s) and the ingredients.

(b) $M^{\mathrm{col}}(C,s)$ satisfies the first-step recursion of Definition A.6, and every family of laws satisfying that recursion with an admissible daughter assignment coincides with M^{col}.

(c) $T_*M^{\mathrm{col}}(C,s)=M^{\mathrm{col}}(TC,s)$ for translations T and for $T\in\{\theta_1,\theta_2\}$.

(d) $(\mathrm{res}_B)_*M^{\mathrm{col}}(C,s)=M^{\mathrm{col}}(B,s)$ for every box $B\subseteq C$ with nonempty interior.

(e) For $z,z'\in C$, the probability that some cell contains both is $e^{-s|z-z'|_1}$; a fixed $z\in\operatorname{int}C$ is almost surely interior to a cell.

Theorem 6.3 (Core-exterior construction and trace decomposition)**.** *For boxes $K\subseteq C$ the core-exterior construction of Definition A.9 produces almost surely a finite colored partition of C with law $M^{\mathrm{col}}(C,s)$, whose restriction to a core K with nonempty interior is the canonical partition of K. For $C=[-L,L]\times J_0$ and $K=\{0\}\times J_0$, the colored partition of the right half-box and the θ_1-mirror image of that of the left half-box are, given the trace data (positions and auxiliary times of the horizontal cuts of straddling cells and colors of the final straddling cells), conditionally independent with the same conditional law; the same holds with coordinates exchanged. Times and positions are both needed in this conditioning (Appendix A).*

Theorem 6.4 (Whole-plane Mondrian neuron)**.** *(a) A consistent colored Mondrian tessellation of budget μ exists; the law of $(P_n)_n$ is unique and invariant under translations and under θ_1,θ_2 acting on closed cells; for a box B with nonempty interior, $\mathrm{res}_B(P_n)$ ($B\subseteq B_n$) does not depend on n almost surely and has law $M^{\mathrm{col}}(B,\mu)$. The lower-left representatives of the cells of any element of $\mathrm{Part}^{\mathrm{col}}(B_n)$ partition $[-n,n)^2$.*

(b) Almost surely each closed cell $R(z)$ is a bounded closed box with nonempty interior, equal to $R_n(z)$ for large n; the distinct closed cells cover $\mathbb{R}^2$, have disjoint interiors and are locally finite; $H(z)=R(z)^{\mathrm{ll}}$, and the distinct half-open representatives partition $\mathbb{R}^2$ and are locally finite.

(c) g is $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{A}$-measurable, $|g|=1/2$ everywhere, almost surely constant on each $H(z)$, $\mathbb{E}g(z)=0$ and $\mathbb{E}g(z)g(z')=C_\mu^{\mathrm{end}}(z-z')$.

(d) Every admissible convention agrees with g off the union of cell boundaries, a null set for every outcome of the consistency event; for open U the law of $g|_U$ in $L^1_{\mathrm{loc}}(U)$ is the same for every realization and every admissible convention; $g \circ \theta_j \overset{d}{=} g$ and $g(\cdot + x) \overset{d}{=} g$ in $L^1_{\mathrm{loc}}(\mathbb{R}^2)$.

For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \geq 1$, a bounded Borel Ω and an $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable $v : \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}$, bounded on each $\Omega \times [-r, r]$, put $U = \int_\Omega v(z, F_N(z))\, dz$, $Z = \mathbb{E}e^{-U}$, $P_N^v(B) = \mathbb{E}[\mathbf{1}_B(\iota F_N)e^{-U}]/Z$ and $M_N = \sup\{|v(z,x)| : z \in \Omega, |x| \leq \sqrt{N}/2\}$, with the convention $\sup \emptyset = 0$; if $|\Omega| = 0$, then $U = 0$ and $Z = 1$.

Theorem 6.5 (Finite-width annealed normalized reflection positivity)**.** *Let $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, $N \geq 1$, $\theta \in \{\theta_1, \theta_2\}$, Ω bounded Borel with $\theta\Omega = \Omega$, and v as above with $v(\theta z, x) = v(z, x)$ for all $z \in \Omega$, $x \in \mathbb{R}$. Then:*

(a) U is a real random variable, $|U| \leq |\Omega| M_N$ and $e^{-|\Omega|M_N} \leq Z \leq e^{|\Omega|M_N}$; P_N^v is a Borel probability on X determined by the law of one neuron, independently of the realization and of the admissible boundary convention;

(b) P_N^v is reflection positive across θ for bounded observables, in the sense of (11)*;*

(c) $\mathbb{E}[e^{-U}\overline{\Phi(S_-^\theta)}\Phi(S_+^\theta)] \geq 0$ for every bounded complex Borel Φ on $L^1_{\mathrm{loc}}(H_+)$; and for every bounded complex Y measurable for $\sigma(F_N(z) : z \in H_+)$, written $Y = \tilde{Y}((F_N(z_k))_{k\in\mathbb{N}})$, the reflected variable $Y^\theta = \tilde{Y}((F_N(\theta z_k))_k)$ is almost surely independent of the representation and $\mathbb{E}[e^{-U}\overline{Y^\theta}Y] \geq 0$.

Corollary 6.6 (Two central lines; the axis-parallel manuscript tilt)**.** *Let $\Lambda = [-l_1, l_1] \times [-l_2, l_2]$ and $\Omega = [-w_1, w_1] \times [-w_2, w_2]$ with $0 \leq l_i \leq w_i$: closed axis-parallel rectangles,* both centered. *Let $q_{\mathrm{in}}, q_{\mathrm{out}}$ be real polynomials, $\kappa \in \mathbb{R}$, and $v(z,x) = \mathbf{1}_\Lambda(z)q_{\mathrm{in}}(x) + \mathbf{1}_{\Omega\setminus\Lambda}(z)q_{\mathrm{out}}(x) - \kappa^2x^2/2$. Then for both architectures and every $N \geq 1$ the conclusions (a)–(c) of Theorem 6.5 hold across θ_1 and across θ_2.* Manuscript tilt. *If Λ is a physical square in $\Omega = [-R, R]^2$ in the sense of Definition 2.1, the integrand v_V of* (7) *agrees off $\partial\Lambda \times \mathbb{R}$ with the displayed v for the rectangles $\bar{\Lambda} = [-l, l]^2$, $\Omega = [-R, R]^2$, with $q_{\mathrm{in}}(x) = u(x^4 - 6c_ax^2 + 3c_a^2)$, $q_{\mathrm{out}}(x) = \varepsilon x^4$, $\kappa = \kappa_a$; for every outcome $U = V_{\varepsilon,\Omega,a;\Lambda}(F_N)$, and $P_N^{v_V} = P^b_{N,\varepsilon,\Omega,a;\Lambda}$. Hence every neural law* (8) *is reflection positive across both central coordinate lines; this covers $\Lambda_n = (-n\ell, n\ell)^2$ with $n\ell \leq R$.*

Corollary 6.7 (General potentials invariant under both reflections)**.** *If $\theta_1\Omega = \theta_2\Omega = \Omega$ and v is $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable, bounded on each $\Omega \times [-r, r]$ and invariant under both central reflections on Ω, then (a)–(c) of Theorem 6.5 hold across both central lines, for both architectures and every $N \geq 1$.*

Remark 6.8 (Geometry, measurability and scope). (i) *Centering alone is not enough.* For $O = 5^{-1/2}\left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$, the centered rotated square $O((-1,1)^2)$ lies in $\Omega = [-2, 2]^2$, and with $m = u = \varepsilon = 1$, $a = 1/2$ the integrand built on it satisfies $v_V(z, 0) = 3uc_a^2 > 0 = v_V(\theta_j z, 0)$ for both j on a disc of positive area. This shows only that the invariance hypothesis of Theorem 6.5 fails for that tilt; no failure of reflection positivity is claimed for it or for any rotated square. (ii) *The general clause keeps its own hypotheses.* A centered square rotated by an odd multiple of $\pi/4$, $\{|z_1| + |z_2| < c\}$, is invariant under both central reflections; a potential of the form (7) built on it, with closure inside an invariant Ω, satisfies the hypotheses of Corollary 6.7, which therefore applies to it, although it is not a physical square in the sense of Definition 2.1. (iii) *Measurability.* "Measurable potential" means $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable. Under the broader reading "Lebesgue measurable on $\mathbb{R}^3$" the integrated potential can be undefined: for a non-measurable $A \subset (0,1)^2$ symmetrized under both reflections

and $v(z,x) = \mathbf{1}_A(z)\mathbf{1}\{x = x_0\}$ with $x_0 \in \{0, 1/(2\sqrt{N})\}$ of the parity of N, v is Lebesgue-null on $\mathbb{R}^3$, but on an event of positive probability $F_N = x_0$ on Ω and $z \mapsto v(z, F_N(z))$ is not measurable. (iv) *Scope.* Positivity is asserted across θ only when Ω and v are θ-invariant; for the two-line statements, invariance under both reflections of both the inner and the outer region is used. Nothing is asserted for shifted or rotated lines, for Euclidean invariance of tilted laws, or for limits in N, ε, Ω or a by this theorem.

7 Gaussian reference fields

Gaussian fields enter in two roles. First, they serve as references for the intermediate laws of the chain: the width limit replaces the untilted neural reference field by the continuous endpoint field γ_μ (Theorem 7.1); its quadratic tilt on a bounded Ω is again Gaussian, with a covariance $C'_{\Omega,a}$ between C^{end}_μ and C_a (Theorems 7.4 and 7.5); and $C'_{\Omega,a} \to C_a$ locally uniformly as Ω grows (Theorem 7.6). Second, Theorems 7.7–7.10 realize all regulated fields and the free field on one probability space, so that the regulator can be removed through L^2 estimates for random variables.

Proofs are in Appendix B. Throughout, a is a regulator unless $a = 0$ is stated, and Ω is a bounded measurable set.

Theorem 7.1 (Endpoint Gaussian)**.** *For every $\mu > 0$ there is a $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{F}$-measurable $\gamma_\mu : \mathbb{R}^2 \times \Gamma_\mu \to \mathbb{R}$ with continuous paths for every outcome, which is a centered Gaussian process with covariance $C^{\mathrm{end}}_\mu(z - z')$ and $\mathbb{E}|\gamma_\mu(z+h) - \gamma_\mu(z)|^2 = (1 - e^{-\mu|h|_1})/2 \le \mu|h|_1/2$. Its law G_μ on $C(\mathbb{R}^2)$ is the unique Borel probability under which the coordinate process is centered Gaussian with this covariance.*

Theorem 7.2 (Regulated covariance and mass shift)**.** *(a) $\hat{C}_a \le \hat{C}_0 \le m^{-2}$, $\hat{C}_a(w,q) \le m^{-2}(1 + a^2w^2)^{-1}(1 + a^2q^2)^{-1}$, $\hat{C}_a \in L^1$, and (2) holds with $c_a \in (0,\infty)$.*

(b) (3) holds pointwise, with $0 \le \kappa_a^2 \hat{C}^{\mathrm{end}}_\mu \le r_a$.

(c) $s_k = \kappa_a^{2k}(C^{\mathrm{end}}_\mu)^{*(k+1)}$ is continuous and positive with $\sup s_k \le r_a^k/4$ and $\|s_k\|_1 = \mu^{-2} r_a^k$; $\sum_k s_k$ converges uniformly and in L^1 to C_a. Hence C_a is continuous, $0 < C^{\mathrm{end}}_\mu \le C_a \le c_a$, $\|C_a\|_1 = m^{-2}$ and $\|C_a\|_2^2 \le 1/(4\pi m^2)$.

(d) As operators on $L^2(\mathbb{R}^2)$: $\|C^{\mathrm{end}}_\mu\| = \mu^{-2}$, $C_a = C^{\mathrm{end}}_\mu(I - \kappa_a^2 C^{\mathrm{end}}_\mu)^{-1}$, and $0 \le \langle f, C_a f\rangle \le m^{-2}\|f\|_2^2$.

(e) For every $h \in \mathbb{R}^2$: $2(c_a - C_a(h)) \le \varpi_a(h)^2 := (2\pi m^2 a^3)^{-1} \sum_{i=1,2} |h_i|[\log(1 + 4a^2/h_i^2) + 2]$ (summand 0 if $h_i = 0$).

(f) For $0 < \alpha < 1$: $2(c_a - C_a(h)) \le M'_{a,\alpha}|h|^\alpha$ for all h, with $M_{a,\alpha} = [2(2a)^{1-\alpha}/(1-\alpha) + 2]/(\pi m^2 a^3)$ and $M'_{a,\alpha} = \max(M_{a,\alpha}, 2c_a)$.

Theorem 7.3 (Restricted endpoint covariance)**.** *$K_\Omega f(x) = \int_\Omega C^{\mathrm{end}}_\mu(x-y) f(y)\,dy$ on $L^2(\Omega;\mathbb{R})$ is positive, self-adjoint and trace class with $\operatorname{tr} K_\Omega = |\Omega|/4$, $\|K_\Omega\| \le \mu^{-2}$ and $\kappa_a^2\|K_\Omega\| \le r_a < 1$; it has an orthonormal eigenbasis with eigenvalues $\lambda_j \in [0, \mu^{-2}]$.*

Theorem 7.4 (Pure quadratic tilt on finite Ω)**.** *(a) $\mathrm{Det}_{\Omega,a} = \mathbb{E}_{G_\mu} e^{T_{\Omega,a}} = \prod_j (1 - \kappa_a^2\lambda_j)^{-1/2} = \det_F(I - \kappa_a^2 K_\Omega)^{-1/2}$, with $1 \le \mathrm{Det}_{\Omega,a} \le \exp(\kappa_a^2|\Omega|/(8a^2m^2))$, nondecreasing in Ω.*

(b) $c_0^\Omega(z,z') = C^{\mathrm{end}}_\mu(z-z')$ and $c_k^\Omega(z,z') = \kappa_a^{2k}\int_{\Omega^k} C^{\mathrm{end}}_\mu(z-x_1)\cdots C^{\mathrm{end}}_\mu(x_k - z')\,dx$ are continuous, symmetric and $0 \le c_k^\Omega \le s_k(z-z')$; $C'_{\Omega,a} = \sum_k c_k^\Omega$ converges uniformly on $\mathbb{R}^2 \times \mathbb{R}^2$, and the operator with this kernel is $C^{\mathrm{end}}_\mu(I - \kappa_a^2 \mathbf{1}_\Omega C^{\mathrm{end}}_\mu)^{-1} = C^{\mathrm{end}}_\mu + \kappa_a^2 C^{\mathrm{end}}_\mu R^*(I - \kappa_a^2 K_\Omega)^{-1} R\, C^{\mathrm{end}}_\mu$.

(c) $G'_{\Omega,a} = \mathrm{Det}^{-1}_{\Omega,a} e^{T_{\Omega,a}} G_\mu$ *is the unique Borel probability on* $C(\mathbb{R}^2)$ *under which the coordinate process at* all *points of the plane is centered Gaussian with covariance* $C'_{\Omega,a}$*, and* $\mathbb{E}_{G_\mu}[Fe^{T_{\Omega,a}}] = \mathrm{Det}_{\Omega,a}\mathbb{E}_{G'_{\Omega,a}}[F]$ *for Borel* $F \geq 0$ *and for Borel* F *integrable under* $G'_{\Omega,a}$*.*

Theorem 7.5 (Covariance order)**.** $0 < C^{\mathrm{end}}_\mu(z-z') \leq C'_{\Omega,a}(z,z') \leq C_a(z-z') \leq c_a$ *pointwise;* $0 \leq C^{\mathrm{end}}_\mu \leq C'_{\Omega,a} \leq C_a$ *as operators;* $C'_{\Omega,a} \leq C'_{\Omega',a}$ *for* $\Omega \subseteq \Omega'$*.*

Theorem 7.6 (The covariance difference)**.** *Let* $\Delta_{\Omega,a} = C_a - C'_{\Omega,a}$ *and* $Q = \mathbf{1}_{\mathbb{R}^2\setminus\Omega}$*.*

(a) $\Delta = \kappa_a^2 C_a Q C' = \kappa_a^2 C' Q C_a = \kappa_a^2 C' Q C' + \kappa_a^4 C' Q C_a Q C'$ *as operators, with the corresponding kernel identities.*

(b) Δ *is continuous and positive semidefinite as an operator and as a kernel.*

(c) $0 \leq \Delta(z,z') \leq \kappa_a^2 \int_{\mathbb{R}^2\setminus\Omega} C_a(z-x)C_a(x-z')\,dx$*; if* Ω *contains every point at distance* $< r$ *from a compact* K*, then* $0 \leq \Delta(z,z') \leq \epsilon_a(r) = \kappa_a^2 \int_{|x|>r} C_a(x)^2 dx$ *for* $z, z' \in K$*, and* $\epsilon_a(r) \downarrow 0$*.*

(d) $\sup_{K\times K} \Delta_{\Omega,a} \to 0$ *along every family of bounded* Ω *that eventually contains each bounded set, for instance* $[-R,R]^2$*,* $R \to \infty$*.*

(e) $\Delta(z+h,z+h)+\Delta(z,z)-2\Delta(z+h,z) \leq 2(c_a - C_a(h))$*, and the same for* C'*; centered Gaussian processes with covariance* $\Delta_{\Omega,a}$ *or* $C'_{\Omega,a}$ *exist and have jointly measurable modifications with continuous paths.*

The spectral Gaussian space: $\mu_0(dp) = (2\pi)^{-2}\hat{C}_0(p)\,dp$, $H_{\mathbb{C}} = \{h \in L^2(\mu_0;\mathbb{C}) : h(-p) = \overline{h(p)}\}$ with the real inner product $\langle g,h\rangle = \int \overline{g}h\,d\mu_0$, $\tau_a = (\hat{C}_a/\hat{C}_0)^{1/2}$ ($\tau_0 = 1$), $e_z(p) = e^{-ip\cdot z}$ and $\rho(p) = (1+|p|^2)^{-1}$. An isonormal process on a real Hilbert space is a real-linear isometry into L^2 whose range consists of centered jointly Gaussian variables.

Theorem 7.7 (One common Gaussian space)**.** *There is a probability space* $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0)$ *carrying an isonormal process* W *on* $H_{\mathbb{C}}$*. With* $\phi_a(f) = W(\hat{f}\tau_a)$ *for* $0 \leq a < 1/m$ *and real* $f \in L^2(\mathbb{R}^2)$*,* $\phi = \phi_0$*, the whole family is centered and jointly Gaussian,* $f \mapsto \phi_a(f)$ *is real-linear, and* $\mathbb{E}[\phi_a(f)\phi_b(g)] = (2\pi)^{-2}\int \overline{\hat{f}}\hat{g}(\hat{C}_a\hat{C}_b)^{1/2}dp$*.*

Theorem 7.8 (Smearing bounds)**.** *For real* $f \in L^2$*:* $\|\phi_a(f)\|_2 \leq \|\phi(f)\|_2 \leq m^{-1}\|f\|_2$ *and* $\|\phi_a(f) - \phi(f)\|_2 \leq \frac{a}{2}\|f\|_2$*. For complex* f *and* $T_a(f) = \phi_a(\mathrm{Re}f) + i\phi_a(\mathrm{Im}f)$*:* $\mathbb{E}|T_a(f)|^2 \leq m^{-2}\|f\|_2^2$ *and* $\mathbb{E}|T_a(f) - T_0(f)|^2 \leq \frac{a^2}{4}\|f\|_2^2$*.*

Theorem 7.9 (Point fields and continuous modifications)**.** *For a regulator* a*,* $\phi_a(z) = W(e_z\tau_a)$ *is a centered Gaussian process with* $\mathbb{E}[\phi_a(z)\phi_b(z')] = C_{ab}(z-z')$*,* $C_{ab} = \mathcal{I}[(\hat{C}_a\hat{C}_b)^{1/2}]$*, and* $\mathbb{E}|\phi_a(z+h) - \phi_a(z)|^2 = 2(c_a - C_a(h)) \leq \varpi_a(h)^2$*. There is a* $\mathcal{B}(\mathbb{R}^2)\otimes\mathcal{F}_0$*-measurable* $\tilde{\phi}_a$ *with continuous paths for every outcome and* $\mathbb{P}_0(\tilde{\phi}_a(z) = \phi_a(z)) = 1$ *for every* z*; for real* $f \in L^1 \cap L^2$*,* $\int |f||\tilde{\phi}_a| < \infty$ *and* $\int f\tilde{\phi}_a = \phi_a(f)$ *almost surely. The exceptional null sets depend on* (a,z) *and on* (a,f)*; for countable families of regulators, points and tests there is one common null set; no common null set for uncountably many* a*, and no measurability or continuity in* a*, is asserted.*

Theorem 7.10 (Borel X-valued versions, including $a = 0$)**.** *For each* $0 \leq a < 1/m$*,* $\psi_a(z) = W(e_z\tau_a\rho)$ *has a modification* $\tilde{\psi}_a$ *with continuous paths, and* $\Phi_a := L(\tilde{\psi}_a(\cdot,\omega))$ *(the map* L *of Proposition 5.1(viii)) is a Borel map* $\Gamma_0 \to X$ *with:*

(a) $\Phi_a(f) = \phi_a(f)$ *almost surely for every real* $f \in C_c^\infty(\mathbb{R}^2)$*, and* $\Phi_a(f) = T_a(f)$ *for complex* f*;*

(b) $\Phi_a = \iota\tilde{\phi}_a$ *outside a null set depending on* a*, for* $a > 0$*;*

(c) if $\Phi' : \Gamma_0 \to X$ *satisfies* $\Phi'(f) = \phi_a(f)$ *almost surely for every* $f \in \mathcal{D}_0$ *(Proposition 5.1(ix)), then* $\Phi' = \Phi_a$ *almost surely;*

(d) for countable families of regulators and tests the identities in (a), (b) hold off one null set;

(e) $\mathbb{E}_0 p_j(\Phi_a)^2 \le m^{-2} B_j$ *and, for* $a > 0$, $\mathbb{E}_0 p_j(\Phi_a - \Phi_0)^2 \le \frac{a^2}{4} B_j$*, with* $B_j = (\|\chi_j\|_2^2/|\mathbb{T}_j|) \sum_k \lambda_{j,k}^{-2} < \infty$*; hence* $\|d(\Phi_a, \Phi_0)\|_{L^2(\mathbb{P}_0)} \le h(a) := \sum_j 2^{-j} \min(1, \frac{a}{2} B_j^{1/2}) \to 0$*;*

(f) for real $f_1, \dots, f_n \in C_c^\infty$ *the vector* $(\Phi_a(f_i))_i$ *is centered Gaussian with covariance* $(2\pi)^{-2} \int \overline{\hat{f}_i} \hat{f}_l \hat{C}_a \, dp$*.*

No interaction functional, Wick power, weight or normalizer enters these constructions.

Corollary 7.11 (Laws of the versions on X)**.** *For* $0 \le a < 1/m$ *the Borel law of* Φ_a *on* X *is the unique Borel probability whose finite-dimensional laws on compact smooth tests are those of (f); at* $a = 0$ *it is the centered Gaussian law with covariance* C_{free}*.*

Proof. By Proposition 5.1(iii) the finite-dimensional cylinder events of compact smooth evaluations form a $\cap$-stable generator of $\mathcal{B}(X)$; apply Dynkin's theorem. □

Remark 7.12 (Scope of the Gaussian foundations). Theorems 7.1–7.10 use only the Gaussian space; they assert nothing about interaction weights, Wick powers, normalizers, interacting laws or reflection positivity. Corollary 7.11 uses the Borel generation of Proposition 5.1(iii), which is proved in this paper. The interaction objects on the same space are supplied by Theorem 8.2 below. Theorem 7.2(e) is proved for every regulator $0 < a < 1/m$; nothing is claimed at $a = 1/m$.

8 Fixed-volume Wick theory

This section asks whether the interaction has a limit when the regulator is removed. The free field is distribution-valued, so its fourth power has no pointwise meaning, and the Wick constant c_a diverges; nevertheless the Wick-ordered integrals $U_{a,\Lambda}$ converge in L^2 to a fourth-chaos variable $U_{0,\Lambda}$ (Theorem 8.2(a),(b)). The key estimate is the bound $\mathbb{E}_0 e^{-pU_a} \le \Xi_p$, uniform in the regulator (Theorem 8.2(e)), which controls the weights and normalizers. These weight and regulator estimates have volume-dependent constants (Remark 8.4); the volume-uniform moment bounds needed later come from classical input (Section 10).

Proofs are in Appendix C. Fix $m, u > 0$, a physical square Λ with half-side $l > 0$ (open or closed; the boundary is null), and $A = |\Lambda| = 4l^2$. There is no weak-coupling restriction. For $c \ge 0$ let $:x^4:_c = x^4 - 6cx^2 + 3c^2$; for a centered Gaussian variable ξ put $:\xi^4: = :\xi^4:_{\operatorname{Var}\xi}$ (own-variance Wick power; zero if $\operatorname{Var}\xi = 0$). Let $H_1 = W(H_{\mathbb{C}})$ be the Gaussian Hilbert space of Theorem 7.7, $H^{:4:}$ its fourth Wiener chaos, $\mathcal{N}$ the $\mathbb{P}_0$-null sets and $\mathcal{F}_{\text{free}} = \sigma(\phi(f) : f \in C_c^\infty(\mathbb{R}^2; \mathbb{R})) \vee \mathcal{N}$.

Definition 8.1 (Interactions and admissible cutoffs)**.** For a regulator a, $U_a = U_{a,\Lambda} = u \int_\Lambda :\tilde{\phi}_a(z)^4:_{c_a} dz$ pathwise, and $U_0 = L^2(\mathbb{P}_0)\text{-}\lim_{a \downarrow 0} U_a$ through all real regulators (existence is part of Theorem 8.2). An *admissible cutoff family* is a family $(\chi_\epsilon)_{\epsilon > 0}$ of measurable functions with $\chi_\epsilon(-p) = \overline{\chi_\epsilon(p)}$, $\sup_\epsilon \|\chi_\epsilon\|_\infty < \infty$, $\chi_\epsilon \to 1$ almost everywhere and $\int |\chi_\epsilon|^2 \hat{C}_0 < \infty$; put $\phi_\epsilon^\chi(z) = W(e_z \chi_\epsilon)$ and $N_\epsilon^\chi = u \int_\Lambda :\phi_\epsilon^\chi(z)^4: dz$ (Bochner integral in $L^2(\mathbb{P}_0)$). Examples: smooth mollifiers $\chi_\epsilon(p) = \hat{\eta}(\epsilon p)$ with real $\eta \in C_c^\infty$, $\int \eta = 1$, for which $\phi_\epsilon^\chi(z) = \phi(\eta_\epsilon(z - \cdot))$;

and sharp cutoffs $\mathbf{1}_{\{|p|\le 1/\epsilon\}}$. The constants are $\kappa_0 = e(1+\sqrt{2})$, $\kappa_2 = (2\pi)^{-1}(17\pi/16)^{1/2}$, $M_6 = (2\pi)^{-5/3}(5\pi)^{5/6}m^{-1/3}$, $K_\Lambda = u(384A\kappa_2 M_6^3)^{1/2}$, $K_\sharp = \max\{12uA, (2/e)K_\Lambda(\kappa_0/m)^{1/2}\}$, $T(K) = \exp(-\exp((\pi/2)(K/(12uA))^{1/2}-1))$ and $\Xi_p = e^{pK_\sharp} + \int_{K_\sharp}^\infty pe^{pK}T(K)\,dK$.

For $\theta \in \{\theta_1, \theta_2\}$ let Θ be the reflection of the Gaussian space induced by $h \mapsto h \circ \theta$ on $H_{\mathbb{C}}$ (Appendix C): it is linear, multiplicative and law-preserving, with $\Theta\phi_a(f) = \phi_a(f \circ \theta)$ and $\Theta\phi_a(z) = \phi_a(\theta z)$. Put $\mathcal{A}_+^{\theta,a} = \sigma(\phi_a(f) : f \in C_c^\infty(H_+^\theta; \mathbb{R})) \vee \mathcal{N}$.

Theorem 8.2 (Fixed-volume Wick theory)**.** *On the common Gaussian space of Theorem 7.7:*

(a) $U_a \in L^2(\mathbb{P}_0)$ *lies in* $H^{:4:}$*, equals the Bochner integral of* $z \mapsto u{:}\phi_a(z)^4{:}$*, and satisfies* $U_a \ge -6uAc_a^2$ *pathwise; the limit* U_0 *exists and lies in* $H^{:4:}$*.*

(b) Construction statement. *For every admissible cutoff family,* $N_\epsilon^\chi \to U_0$ *in* $L^2(\mathbb{P}_0)$*;* U_0 *is the unique* $Y \in H^{:4:}$ *with* $\mathbb{E}_0[Y{:}\phi(f_1)\cdots\phi(f_4){:}] = 4!\,u\int_\Lambda \prod_{j\le 4}(C_{\text{free}} * f_j)(z)\,dz$ *for all real* $f_j \in C_c^\infty$*; and* U_0 *is* $\mathcal{F}_{\text{free}}$*-measurable. The construction uses only the Gaussian variables* $W(h)$*,* $h \in H_{\mathbb{C}}$*, their own-variance Wick polynomials, Bochner integrals in* $L^2(\mathbb{P}_0)$ *and* $L^2(\mathbb{P}_0)$*-limits; it uses no point variable at* $a = 0$ *and forms no pointwise product of the distribution-valued free field. Nothing is asserted about other representations of* U_0*.*

(c) $\mathbb{E}_0 U_a = 0$ *for* $0 \le a < 1/m$*.*

(d) $\|U_a - U_0\|_2 \le K_\Lambda\sqrt{a}$ *for every regulator* a*.*

(e) For every real $p \ge 1$*,* $\Xi_p < \infty$ *and* $\mathbb{E}_0 e^{-pU_a} \le \Xi_p$ *for* $0 \le a < 1/m$*, including* $a = 0$*.*

(f) For real $p \ge 1$ *and regulators* a*,* $\|e^{-U_a} - e^{-U_0}\|_p \le 2(2p-1)^2 K_\Lambda \Xi_{2p}^{1/(2p)}\sqrt{a}$*.*

(g) $1 \le Z_a := \mathbb{E}_0 e^{-U_a} \le \Xi_1$ *for* $0 \le a < 1/m$*.*

(h) For $\theta \in \{\theta_1, \theta_2\}$*, a regulator* a *and* $F \in L^2(\mathcal{A}_+^{\theta,a})$*:* $\mathbb{E}_0[\overline{\Theta F} F e^{-U_a}] \ge 0$*.*

(i) For $\theta \in \{\theta_1, \theta_2\}$ *and bounded* $\mathcal{A}_+^{\theta,0}$*-measurable* F*:* $\mathbb{E}_0[\overline{\Theta F} F e^{-U_0}] \ge 0$*.*

(j) For $0 \le a < 1/m$ *and any measurable* $\Phi' : \Gamma_0 \to X$ *with* $\Phi'(f) = \phi_a(f)$ *almost surely for real* $f \in C_c^\infty$*:* $B \mapsto \mathbb{E}_0[\mathbf{1}_B(\Phi')e^{-U_a}]/Z_a$ *is a Borel probability on* X*, reflection positive across* θ_1 *and* θ_2 *for bounded observables.*

(k) No weak-coupling restriction is used; no statement is uniform in Λ*.*

Moreover, for $K \ge K_\sharp$*:* $\sup_{0<a<1/m} \mathbb{P}_0(U_a \le -K) \le T(K)$ *and* $\mathbb{P}_0(U_0 < -K) \le T(K)$*.*

Corollary 8.3 (The physical laws)**.** *For* $0 \le a < 1/m$*,* $\nu_{a,\Lambda}$ *of Definition 2.10(d), built with* $\Phi_a = L\tilde{\psi}_a$ *of Theorem 7.10, is a Borel probability on* X *with* $1 \le Z_{a,\Lambda} \le \Xi_1(\Lambda)$*; it is reflection positive across both central coordinate lines for bounded observables; and any Borel* Φ' *satisfying the identity of Theorem 7.10(a) gives the same law. At* $a = 0$ *this is the finite-volume Wick-quartic law on* X*; no interface hypothesis remains.*

Proof. Theorem 7.10(a) supplies the hypothesis of Theorem 8.2(j) for every $0 \le a < 1/m$, including $a = 0$. Uniqueness: Theorem 7.10(c) gives $\Phi' = \Phi_a$ almost surely. □

Remark 8.4 (Scope of the Wick theory). Every constant depends on Λ through $A = |\Lambda|$: $K_\Lambda \propto uA^{1/2}m^{-1/2}$, $K_\sharp \geq 12uA$ and $\Xi_p \geq e^{12puA}$. The only volume-uniform statement is $Z_{a,\Lambda} \geq 1$. The $\sqrt{a}$ bounds are fixed-volume upper bounds, not practical convergence rates. The Gaussian input is used only through Theorems 7.7, 7.9 and 7.10; the weight, the normalizer and reflection positivity enter only after that input is fixed. Exceptional null sets are those of Theorems 7.9 and 7.10: one per fixed regulator value and test, one for countable families, none for uncountably many regulators. The classical inputs C6(a)–(c) are used in their stated forms; their original sources were not inspected.

9 The fixed-volume chain

Throughout this section a physical square Λ (Definition 2.1) is fixed; only the width, the stabilizer, the auxiliary square and the regulator move, one at a time, with all other parameters fixed. Every statement holds for all $m, u > 0$; no weak-coupling restriction is used, and no constant is uniform in the parameters held fixed. The inputs are the theorems of Sections 5–8. Appendix D records the argument lineage.

Lemma 9.1 (Open and closed physical squares)**.** *The neural, Gaussian, residual and physical laws built with $\Lambda = (-l, l)^2$ coincide with those built with $\bar{\Lambda} = [-l, l]^2$.*

Proof. For $v \in E$ the integrals of $v^4 - 6c_a v^2 + 3c_a^2$ and of v^4 over Λ and $\bar{\Lambda}$ (and over the corresponding parts of Ω) coincide because $\partial\Lambda$ is Lebesgue-null; $T_{\Omega,a}$ does not involve Λ. Hence every ordinary-field law coincides. On the common Gaussian space $U_{a,\Lambda} = U_{a,\bar{\Lambda}}$ for every regulator, so their $L^2(\mathbb{P}_0)$ limits coincide. □

The four arrows below remove the width, the stabilizer, the auxiliary square and the regulator, in that order. The width arrow combines a central limit theorem for finitely many smearings with compactness in E, obtained from the increment bounds (17); the bounded continuous weight e^{-V} then passes to the limit. The stabilizer arrow is a monotone limit of weights, the auxiliary-square arrow couples two Gaussian fields whose covariances differ by $\Delta_{\Omega,a}$, and the regulator arrow uses the estimates of Theorem 8.2. Remark 9.7 summarizes the resulting chain.

9.1 The width arrow

Proposition 9.2 (Width arrow)**.** *Fix $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, a regulator a, $\varepsilon > 0$, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. Then Z_N^b and Z_{full} are positive and finite, $Z_N^b \to Z_{\mathrm{full}}$, and $P^b_{N,\varepsilon,\Omega,a;\Lambda} \Rightarrow \nu^v_{\varepsilon,\Omega,a;\Lambda}$ weakly on X as $N \to \infty$. The limit is the same full Gibbs tilt of the Gaussian reference for both architectures.*

Proof. Write $F_N = F^b_{N,a}$, Q_N for its law on E and Q for the law of γ_μ on E; both are Borel by Proposition 5.1(i), since $|F_N| \leq \sqrt{N}/2$ and γ_μ is continuous and jointly measurable (Theorems 6.1, 6.4, 7.1).

Moments. By Theorems 6.1(b) and 6.4(c), each neuron has $|g| = 1/2$ everywhere, $\mathbb{E}g = 0$ and $\mathbb{E}g(z)g(z') = C^{\mathrm{end}}_\mu(z - z')$. For $r = |z - z'|_1$ put $b(r) = \frac{1}{2}(1 - e^{-\mu r}) \leq \mu r/2$. Then $d_i = g_i(z) - g_i(z') \in \{0, \pm 1\}$, $\mathbb{E}d_i = 0$ and $\mathbb{E}d_i^4 = \mathbb{E}d_i^2 = b(r)$; independence across neurons gives

$$\mathbb{E}|F_N(z) - F_N(z')|^4 = \frac{b(r)}{N} + 3\Big(1 - \frac{1}{N}\Big)b(r)^2 \leq \frac{\mu r}{2} + \frac{3\mu^2 r^2}{4}, \qquad \mathbb{E}F_N(z)^4 = \frac{3 - 2/N}{16} \leq \frac{3}{16}. \tag{17}$$

For γ_μ the increment has variance $b(r)$ and fourth moment $3b(r)^2$, and $\mathbb{E}\gamma_\mu(z)^4 = 3/16$, so the same bounds hold.

Compact containment in E. For a box B and the cell-average projection P_δ onto a partition of B into squares of ℓ^1-diameter at most δ, Jensen on each cell C gives $\int_C |v - |C|^{-1}\int_C v|^4 \le |C|^{-1}\int_C\int_C |v(x)-v(y)|^4$, so by (17) and Tonelli $\mathbb{E}\|v - P_\delta v\|^4_{L^4(B)} \le |B|(\mu\delta/2 + 3\mu^2\delta^2/4)$ and $\mathbb{E}\|v\|^4_{L^4(B)} \le 3|B|/16$ under every Q_N and under Q. Fix $\rho > 0$. For $B_j = (-j, j)^2$ choose R_j with Markov tail $\mathbb{P}(\|v\|_{L^4(B_j)} > R_j) \le \rho 2^{-j-1}$, and for $l \ge 1$ a dyadic projection $P_{j,l}$ with $\mathbb{P}(\|v - P_{j,l}v\|_{L^4(B_j)} > 2^{-l}) \le \rho 2^{-j-l-1}$, uniformly in N. The set $K_j = \{v : \|v\|_4 \le R_j,\ \|v - P_{j,l}v\|_4 \le 2^{-l}\ \forall l\} \subset L^4(B_j)$ is closed and totally bounded (each $P_{j,l}K_j$ is bounded in a finite-dimensional space), hence compact; $K = \{v \in E : v|_{B_j} \in K_j\ \forall j\}$ is compact in E, and $\inf_N Q_N(K) \ge 1 - \rho$, $Q(K) \ge 1 - \rho$.

Finite vectors. For real $f_1, \dots, f_r \in C_c^\infty$, the vectors $Y_i = (\int g_i f_1, \dots, \int g_i f_r)$ are independent, identically distributed, bounded and centered, with covariance $\Sigma_{kl} = \iint f_k(z) C_\mu^{\mathrm{end}}(z - z') f_l(z')\, dz\, dz'$ (Fubini). A Taylor expansion of $\mathbb{E}e^{it\cdot Y_i/\sqrt{N}}$ and Lévy's continuity theorem give $(F_N(f_1), \dots, F_N(f_r)) \Rightarrow (\gamma_\mu(f_1), \dots, \gamma_\mu(f_r))$.

Identification on E. Let $\mathcal{D}_E$ be a countable family of real compact smooth tests, dense in each $\mathcal{D}_K$; the evaluations $v \mapsto \int v d$, $d \in \mathcal{D}_E$, are continuous on E and separate its points. Let $H \in C_b(E)$, $\eta > 0$, and K as above. Around each $x \in K$ choose a relatively open $U_x \ni x$ in K on which $|H - H(x)| < \eta$. Some finite-coordinate cylinder $\{v \in K : |\int (v-x)d_i| < \epsilon_x,\ i \le r_x\}$ lies in U_x; otherwise a sequence in the closed set $K \setminus U_x$ would agree ever more closely with x on the first k coordinates, and a convergent subsequence would have the same coordinates as x, contradicting separation. Cover K by finitely many half-size cylinders, let π be the finite vector of all coordinates used, choose continuous $\alpha_i \ge 0$ on its range space supported in the full cylinders and positive on the half-size ones, put $S = \sum \alpha_i$, $A = \pi(K)$ and $G(y) = \sum_i H(x_i)\alpha_i(y)/(S(y) + \mathrm{dist}(y, A))$. Then G is continuous, $\|G\|_\infty \le \|H\|_\infty$ and $\sup_K |H - G\circ\pi| \le \eta$. Hence $|Q_N(H) - Q(H)| \le |Q_N(G\circ\pi) - Q(G\circ\pi)| + 2\eta + 4\|H\|_\infty\rho$, and the finite-vector convergence gives $Q_N \Rightarrow Q$ on E.

Full weight. Completing squares in x^2 gives $ux^4 - b_a x^2 + 3uc_a^2 \ge 3uc_a^2 - b_a^2/(4u)$ with $b_a = 6uc_a + \kappa_a^2/2$ on Λ, and $\varepsilon x^4 - \kappa_a^2 x^2/2 \ge -\kappa_a^4/(16\varepsilon)$ on $\Omega \setminus \Lambda$, so

$$0 < e^{-V_{\varepsilon,\Omega,a;\Lambda}} \le M_{\mathrm{full}} = \exp\{|\Lambda|[b_a^2/(4u) - 3uc_a^2] + |\Omega \setminus \Lambda|\kappa_a^4/(16\varepsilon)\}. \tag{18}$$

V is continuous on E (Proposition 5.1(vi)) and finite on every element. Applying $Q_N \Rightarrow Q$ to the bounded continuous functions e^{-V} and He^{-V} gives $Z_N^b \to Z_{\mathrm{full}} \in (0, \infty)$ and convergence of the normalized tilts on E; composing with the continuous ι gives the statement on X. □

9.2 The mass tilt and the residual laws

Proposition 9.3 (Whole-field mass-tilt identities)**.** *Fix a regulator a, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. For $\varepsilon \ge 0$, the residual weight satisfies $0 < e^{-W_{a,\Lambda} - \varepsilon J_{\Omega,\Lambda}} \le M_{\mathrm{res}} = e^{6u|\Lambda|c_a^2}$ on $C(\mathbb{R}^2)$, and $e^{-3u|\Lambda|c_a^2 - 3\varepsilon|\Omega\setminus\Lambda|c_a^2} \le z_{\varepsilon,\Omega,a;\Lambda} \le M_{\mathrm{res}}$. For $\varepsilon > 0$, $Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a} z_{\varepsilon,\Omega,a;\Lambda}$ and $\nu^v_{\varepsilon,\Omega,a;\Lambda} = \iota_* \mathrm{Res}_{\varepsilon,\Omega,a;\Lambda}$. These identities concern the whole field, including observations outside Ω.*

Proof. $x^4 - 6cx^2 + 3c^2 = (x^2 - 3c)^2 - 6c^2$ gives $W_{a,\Lambda} \ge -6u|\Lambda|c_a^2$, and $J \ge 0$. Under $G'_{\Omega,a}$ the coordinate $\gamma'(z)$ is centered Gaussian with variance $r(z) = C'_{\Omega,a}(z,z) \in [1/4, c_a]$ (Theorem 7.5), so $\mathbb{E}W_{a,\Lambda} = 3u\int_\Lambda (r - c_a)^2 \le 3u|\Lambda|c_a^2$ and $\mathbb{E}J = 3\int_{\Omega\setminus\Lambda} r^2 \le 3|\Omega \setminus \Lambda|c_a^2$; Jensen gives the lower bound. The functionals are continuous on $C(\mathbb{R}^2)$, hence Borel. Since $e^{-V} = e^{T_{\Omega,a}} e^{-W - \varepsilon J}$, Theorem 7.4(c)

with $F = e^{-W-\varepsilon J}\mathbf{1}_B \circ \iota \geq 0$ gives $\mathbb{E}_{G_\mu}[\mathbf{1}_B(\iota\gamma_\mu)e^{-V}] = \mathrm{Det}_{\Omega,a}\mathbb{E}_{G'}[\mathbf{1}_B(\iota\gamma')e^{-W-\varepsilon J}]$; $B = X$ gives the normalizer identity. □

9.3 The stabilizer and auxiliary-square arrows

Proposition 9.4 (Stabilizer arrow). *With a, Ω, Λ fixed, $\|\mathrm{Res}_{\varepsilon,\Omega,a;\Lambda} - \mathrm{Res}_{0,\Omega,a;\Lambda}\|_{\mathrm{var}} \to 0$ as $\varepsilon \downarrow 0$, where $\|\rho - \eta\|_{\mathrm{var}} = \sup_{|H|\leq 1 \text{ Borel}} |\rho(H) - \eta(H)|$; hence $\|\nu^{v}_{\varepsilon,\Omega,a;\Lambda} - \nu^{(2)}_{\Omega,a;\Lambda}\|_{\mathrm{var}} \to 0$, $z_\varepsilon \to z_0$ and $Z_{\mathrm{full}} \to \mathrm{Det}_{\Omega,a} z_0$.*

Proof. Under $G'_{\Omega,a}$ the weights $h_\varepsilon = e^{-W-\varepsilon J}$ increase pointwise to h_0 as $\varepsilon \downarrow 0$ and are bounded by M_{res}, so $\|h_\varepsilon - h_0\|_1 = z_0 - z_\varepsilon \to 0$ and $\|h_\varepsilon/z_\varepsilon - h_0/z_0\|_1 \leq 2\|h_\varepsilon - h_0\|_1/z_0$. The variation distance of the path laws equals this L^1 distance, and a measurable pushforward does not increase it. □

Proposition 9.5 (Auxiliary-square arrow). *With a regulator a and Λ fixed, along every increasing family of squares $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$ with $R \to \infty$, $\beta(\nu^{(2)}_{\Omega,a;\Lambda}, \nu_{a,\Lambda}) \leq e^{3u|\Lambda|c_a^2}(2e_\Omega + M_{\mathrm{res}} r_\Omega) \to 0$ and $|Z_{a,\Lambda} - z_{0,\Omega,a;\Lambda}| \leq e_\Omega \to 0$, where e_Ω and r_Ω are defined in the proof. For each Ω a separate coupling is used.*

Proof. Fix Ω and write $\Delta = \Delta_{\Omega,a}$. On a product space let γ' have law $G'_{\Omega,a}$ and let ψ_Ω be an independent centered Gaussian field with covariance Δ and continuous paths (Theorem 7.6(e)); independence of the two continuous paths follows from their countable dense point evaluations. Put $X_\Omega = \gamma' + \psi_\Omega$ and $Y_\Omega = \gamma'$. Then X_Ω is a continuous centered Gaussian field with covariance $C' + \Delta = C_a$, so its law on $C(\mathbb{R}^2)$ is that of $\tilde{\phi}_a$ (Theorem 7.9). On the common Gaussian space $\Phi_a = \iota\tilde{\phi}_a$ almost surely (Theorem 7.10(b)) and $U_{a,\Lambda} = W_{a,\Lambda}(\tilde{\phi}_a)$, so $\nu_{a,\Lambda}(B) = \mathbb{E}[\mathbf{1}_B(\iota X_\Omega)e^{-W(X_\Omega)}]/\mathbb{E}e^{-W(X_\Omega)}$ and $Z_{a,\Lambda} = \mathbb{E}e^{-W(X_\Omega)}$, while $\nu^{(2)}_{\Omega,a;\Lambda}(B) = \mathbb{E}[\mathbf{1}_B(\iota Y_\Omega)e^{-W(Y_\Omega)}]/z_0$.

Let $\delta_\Omega(K) = \sup_{z\in K} \Delta(z,z)$ for compact K. By Parseval, $p_j(\iota\psi) \leq \|\chi_j\psi\|_{L^2}$, so $\mathbb{E}p_j(\iota X_\Omega - \iota Y_\Omega)^2 \leq \|\chi_j\|_2^2 \delta_\Omega(\mathrm{supp}\,\chi_j)$ and $\mathbb{E}d(\iota X_\Omega, \iota Y_\Omega) \leq r_\Omega := \sum_j 2^{-j}\min(1, \|\chi_j\|_2 \delta_\Omega(\mathrm{supp}\,\chi_j)^{1/2})$, which tends to 0 by Theorem 7.6(d) and dominated convergence. For $p_c(x) = x^4 - 6cx^2 + 3c^2$, $|p_c(x) - p_c(y)| \leq |x-y|(|x|+|y|)(x^2+y^2+6c) \leq |x-y|[2(|x|^3+|y|^3) + 6c(|x|+|y|)]$. Each $X_\Omega(z)$, $Y_\Omega(z)$ has variance at most $c = c_a$, so the bracket has L^2 norm at most $L_c = (4\sqrt{15}+12)c^{3/2}$, and Cauchy–Schwarz gives $\mathbb{E}|W(X_\Omega) - W(Y_\Omega)| \leq u|\Lambda|L_c\,\delta_\Omega(\bar{\Lambda})^{1/2}$. Both interactions are at least $-6u|\Lambda|c^2$, where $x \mapsto e^{-x}$ is M_{res}-Lipschitz, so $\mathbb{E}|e^{-W(X_\Omega)} - e^{-W(Y_\Omega)}| \leq e_\Omega := M_{\mathrm{res}} u|\Lambda|L_c\delta_\Omega(\bar{\Lambda})^{1/2} \to 0$. For $|H| \leq 1$ and $\mathrm{Lip}_d(H) \leq 1$, adding and subtracting the cross terms gives $|\nu^{(2)}(H) - \nu_{a,\Lambda}(H)| \leq [2e_\Omega + \mathbb{E}(e^{-W(X_\Omega)}d(\iota X_\Omega, \iota Y_\Omega))]/z_0$, and $z_0 \geq e^{-3u|\Lambda|c^2}$ by Proposition 9.3. □

9.4 The regulator arrow

Proposition 9.6 (Regulator arrow). *With Λ fixed, $Z_{a,\Lambda} \to Z_{0,\Lambda} \geq 1$ and $\beta(\nu_{a,\Lambda}, \nu_{0,\Lambda}) \leq 2e_a + \Xi_2(\Lambda)^{1/2}h(a) \to 0$ as $a \downarrow 0$, where $e_a = \|e^{-U_{a,\Lambda}} - e^{-U_{0,\Lambda}}\|_{L^1(\mathbb{P}_0)} \leq 2K_\Lambda \Xi_2(\Lambda)^{1/2}\sqrt{a}$ and $h(a)$ is as in Theorem 7.10(e).*

Proof. Theorem 8.2(f) with $p = 1$ bounds e_a, so $|Z_a - Z_0| \leq e_a$, and $Z_a, Z_0 \geq 1$ by Theorem 8.2(g). For $|H| \leq 1$, $\mathrm{Lip}_d(H) \leq 1$, with $w_a = e^{-U_a}$, $|\mathbb{E}[H(\Phi_a)w_a - H(\Phi_0)w_0]| \leq e_a + \mathbb{E}[w_0 d(\Phi_a, \Phi_0)] \leq e_a + \Xi_2^{1/2}h(a)$ by Cauchy–Schwarz, Theorem 8.2(e) with $p = 2$ and Theorem 7.10(e); the normalizer correction is at most e_a. □

Remark 9.7 (The chain of laws). For fixed parameters the five laws $P^b_{N,\varepsilon,\Omega,a;\Lambda}$, $\nu^v_{\varepsilon,\Omega,a;\Lambda} = \iota_*\mathrm{Res}_\varepsilon$, $\nu^{(2)}_{\Omega,a;\Lambda} = \iota_*\mathrm{Res}_0$, $\nu_{a,\Lambda}$ and $\nu_{0,\Lambda}$ are joined by the four arrows of Propositions 9.2, 9.4, 9.5 and 9.6; there is no mass change, no factorial change, no omitted quadratic term and no extra normalization between them. The quadratic-tilt normalizer $\mathrm{Det}_{\Omega,a}$ is nondecreasing in Ω (Theorem 7.4(a)). No monotonicity in Ω is asserted for the full Gaussian normalizer $Z_{\mathrm{full}} = \mathrm{Det}_{\Omega,a} z_{\varepsilon,\Omega,a;\Lambda}$, and none holds in general at a fixed $\varepsilon > 0$. For $\Omega_1 = \bar{\Lambda}$ the outer integral $J_{\Omega_1,\Lambda}$ vanishes, so Z_{full} for Ω_1 does not depend on ε. For a strictly larger centered square Ω_2, the continuous endpoint Gaussian is almost surely nonzero at an interior point of $\Omega_2 \setminus \bar{\Lambda}$, so $J_{\Omega_2,\Lambda} > 0$ almost surely; since $e^{-W_{a,\Lambda}} e^{T_{\Omega_2,a}} \le e^{6u|\Lambda|c_a^2} e^{T_{\Omega_2,a}}$ is integrable, dominated convergence gives $Z_{\mathrm{full}} \to 0$ for Ω_2 as $\varepsilon \to \infty$. Only the residual normalizers $z_{0,\Omega,a;\Lambda}$, which satisfy $e^{-3u|\Lambda|c_a^2} \le z_{0,\Omega,a;\Lambda} \le e^{6u|\Lambda|c_a^2}$ for every Ω (Proposition 9.3), enter the auxiliary arrow. The identification of $\nu_{0,\Lambda}$ with the classical free-boundary law is Proposition 10.4.

10 The continuum target: reduced lemmas

The growth of the physical square is handled with classical input in three steps. Proposition 10.4 identifies the continuum law $\nu_{0,\Lambda}$ on compact smooth tests with the classical free-boundary law, so the classical theorems apply. Theorem 10.8 uses dilation (Lemma 10.5) to move (m, u) to the reference mass, where C2 makes the two classical states coincide and C1 gives convergence for finitely many smearings. Theorem 10.10 upgrades this to weak convergence on X with all compact moments, and Theorem 10.16 records the properties of the target.

This section proves, directly from C1–C6 and the foundations of Sections 5–9, the three continuum statements used by the construction: the continuum limit with its Wick identification (Theorem 10.8), the whole-field volume limit with all compact moments (Theorem 10.10) and the target properties (Theorem 10.16). Appendix D records the argument lineage.

10.1 Classical finite-volume laws and the Wick identification

For $\psi \in \mathcal{S}'(\mathbb{R}^2)$ and a real even nonnegative $\rho_1 \in C_c^\infty(\mathbb{R}^2)$ with $\int \rho_1 = 1$, put $\rho_\delta = \delta^{-2}\rho_1(\cdot/\delta)$ and $\psi_\delta(z) = \psi(\rho_\delta(z - \cdot))$.

Lemma 10.1 (Classical Wick existence and finite normalizers from standard tools)**.** *Let $r, b > 0$ and $B \subset \mathbb{R}^2$ be bounded measurable, with $A = |B|$. On the free Gaussian space $(\mathcal{S}', \mathcal{C}, P_r)$ the own-variance mollified quartics $b\int_B :\psi_\delta(z)^4: dz$ have a mollifier-independent limit $U^S_{r,b,B}$ in the fourth homogeneous Gaussian chaos in $L^2(P_r)$. It has mean zero and, for every real $t > 0$,*

$$\mathbb{E}_{P_r} e^{-tU^S_{r,b,B}} < \infty, \qquad 1 \le Z^S_{r,b,B} := \mathbb{E}_{P_r} e^{-U^S_{r,b,B}} < \infty.$$

The normalized density defines the classical finite-volume law in the convention of Section 3. If B is a centered axis-parallel square, this law is invariant under field sign and θ_1, θ_2. This lemma uses only the free covariance and C6; it does not use the Gaussian or Wick foundations of Sections 7–8. Every constant below is fixed-volume. If $A = 0$, the interaction is zero and all conclusions hold with $Z^S = 1$.

Proof. Assume $A > 0$ and write $c(p) = (r^2 + |p|^2)^{-1}$ and $C = \mathcal{I}c$. Polar integration and Hausdorff–Young give

$$\int_{\mathbb{R}^2} c(p)^{4/3} dp = 3\pi r^{-2/3}, \qquad \|C\|_4 \le M_4(r) := (2\pi)^{-3/2}(3\pi)^{3/4} r^{-1/2}. \tag{19}$$

For the real even mollifier fixed above, the cross-covariance of $\psi_\delta(z)$ and $\psi_\epsilon(z')$ is $C_{\delta\epsilon}(z-z') = (\rho_\delta * C * \rho_\epsilon)(z-z')$. Approximate-identity convergence gives $C_{\delta\epsilon} \to C$ in L^4; equivalently apply Hausdorff–Young and dominated convergence to $\hat\rho_1(\delta p)\hat\rho_1(\epsilon p)c(p)$. The multipliers are bounded uniformly, converge pointwise to one and are dominated by a constant times $c \in L^{4/3}$. The same argument works with two different real mollifiers of integral one. Since

$$\|f^4 - g^4\|_1 \le \|f-g\|_4(\|f\|_4^3 + \|f\|_4^2\|g\|_4 + \|f\|_4\|g\|_4^2 + \|g\|_4^3),$$

the fourth powers converge in L^1. The Wick covariance identity gives

$$\mathbb{E}J_\delta J_\epsilon = 4!b^2 \int_{B^2} C_{\delta\epsilon}(z-z')^4\,dz\,dz', \qquad J_\delta = b\int_B :\psi_\delta(z)^4:\,dz. \tag{20}$$

For integrable H, $\int_{B^2}|H(z-z')| \le A\|H\|_1$. Thus J_δ is Cauchy in L^2, with the same limit for every such mollifier. The integrands are continuous L^2 fourth-chaos variables; their Bochner integrals equal the pathwise integrals by Fubini, using finite Gaussian moments on B. Chaos closure proves fourth-chaos membership and mean zero. No point variable or product of distributions at zero cutoff is formed.

A cutoff with a lower bound. Fix a real radial $\chi \in C_c^\infty(\mathbb{R}^2)$ with $0 \le \chi \le 1$, $\chi = 1$ on the unit ball and $\chi = 0$ outside the ball of radius two. For $R \ge R_0 := \max(1, r)$, put $\chi_R(p) = \chi(p/R)$, $\psi_R(z) = \psi(\mathcal{I}\chi_R(z-\cdot))$ and $U_R = b\int_B :\psi_R(z)^4:_{c_R} dz$. The inverse transform is a real Schwartz function, so these are ordinary smoothed evaluations of ψ; their paths are smooth and jointly measurable. Their variance and lower bound are

$$c_R = (2\pi)^{-2}\int \chi_R^2 c \le \frac{1}{4\pi}\log(1+4R^2/r^2), \qquad U_R \ge -6bAc_R^2. \tag{21}$$

The last inequality is $(x^2-3c)^2 - 6c^2 \ge -6c^2$. Let $C_R^1 = \mathcal{I}(\chi_R c)$, $C_R^2 = \mathcal{I}(\chi_R^2 c)$ and $a_4 = (2\pi)^{-3/2}(3\pi)^{3/4}$. Polar integration of the tail gives, for $j = 1, 2$,

$$\|C - C_R^j\|_4 \le a_4 R^{-1/2}, \qquad \|C_R^j\|_4 \le M_4(r). \tag{22}$$

Indeed $|1-\chi_R^j| \le \mathbf{1}_{\{|p|>R\}}$ and $\int_{|p|>R} c^{4/3} \le 3\pi R^{-2/3}$. Taking the mollifier limit in the Gaussian cross-covariances yields

$$\begin{aligned}\|U_{r,b,B}^S - U_R\|_2^2 &= 4!b^2\int_{B^2}[C^4 + (C_R^2)^4 - 2(C_R^1)^4](z-z')\,dz\,dz' \\ &\le 288b^2AM_4(r)^3a_4R^{-1/2}.\end{aligned} \tag{23}$$

Here each fourth-power difference is bounded by four times $M_4(r)^3$ times its L^4 difference, and the coefficients of the two differences are one and two. Put $a_2 = b[288AM_4(r)^3a_4]^{1/2}$. Both interactions lie in the fourth chaos, so

$$\|U_{r,b,B}^S - U_R\|_p \le (p-1)^2 a_2 R^{-1/4} \quad (p \ge 2). \tag{24}$$

Negative tails and every needed normalization. The following explicit choices make the fixed-volume character of the constants unambiguous:

$$c_0 = \frac{1}{4\pi}\log(1+4R_0^2/r^2), \qquad d = \frac{\pi}{\sqrt{24bA}}, \qquad R(K) = R_0 e^{d\sqrt{K}},$$

$$K_0 = \max\{1, 32bAc_0^2, (16\log 2/d)^2, 2ea_2\}.$$

For $K \geq K_0$, the logarithmic variance bound implies $c_{R(K)} \leq c_0 + d\sqrt{K}/(2\pi)$ and hence $6bAc_{R(K)}^2 \leq 12bAc_0^2 + 3bAd^2K/\pi^2 \leq K/2$. On $\{U^S_{r,b,B} < -K\}$ the difference from $U_{R(K)}$ therefore has absolute value greater than $K/2$. Set $p = R(K)^{1/16} \geq 2$. Markov and (24) give

$$P_r(U^S_{r,b,B} < -K) \leq \left[\frac{2a_2}{K} R(K)^{-1/8}\right]^{R(K)^{1/16}} \leq \exp[-R(K)^{1/16}]. \tag{25}$$

The bracket is at most e^{-1} by the choice of K_0. For every fixed $t > 0$, the layer-cake formula bounds the exponential moment by

$$\mathbb{E}e^{-tU^S_{r,b,B}} \leq e^{tK_0} + \int_{K_0}^{\infty} te^{tK} \exp[-R_0^{1/16} e^{d\sqrt{K}/16}]\, dK < \infty. \tag{26}$$

The double exponential in $\sqrt{K}$ dominates every e^{tK}: for example $e^{d\sqrt{K}/16}/K \to \infty$. Jensen and $\mathbb{E}U^S_{r,b,B} = 0$ give $Z^S \geq 1$. Thus both the L^2 and finite-normalizer provisos in the source convention are proved before that convention is used. These estimates are separate from the constants K_Λ, Ξ_p of Theorem 8.2.

Finally P_r is invariant under field sign and Euclidean isometries, by its centered Gaussian covariance and cylinder-law uniqueness. With radial χ_R, the interaction U_R on a centered axis-parallel square is unchanged by sign and θ_1, θ_2. The same identities pass in L^2 to U^S, then to its exponential and its normalized law. This symmetry proof uses no foundation interaction or normalizer. □

Lemma 10.2 (Transfer between Gaussian realizations)**.** *Let $(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0, W)$ be the common space of Theorem 7.7 at mass m and P_m the free measure on $(\mathcal{S}', \mathcal{C})$ of Section 3. There is a unique map Ξ from $L^0(\Gamma_0, \mathcal{F}^{\mathcal{S}}_{\text{free}}, \mathbb{P}_0)$, $\mathcal{F}^{\mathcal{S}}_{\text{free}} = \sigma(\phi(f) : f \in \mathcal{S}(\mathbb{R}^2;\mathbb{R})) \vee \mathcal{N}$, to $L^0(\mathcal{S}', \mathcal{C}, P_m)$ with $\Xi[g((\phi(f_j))_j)] = g((\psi(f_j))_j)$ for every sequence of real Schwartz f_j and every Borel $g : \mathbb{R}^{\mathbb{N}} \to \mathbb{C}$. It is linear and multiplicative, preserves joint laws of countable families, is an isometry of every L^r, commutes with L^r-limits and with Bochner integrals in L^2, and maps $:\xi^4:_{\operatorname{Var}\xi}$ to $:(\Xi\xi)^4:_{\operatorname{Var}\xi}$.*

Proof. For real Schwartz f, g, $\mathbb{E}_0[\phi(f)\phi(g)] = (2\pi)^{-2} \int \overline{\hat{f}}\hat{g}\hat{C}_0 = \mathbb{E}_{P_m}[\psi(f)\psi(g)]$ (Theorem 7.7), so for any two sequences the vectors $((\phi(f_j)), (\phi(f'_j)))$ and $((\psi(f_j)), (\psi(f'_j)))$ are centered Gaussian on $\mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}}$ with the same covariances, hence have the same law. If $g((\phi(f_j))) = g'((\phi(f'_j)))$ almost surely, the Borel set $\{g \neq g'\}$ therefore has measure zero for the ψ-vectors too, so Ξ is well defined. Every $Y \in L^0(\mathcal{F}^{\mathcal{S}}_{\text{free}})$ has such a representation: for a countable family $\mathcal{D}_S$ dense in $\mathcal{S}$, $\|\phi(f) - \phi(d)\|_2 \leq m^{-1}\|f - d\|_2$ (Theorem 7.8), so each $\phi(f)$ is an almost sure limit of $\phi(d_k)$, and Doob–Dynkin applies. Merging families gives linearity and multiplicativity; law preservation gives the isometries; a bounded linear map commutes with Bochner integrals; variances are preserved. □

Definition 10.3 (Classical finite-volume laws)**.** For a physical square Λ, the mollified own-variance Wick integrals $N^S_\delta = u\int_\Lambda :\psi_\delta(z)^4: dz$ (pathwise integrals of the continuous functions $z \mapsto \psi_\delta(z)$, equal to the corresponding Bochner integrals in $L^2(P_m)$) converge in $L^2(P_m)$ as $\delta \downarrow 0$ (Lemma 10.1); put $U^S_{m,u,\Lambda} = \lim_\delta N^S_\delta$, $Z^S_{m,u,\Lambda} = \mathbb{E}_{P_m} e^{-U^S}$ and $d\nu_{m,u,\Lambda} = (Z^S)^{-1} e^{-U^S} dP_m$. The normalizer is positive and finite by Lemma 10.1, independently of the project Wick foundations. By the convention paragraph of Section 3, this is the free-boundary finite-volume law of [8, (1.1), pp. 496–497] for $P(X) = uX^4$ at mass m, and the continuum law of [10, §IX] with boundary condition F.

Proposition 10.4 (Identification of the zero-regulator law)**.** *For every $m, u > 0$ and physical square Λ:*

(i) $U^S_{m,u,\Lambda}$ *exists, does not depend on* ρ_1*, and* $U^S = \Xi U_{0,\Lambda}$*; also* $Z^S_{m,u,\Lambda} = Z_{0,\Lambda} \in [1, \Xi_1(\Lambda)]$ *and* $\|d\nu_{m,u,\Lambda}/dP_m\|_{L^2(P_m)} \le \Xi_2(\Lambda)^{1/2}$*;*

(ii) for every finite tuple of real $f_1, \dots, f_r \in C_c^\infty(\mathbb{R}^2)$*, the law of* $(v(f_1), \dots, v(f_r))$ *under* $\nu_{0,\Lambda}$ *equals the law of* $(\psi(f_1), \dots, \psi(f_r))$ *under* $\nu_{m,u,\Lambda}$*;*

(iii) $\nu_{m,u,\Lambda}$ *is invariant under* $\psi \mapsto -\psi$ *and under* θ_1*,* θ_2*.*

Proof. (i) The family $\chi_\delta(p) = \overline{\hat\rho_1(\delta p)}$ is admissible (Definition 8.1), with $\phi^\chi_\delta(z) = \phi(\rho_\delta(z - \cdot))$, so by Theorem 8.2(b) the corresponding Bochner integrals N_δ on Γ_0 converge to $U_0 = U_{0,\Lambda}$ in $L^2(\mathbb{P}_0)$. Each N_δ and U_0 is $\mathcal{F}^{\mathcal{S}}_{\mathrm{free}}$-measurable (Theorem 8.2(b)), and $\Xi N_\delta = N^S_\delta$ by Lemma 10.2. Hence $N^S_\delta \to \Xi U_0$ in $L^2(P_m)$; the limit does not depend on ρ_1 because U_0 does not. Multiplicativity and law preservation give $\mathbb{E}_{P_m} e^{-pU^S} = \mathbb{E}_0 e^{-pU_0}$, and Theorem 8.2(e),(g) give the bounds. (ii) By Theorem 7.10(a), $\Phi_0(f_i) = \phi(f_i)$ almost surely; for bounded Borel G, $\mathbb{E}_0[G(\phi(f))e^{-U_0}] = \mathbb{E}_{P_m}[G(\psi(f))e^{-U^S}]$ by the lemma. (iii) P_m is invariant under $\psi \mapsto -\psi$ and under θ_j, and the mollified quartics are even. By (i) we may take ρ_1 radial; then $(\theta_j\psi)_\delta(z) = \psi_\delta(\theta_j z)$, and since Λ is θ_j-invariant (a centered axis-parallel square) the change of variables $z \mapsto \theta_j z$ shows that $N^S_\delta \circ \theta_j = N^S_\delta$; hence $U^S \circ \theta_j = U^S$ almost surely. □

Lemma 10.5 (Dilation)**.** *For* $s > 0$ *let* $(L_s f)(x) = s^{-2} f(x/s)$ *and* $(T_s\psi)(f) = \psi(L_s f)$ *on* $\mathcal{S}'$*. Then* $(T_s)_* P_M = P_{sM}$*; for* $s = m/M$*,* $\lambda = u/s^2$ *and every physical square* Λ*,* $U^S_{m,u,\Lambda} \circ T_s = U^S_{M,\lambda,s\Lambda}$ P_M*-almost surely and* $\nu_{m,u,\Lambda} = (T_s)_* \nu_{M,\lambda,s\Lambda}$*. With* $(A_b\psi)(f) = \psi(f(\cdot - b))$ *and* $(R\psi)(f) = \psi(f \circ R^{-1})$ *for orthogonal* R*:* $A_b T_s = T_s A_{sb}$ *and* $RT_s = T_s R$*;* T_s *commutes with* θ_1, θ_2 *and maps* $\mathcal{C}_{H^\theta_+}$ *into itself.*

Proof. This proof uses the classical L^2 limits and finite normalizers of Lemma 10.1, without the common-space identification. $\widehat{L_s f}(p) = \hat f(sp)$, so

$$\begin{aligned}\operatorname{Var}_{P_M} \psi(L_s f) &= (2\pi)^{-2} \int |\hat f(sp)|^2 (M^2 + |p|^2)^{-1} dp \\ &= (2\pi)^{-2} \int |\hat f(q)|^2 (s^2 M^2 + |q|^2)^{-1} dq.\end{aligned}$$

Next, $L_s[\rho_\delta(x - \cdot)] = \rho_{s\delta}(sx - \cdot)$, so $(T_s\psi)_\delta(x) = \psi_{s\delta}(sx)$; since $T_s\psi$ has law P_m under P_M, the Wick constants agree, $\operatorname{Var}_{P_m} \phi_\delta(x) = \operatorname{Var}_{P_M} \psi_{s\delta}(sx)$. The change of variables $y = sx$ in the Bochner integral gives $u \int_\Lambda {:}(T_s\psi)_\delta(x)^4{:}\, dx = \lambda \int_{s\Lambda} {:}\psi_{s\delta}(y)^4{:}\, dy$; as T_s is measure preserving from $(\mathcal{S}', P_M)$ to $(\mathcal{S}', P_m)$, the L^2 limits agree, and the normalized laws correspond. The intertwining relations follow from $L_s[f(\cdot - b)] = (L_s f)(\cdot - sb)$ and $L_s(f \circ R^{-1}) = (L_s f) \circ R^{-1}$; L_s preserves supports in $\{z_2 > 0\}$ and $\{z_1 > 0\}$. □

Lemma 10.6 (Readings of the convergence in C1(iv))**.** *Let* μ_n, μ *be probabilities on* $(\mathcal{S}', \mathcal{C})$ *and* $g \in C_c^\infty$ *real, and suppose* $\mathbb{E}_\mu e^{t\psi(g)} < \infty$ *for all real* t*. If* $\mathbb{E}_{\mu_n} e^{it\psi(g)} \to \mathbb{E}_\mu e^{it\psi(g)}$ *for every real* t*, or all moments of* $\psi(g)$ *converge, or* $\mathbb{E}_{\mu_n} e^{t\psi(g)} \to \mathbb{E}_\mu e^{t\psi(g)}$ *for all real* t*, then the laws of* $\psi(g)$ *converge weakly. For the states of C1 at mass* M*,* $\mathbb{E} e^{t\psi(g)} \le \|f_B\|_{L^2(P_M)} \exp(t^2 \langle g, (-\Delta + M^2)^{-1} g\rangle)$ *for every real* t *and every bounded open* $B \supseteq \operatorname{supp} g$*, by C1(ii) and Cauchy–Schwarz.*

Proof. Characteristic functions: Lévy's continuity theorem, applied with convergence at every real frequency t to the characteristic function of the law of $\psi(g)$ under μ. Moments: a law with finite exponential moments is determined by its moments, and moment convergence then implies weak convergence. Laplace transforms: convergence of moment generating functions on a neighbourhood of 0 implies weak convergence. □

Remark 10.7 (One frequency does not suffice). Convergence of $\mathbb{E}_{\mu_n} e^{i\psi(g)}$ at the single frequency $t = 1$ does not imply weak convergence of the laws of $\psi(g)$: for a nonzero real $g \in C_c^\infty$ and the regular tempered distribution $h = 2\pi g / \int g^2$, the probabilities $\mu_n = \delta_h$ and $\mu = \delta_0$ have equal characteristic values at frequency one, and μ has every exponential moment, while the laws of $\psi(g)$ are $\delta_{2\pi}$ and δ_0. An earlier version of Lemma 10.6 admitted that single-frequency alternative; it is withdrawn. Where the characteristic-function reading of C1(iv) is used (proof of Theorem 10.8), it is applied to $g = \sum_j t_j L_s f_j$ for every $t \in \mathbb{R}^r$, which supplies every real frequency.

Theorem 10.8 (Continuum limit and target; reduced lemma LR-R)**.** *Fix the admissible pair of Definition 3.2 and (m, u) in the regime; let $s = m/M_{\mathrm{ref}}$ and $\lambda = uM_{\mathrm{ref}}^2/m^2 \in (0, \lambda_*)$. The two states $\nu_\pm$ of C1(i) for λX^4 at mass M_{ref} coincide; call the common state v, and put $\nu^S = (T_s)_* v$. Then for every $\ell > 0$ and every finite tuple of real $f_1, \dots, f_r \in C_c^\infty(\mathbb{R}^2)$,*

$$\mathrm{Law}_{\nu_{m,u,\Lambda_n}}(\psi(f_1), \dots, \psi(f_r)) \Rightarrow \mathrm{Law}_{\nu^S}(\psi(f_1), \dots, \psi(f_r)), \qquad \Lambda_n = (-n\ell, n\ell)^2,$$

and therefore, by Proposition 10.4(ii), the laws of $(v(f_1), \dots, v(f_r))$ under $\nu_n = \nu_{0,\Lambda_n}$ converge to the same limit. The same ν^S serves all tuples and all ℓ.

Proof. At mass $M_{\mathrm{ref}} > M_0$ and $0 < \lambda < \lambda_* \le \lambda_0$, both ν_+ and ν_- belong to $\mathcal{G}_{\mathrm{ur}}$ (C2(i)), which is a singleton (C2(ii)); the states coincide as measures on $\mathcal{C}$ (only $\mathcal{C}$-measurable observables are used, so only the restrictions of the AHKZ members to $\mathcal{C}$ enter). By C1(iii) the pressure is differentiable at zero external field, and by C1(iv) the free measures $\nu_{M_{\mathrm{ref}},\lambda,\Lambda'}$ converge to v along every exhaustion by centered squares Λ', on characteristic functions of compact smooth tests (under the moment or Laplace readings, Lemma 10.6 gives weak convergence of the law of $\psi(g)$, hence convergence of $\mathbb{E}e^{i\psi(g)}$, for every real compact smooth g). For $t \in \mathbb{R}^r$ put $g = \sum_j t_j L_s f_j$, a real compact smooth test. By Lemma 10.5, $\mathbb{E}_{\nu_{m,u,\Lambda_n}} e^{i\sum_j t_j \psi(f_j)} = \mathbb{E}_{\nu_{M_{\mathrm{ref}},\lambda,s\Lambda_n}} e^{i\psi(g)} \to \mathbb{E}_v e^{i\psi(g)} = \mathbb{E}_{\nu^S} e^{i\sum_j t_j \psi(f_j)}$, since $s\Lambda_n = (-ns\ell, ns\ell)^2$ is an exhaustion by centered squares. Since t ranges over all of $\mathbb{R}^r$, this is convergence of the joint characteristic functions at every frequency; the limit is the characteristic function of $\mathrm{Law}_{\nu^S}(\psi(f_1), \dots, \psi(f_r))$, and Lévy's continuity theorem concludes. The classical finite-volume laws used here exist with finite normalizers by Lemma 10.1; that direct lemma, dilation and the readings argument use only C6 and the stated classical sources. Finally Proposition 10.4(ii) supplies the separate identification with ν_n on the common project space. Only this last bridge uses the Gaussian/Wick foundations. □

Remark 10.9 (Scope of Theorem 10.8). The classical target and finite-joint classical limit use only C1, C2, C6 and the direct classical Lemma 10.1. Its exponential normalizers follow from the explicit negative-tail proof, separately from L^2 convergence. The identification with the project common-space law additionally uses Proposition 10.4 and the Gaussian/Wick foundations. No neural law, stagewise neural bridge or midpoint approximation is used. The convergence asserted is finite-joint weak convergence for compact smooth tests; no moment, density, total-variation or rate conclusion is drawn from C1(iv).

10.2 The volume limit on X

Theorem 10.10 (Volume estimates and whole-field limit; reduced lemma V1-R)**.** *(i)* (E, fixed volume.) *For all $m, u > 0$, every physical square Λ, every nonnegative real $g \in C_c^\infty(\Lambda)$ and real t, $\mathbb{E}_{\nu_{m,u,\Lambda}} e^{t\psi(g)} \le \exp(t^2 \mathbb{E}_{\nu_{m,u,\Lambda}} \psi(g)^2/2)$.*

(ii) (E, uniform.) *Let (m,u) be in the regime, $g \geq 0$ real compact smooth and $N_g \geq 2$ an integer such that* $\operatorname{supp} g \subset \Lambda_n$ *for $n \geq N_g$ (the choice $N_g \geq 2$ makes the early-volume maximum below nonempty). With $R_g > 0$ such that* $\sup_{n \geq N_g} \nu_n(|v(g)| > R_g) < 1/12$ *and $V_g = 2R_g^2$: for $n \geq N_g$, real smooth f with $|f| \leq g$ and real t, $\mathbb{E}_{\nu_n} e^{tv(f)} \leq e^{V_g t^2/2}$, and $\mathbb{E}_{\nu_n}|v(f)|^p \leq 2(2V_g)^{p/2}\Gamma(1+p/2)$ for $p > 0$. Including the finitely many $n < N_g$, $\sup_{n \geq 1, |f| \leq g} \mathbb{E}_{\nu_n}|v(f)|^p \leq C_{p,g} < \infty$, with $C_{p,g} = \max\{2(2V_g)^{p/2}\Gamma(1+\frac{p}{2}),\ [2^p\Gamma(p+\frac{1}{2})/\sqrt{\pi}]^{1/2} m^{-p}\|g\|_2^p \max_{n<N_g} \Xi_2(\Lambda_n)^{1/2}\}$.*

(iii) (F.) *For $j \geq 1$ and $1 < s < 2$, $\sup_n \mathbb{E}_{\nu_n}\|\chi_j v\|^2_{H^{-s}(\mathbb{T}_j)} \leq B_{j,s} < \infty$, and every finite positive moment of this norm is bounded uniformly in n. The family (ν_n) is tight on X. There is a Borel probability $\nu = \nu_\infty^X$ on X and a $\mathcal{C}$-measurable map $\mathcal{R}$ from a set of full ν^S-measure into X with $\mathcal{R}(\psi)(f) = \psi(f)$ for all compact smooth f and $\mathcal{R}_*\nu^S = \nu$ (ν represents ν^S); and $\nu_n \Rightarrow \nu$ weakly on X. The target is almost surely locally H^{-s} for every $s > 1$.*

(iv) (M.) *For every finite tuple of real compact smooth tests and every real or complex polynomial Q, $\mathbb{E}_{\nu_n} Q(v(f_1), \dots) \to \mathbb{E}_\nu Q(v(f_1), \dots)$; all these expectations are finite, and every fixed positive power of $|Q|$ has expectation bounded uniformly in n.*

Proof. (i) Fix Λ of side L and meshes $\delta = L/(2k+1)$ as in C4. By C4 the lattice density is $\exp[-\sum_i (a_i q_i^4 + b_i q_i^2) + \sum_{i<j} J_{ij} q_i q_j]$ up to normalization, with $a_i = u\delta^2 > 0$, $b_i = (B_\delta)_{ii}/2 - 6u\delta^2 c_\delta \in \mathbb{R}$ and $J_{ij} = -(B_\delta)_{ij} \geq 0$. That the off-diagonal entries are nonpositive can be checked directly: with I the sites in Λ and O the others, $A_{OO}^{-1} = \sum_k (4 + m^2\delta^2)^{-1-k} \mathrm{adj}^k_{OO}$ has nonnegative entries, $A_{IO} \leq 0$, and $B_\delta = A_{II} - A_{IO}A_{OO}^{-1}A_{OI}$. In the ordered-pair form of C5 the pair coefficients are $a_{ij} = a_{ji} = J_{ij}/2$. For $\epsilon \in (0,1]$ replace them by $(J_{ij} + \epsilon)/2$; every fixed polynomial moment and the normalizer converge as $\epsilon \to 0$ by dominated convergence with the integrable majorant $|q|^k \prod_i e^{-a_i q_i^4 - b_i q_i^2} \exp(\sum_{i<j}(J_{ij}+1)|q_i||q_j| + \sum_i h_i|q_i|)$, so the third-cumulant inequality of C5 holds at $\epsilon = 0$. Let $Y_\delta = \delta^2 \sum_i g(\delta i) q_i$ and $F_\delta(t) = \log \mathbb{E}_\delta e^{tY_\delta}$, smooth by quartic confinement. For $t \geq 0$ the t-tilt adds the fields $h_i = t\delta^2 g(\delta i) \geq 0$, and $F_\delta'''(t)$ is the third cumulant of Y_δ, a nonnegative combination of mixed third cumulants, hence ≤ 0. Symmetry gives $F_\delta(0) = F_\delta'(0) = 0$ and $F_\delta''(0) = v_\delta = \mathbb{E}_\delta Y_\delta^2$; integrating three times, $F_\delta(t) \leq v_\delta t^2/2$ for $t \geq 0$, and for all t by evenness. By C4(i), on the common Gaussian realization the smearings and the interaction densities converge in every Gaussian L^p with positive normalizer limits; for bounded continuous H the difference of weighted expectations is controlled by the L^1 density error plus bounded convergence in probability, so the normalized lattice laws of Y_δ converge weakly to that of $\psi(g)$ under $\nu_{m,u,\Lambda}$ (Definition 10.3), and $v_\delta \to \mathbb{E}\psi(g)^2$. For the nonnegative continuous $y \mapsto e^{ty}$, the portmanteau inequality gives $\mathbb{E}e^{t\psi(g)} \leq \liminf_\delta \mathbb{E}_\delta e^{tY_\delta} \leq e^{t^2 \mathbb{E}\psi(g)^2/2}$.

(ii) By Theorem 10.8 the laws of $v(g)$ under ν_n converge, hence are tight, and R_g exists. Let $n \geq N_g$ and $v_n = \mathbb{E}_{\nu_n} v(g)^2$; the laws on compact smooth tests are those of ν_{m,u,Λ_n} (Proposition 10.4). By C4(ii) and the vanishing of odd moments, $\mathbb{E}v(g)^4 \leq 3v_n^2$. If $v_n > 0$, Paley–Zygmund gives $\nu_n(v(g)^2 > v_n/2) \geq \frac{1}{4}v_n^2/\mathbb{E}v(g)^4 \geq 1/12$; if $v_n > 2R_g^2$ this event lies in $\{|v(g)| > R_g\}$, a contradiction. So $v_n \leq V_g$, and (i) gives the bound for $f = g$. For $|f| \leq g$ put $a = (g+f)/2$, $b = (g-f)/2 \geq 0$; then $\mathbb{E}v(g)^{2r} - \mathbb{E}v(f)^{2r} = 2\sum_{k \text{ odd}} \binom{2r}{k} \mathbb{E}[v(a)^{2r-k} v(b)^k] \geq 0$ by the first Griffiths inequality C4(ii), and comparing the series of $\mathbb{E}\cosh(tv(f)) = \mathbb{E}e^{tv(f)}$ and $\mathbb{E}\cosh(tv(g))$ gives the bound for f. Chernoff gives $\nu_n(|v(f)| > z) \leq 2e^{-z^2/(2V_g)}$, and tail integration the moment bound. For $n < N_g$, Hölder with $\|d\nu_{m,u,\Lambda_n}/dP_m\|_2 \leq \Xi_2(\Lambda_n)^{1/2}$ (Proposition 10.4(i)) and the Gaussian moments with $\mathrm{Var}_{P_m}\psi(f) \leq m^{-2}\|g\|_2^2$ give the second entry of $C_{p,g}$.

(iii) Let $a_{j,k}(v) = v(\chi_j \overline{e_{j,k}})$ and $g_j = |\mathbb{T}_j|^{-1/2}\chi_j$; the real and imaginary parts of $\chi_j\overline{e_{j,k}}$ are dominated by g_j. Choose an integer $N_j \geq 2$ with $\operatorname{supp}\chi_j \subset \Lambda_n$ for $n \geq N_j$, and use $N_{g_j} = N_j$ in

(ii). For $n \geq N_j$, (ii) gives $\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq 2V_{g_j}$; for $n < N_j$, $\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq \sqrt{3}\,\Xi_2(\Lambda_n)^{1/2}m^{-2}\|\chi_j\|_2^2/|\mathbb{T}_j|$. Hence $\sup_{n,k}\mathbb{E}_{\nu_n}|a_{j,k}|^2 \leq A_j/|\mathbb{T}_j|$ with $A_j = \max\{2V_{g_j}|\mathbb{T}_j|, \sqrt{3}m^{-2}\|\chi_j\|_2^2 \max_{n<N_j}\Xi_2(\Lambda_n)^{1/2}\}$. The shell $|k|_\infty = r$ has $8r$ points with $\lambda_{j,k} \geq \pi^2 r^2/(j+1)^2$, so $S_{j,s} = \sum_k \lambda_{j,k}^{-s} < \infty$ exactly when $s > 1$, and Tonelli gives $\sup_n \mathbb{E}_{\nu_n}\|\chi_j v\|^2_{H^{-s}(\mathbb{T}_j)} \leq B_{j,s} := A_j S_{j,s}/|\mathbb{T}_j|$. For $p \geq 2$, (ii) gives $\|a_{j,k}\|_{L^p(\nu_n)} \leq 2C_{p,g_j}^{1/p}$ uniformly, and Minkowski in $L^{p/2}$ bounds $\mathbb{E}_{\nu_n}\|\chi_j v\|^p_{H^{-s}}$ by $2^p C_{p,g_j} S_{j,s}^{p/2}$; for $p < 2$ use Jensen.

Fix $\epsilon > 0$ and $K_\epsilon = \{v \in X : \|\chi_j v\|^2_{H^{-3/2}(\mathbb{T}_j)} \leq 2^j(B_{j,3/2}+1)/\epsilon \ \forall j\}$. Each extended norm is a supremum of continuous finite sums, hence lower semicontinuous, so K_ϵ is closed; Markov gives $\nu_n(X \setminus K_\epsilon) \leq \epsilon$. Let $(v_l) \subset K_\epsilon$. For each j the localized sequence is bounded in $H^{-3/2}(\mathbb{T}_j)$, and the embedding into $H^{-2}(\mathbb{T}_j)$ is compact (finite Fourier truncations, and a tail bounded by the $H^{-3/2}$ radius times $\sup \lambda^{-1/2}$ on the tail); a diagonal subsequence makes $(\chi_j v_l)_l$ convergent in $H^{-2}(\mathbb{T}_j)$ for every j, hence d-Cauchy, hence convergent in X by completeness (Proposition 5.1(ii)), with limit in K_ϵ. Thus K_ϵ is compact and (ν_n) is tight. By Prokhorov every subsequence has a further subsequence converging weakly to some μ; by continuity of evaluations and Theorem 10.8, μ has the finite-dimensional compact-test laws of ν^S.

Let $J_X : X \to \mathbb{R}^{\mathbb{N}}$ list the real and imaginary parts of all $a_{j,k}$; it is continuous and injective (Proposition 5.1(iii)), so by Lusin–Souslin $J_X(X)$ is Borel and J_X^{-1} is Borel on it. Let J_S be the same evaluations on $\mathcal{S}'$. Equal finite-dimensional laws give $(J_X)_*\mu = (J_S)_*\nu^S$, so $\nu^S(J_S \in J_X(X)) = 1$. Put $\mathcal{R} = J_X^{-1} \circ J_S$ on that set. For ψ in it, $\mathcal{R}(\psi)$ and ψ have the same localized Fourier coefficients, and the expansion in the proof of (16), valid for every distribution, gives $\mathcal{R}(\psi)(f) = \psi(f)$ for compact smooth f; and $\mathcal{R}_*\nu^S = \mu$. Since $\mathcal{B}(X) = \sigma(J_X)$, $(J_X)_*\mu$ determines μ; all subsequential limits coincide, and $\nu_n \Rightarrow \nu := \mathcal{R}_*\nu^S$. Lower semicontinuity and the portmanteau inequality transfer the H^{-s} bounds to ν.

(iv) Choose a real nonnegative compact smooth g with $|f_i| \leq g$ for all i. For a polynomial of degree d, $|Q(x)| \leq c_Q(1 + \sum_i |x_i|^d)$, so (ii) bounds $\sup_n \mathbb{E}_{\nu_n}|Q|^q$ for every $q > 0$. The finite-vector laws converge, the target has the corresponding moments by the portmanteau inequality, and continuous truncation at level T with tails at most a constant times T^{1-q} $(q > 1)$, uniformly in n and at the target, gives the convergence. □

Remark 10.11 (Scope of Theorem 10.10). Part (i) holds for all $m, u > 0$. Every volume-uniform statement uses the weak-coupling regime only through the tightness supplied by Theorem 10.8. The early volumes use their individual finite density norms; no bound uniform in n on the densities $d\nu_{m,u,\Lambda_n}/dP_m$ is used. The volume limit is classical continuum input to the construction, not a neural-estimate thermodynamic limit.

10.3 Target properties and reconstruction

Clause (D) of Theorem 4.1 identifies the limiting correlations with Schwinger distributions, and reconstruction C3(i) produces a relativistic quantum field from a measure satisfying four conditions: Euclidean invariance, time-reflection positivity, an exponential regularity bound and translation ergodicity. Theorem 10.16 verifies them in parts (i), (ii), (iv) and (iii), proves temperedness in (v) and non-Gaussianity in (vi), and applies reconstruction only afterwards, in (vii). Non-Gaussianity is the sense in which the reconstructed field is interacting (Remark 10.17).

Throughout, (m, u) is in the regime, s, λ, v and $\nu^S = (T_s)_*v$ are as in Theorem 10.8, and $M = M_{\mathrm{ref}}$. Time is z_2; Θ is the reflection $z_2 \mapsto -z_2$ acting on observables, $\Sigma_+ = \mathcal{C}_{\{z_2>0\}}$, and τ_r shifts an observable by r in the positive time direction.

Lemma 10.12 (Determining family)**.** *There is a countable $D \subset C_c^\infty(\mathbb{R}^2;\mathbb{R})$, independent of all parameters and dense in $\mathcal{S}(\mathbb{R}^2;\mathbb{R})$, with $\sigma\{T \mapsto T(d) : d \in D\} = \mathcal{C}$. Two probabilities on $\mathcal{C}$ with the same finite-dimensional D-laws coincide.*

Proof. Fix real smooth η, ζ with $\eta = 1$ on $[-1,1]^2$, $\operatorname{supp}\eta \subset (-2,2)^2$, $\zeta = 1$ on $[-2,2]^2$ and $\operatorname{supp}\zeta \subset (-3,3)^2$. Put $\eta_n(x) = \eta(x/n)$ and $\zeta_n(x) = \zeta(x/n)$. On the numerical torus with cell $(-3n,3n)^2$ let Q_n be the real trigonometric polynomials with rational coefficients in the sine/cosine basis of frequencies $\pi k/(3n)$, $k \in \mathbb{Z}^2$. Then $D = \bigcup_{n\ge1}\{\zeta_n P : P \in Q_n\}$ is countable, real compact smooth and independent of the model and localization parameters. For every real Schwartz f, the product rule and decay outside $[-n,n]^2$ give $\|(1-\eta_n)f\|_{(N)} \le C_N n^{-1}\|f\|_{(2N+1)}$ for $N \ge 1$. Thus $\eta_n f \to f$ in $\mathcal{S}$. At fixed n, the zero periodic extension of $\eta_n f$ is smooth; Fourier sums and rational approximation of their finite coefficients approximate it in C^∞. Multiplication by ζ_n fixes a compact support and the limit $\zeta_n\eta_n f = \eta_n f$. Choose P_n so that $\zeta_n P_n - \eta_n f$ is less than $1/n$ in the first n Schwartz seminorms. Then $d_n = \zeta_n P_n \to f$ in $\mathcal{S}$. Every $T \in \mathcal{S}'$ satisfies $T(f) = \lim_n T(d_n)$, so all Schwartz evaluations are measurable for $\sigma(D)$; the reverse inclusion is immediate. Finite D-cylinders form a generating π-system, so probability uniqueness follows by π–λ. □

Lemma 10.13 (Source-defined target and symmetries)**.** *The source-defined v and $\nu^S = (T_s)_* v$ of Theorem 10.8 are well defined from C1, C2, C6 alone, with the definitions of the admissible pair and dilation. They are invariant under field sign and the central coordinate reflections, and ν^S is invariant under the full Euclidean group. These conclusions hold under each admitted reading of C1. No foundation input is used here.*

Proof. At $M = M_{\mathrm{ref}} > M_0$ and $0 < \lambda = uM^2/m^2 < \lambda_* \le \lambda_0$, C2(i),(ii) make the two Gibbs extensions of the C1 states the same member μ of $\mathcal{G}_{\mathrm{ur}}$. Their cylinder restrictions therefore coincide as v. This identity uses neither a finite-volume neural law nor a volume-limit estimate. By C1(iii) the specified pressure is differentiable. Lemma 10.1 has independently proved the two provisos that license the classical free-boundary convention. Apply C1(iv) to those laws along centered axis-parallel square exhaustions. For its characteristic reading, apply the assertion to tg for every real t; Lévy's theorem gives the scalar weak limit. For the moment reading, C1(ii) and Gaussian Cauchy–Schwarz give

$$\mathbb{E}_v e^{t\psi(g)} \le \|f_B\|_2 e^{t^2\langle g,(-\Delta+M^2)^{-1}g\rangle} < \infty$$

for every real compact smooth g, all real t, and bounded open B containing its support. Moment determinacy and convergence then give that same scalar weak limit. For the Laplace reading, the moment-generating function theorem applies on a neighborhood of zero. Applying these arguments to every real linear combination of a fixed tuple, and then the finite-dimensional Lévy theorem, gives joint compact-test convergence in all three readings. This is exactly the classical portion of Theorem 10.8; the later common-space identification is not needed.

Lemma 10.1 gives finite-volume field-sign and central reflection invariance. Apply joint convergence to the union of a compact tuple and its transformed tuple; the symmetries pass to the corresponding v-laws. For real Schwartz tests, compact cutoffs converge in $\mathcal{S}$ and their evaluations converge pointwise on every tempered distribution. Bounded convergence of characteristic functions extends the identities to Schwartz cylinders; π–λ gives equality on $\mathcal{C}$. The positivity readings of C1 are unused in this step, and the three convergence readings were treated separately above.

Proper Euclidean invariance of v is directly C1(i)(a). The elementary identities $A_b T_s = T_s A_{sb}$ and $RT_s = T_s R$ of Lemma 10.5 transport it to ν^S. That lemma's Gaussian and Wick dilation identities use the direct construction of Lemma 10.1 and no foundation. Dilation commutes with

sign and θ_1, θ_2, so those symmetries transfer too. Proper motions and one central reflection generate every Euclidean isometry. In particular translation invariance has been proved before any translated local-density norm is used. □

Lemma 10.14 (Local densities of the target)**.** *For every bounded open B, the restriction of ν^S to $\mathcal{C}_B$ has density $F_B = f_{sB} \circ T_{1/s}$ with respect to P_m, and $\|F_B\|_{L^p(P_m)} = \|f_{sB}\|_{L^p(P_M)} < \infty$ for $1 \le p < \infty$. For an open centered axis-parallel square,*

$$F_B = e^{-U^S_{m,u,B} - \alpha_\infty |sB|} g_B, \qquad g_B = g_{\partial(sB)} \circ T_{1/s} > 0 \quad P_m\text{-a.s.}, \tag{27}$$

where g_B is measurable for either stated boundary convention, with that convention retained. Moreover $F_{B+k} = F_B \circ A_k$ P_m-a.s., so their finite L^p norms agree. For real $f_1, \dots, f_r$ supported in B and $t \in \mathbb{R}^r$, writing $q = \sum_i t_i f_i$,

$$\mathbb{E}_{\nu^S} e^{\psi(q)} \le \|F_B\|_2 e^{\langle q, C_{\mathrm{free}} q \rangle} < \infty. \tag{28}$$

Proof. T_r is a cylinder-measurable linear bijection with inverse $T_{1/r}$; L_r acts continuously on $\mathcal{S}$ and scales test supports from U to rU. For bounded $\mathcal{C}_B$-measurable H, Gaussian dilation gives

$$\int H F_B \, dP_m = \int (H \circ T_s) f_{sB} \, dP_M = \int H \circ T_s \, dv = \int H \, d\nu^S.$$

The same change of Gaussian law gives the exact equality of pth absolute integrals, with no probability-norm Jacobian. For a square, compose the C1(ii) factorization with $T_{1/s}$ and use $U^S_{M,\lambda,sB} \circ T_{1/s} = U^S_{m,u,B}$ from the direct mollified dilation in Lemma 10.5. The pressure factor is $\alpha_\infty |sB|$ at mass M, and positivity of g_B transports with Gaussian null sets.

BC1. For every open neighborhood V of ∂B, sV is an open neighborhood of $\partial(sB)$. The boundary factor pulled back by $T_{1/s}$ is therefore $\mathcal{C}_V$-measurable for every such V, hence measurable for their intersection. *BC2.* For real Schwartz h and $r > 0$,

$$\|L_r h\|^2_{H^{-1}} = (2\pi)^{-2} \int \frac{|\hat{h}(q)|^2}{r^2 + |q|^2} \, dq \le \max(1, r^{-2}) \|h\|^2_{H^{-1}}. \tag{29}$$

Thus dilation extends continuously on H^{-1} and scales distribution support by r. Gaussian evaluation at mass M is continuous $H^{-1} \to L^2(P_M)$, since its multiplier is at most $\max(1, M^{-2})(1 + |p|^2)^{-1}$; similarly at mass m. For h supported in $\partial(sB)$ and real Schwartz $h_j \to h$ in H^{-1}, the exact evaluation identity $\psi(h_j) \circ T_{1/s} = \psi(L_{1/s} h_j)$ passes in $L^2(P_m)$ to the extended Gaussian evaluations. The transformed test is supported in ∂B. Gaussian null sets also pull back to Gaussian null sets. This proves transport of the completed BC2 algebra, without identifying it with the uncompleted BC1 algebra.

Translated norms, after invariance. Lemma 10.13 already proves translation invariance of ν^S; the Gaussian law is stationary by its covariance. For bounded $\mathcal{C}_{B+k}$-measurable H,

$$\int H (F_B \circ A_k) \, dP_m = \int (H \circ A_{-k}) F_B \, dP_m = \int H \circ A_{-k} \, d\nu^S = \int H \, d\nu^S.$$

Local Radon–Nikodym uniqueness gives $F_{B+k} = F_B \circ A_k$ and hence the exact norm equality. No part of the preceding invariance proof used it. Finally monotone convergence extends the local density identity to the nonnegative exponential. Cauchy–Schwarz gives $\mathbb{E} F_B e^{\psi(q)} \le \|F_B\|_2 (\mathbb{E}_{P_m} e^{2\psi(q)})^{1/2} = \|F_B\|_2 e^{\langle q, C_{\mathrm{free}} q \rangle}$, with $\langle q, C_{\mathrm{free}} q \rangle \le m^{-2} \|q\|_2^2$. □

Lemma 10.15 (Local L^2 kernels)**.** *If a probability π on $(\mathcal{S}',\mathcal{C})$ has a density $F \in L^2(P_m)$ on $\mathcal{C}_B$, B bounded open, then for every integer $n \geq 1$ its local moment and truncated kernels on B^n are represented by real functions in $L^2(B^n)$, with complex-linear test pairings.*

Proof. Choose the local measurable version $F \geq 0$ with $\mathbb{E}_{P_m} F = 1$. Local moments exist by Cauchy–Schwarz and Gaussian polynomial moments; no moment outside B is required. Plancherel and polar integration give

$$\|C_{\mathrm{free}}\|^2_{L^2(\mathbb{R}^2)} = \frac{1}{4\pi m^2}, \qquad R_B^2 := \|C_{\mathrm{free}}(x-y)\|^2_{L^2(B^2)} \leq \frac{|B|}{4\pi m^2}. \tag{30}$$

The restricted covariance operator K on real $L^2(B)$ is positive self-adjoint with $\|K\| \leq m^{-2}$ by its Fourier form; its integral kernel is $C_{\mathrm{free}}(x-y)$ by Plancherel, first on smooth tests and then by continuity. Empty B is vacuous, so assume $|B| > 0$.

For a finite sum Q of real compact smooth product tests on B^k, define $W_k(f_1 \otimes \cdots \otimes f_k) = {:}\psi(f_1)\cdots\psi(f_k){:}$ linearly. The Wick covariance identity yields

$$\|W_k(Q)\|^2_{L^2(P_m)} = k!\langle \mathrm{Sym}\, Q, K^{\otimes k} \mathrm{Sym}\, Q\rangle \leq k! m^{-2k} \|Q\|^2_{L^2(B^k)}. \tag{31}$$

This also proves independence of the product-sum representation. Density extends W_k boundedly to $L^2(B^k)$. The functional $Q \mapsto \mathbb{E}_{P_m}[F W_k(Q)]$ has norm at most $\sqrt{k!}m^{-k}\|F\|_2$, so real L^2 duality produces a symmetric $h_k \in L^2(B^k)$ representing it, with that norm bound. Normalize the finite Lebesgue measure to a probability when applying C6 duality, then undo the normalization; the displayed bound is unchanged. Put $h_0 = 1$.

For a partial pairing P of $\{1,\dots,n\}$, write $U(P)$ for the unpaired indices. The ordinary moment kernel is

$$M_n(x) = \sum_P \left(\prod_{\{i,j\}\in P} C_{\mathrm{free}}(x_i - x_j) \right) h_{|U(P)|}(x_{U(P)}). \tag{32}$$

Each factor uses a disjoint set of coordinates, so its L^2 norm is $R_B^{|P|}\|h_{|U(P)|}\|_2$. In particular

$$\|M_n\|_2 \leq \|F\|_2 \sum_{p=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^p p!(n-2p)!} R_B^p \sqrt{(n-2p)!}\, m^{-(n-2p)}.$$

To identify it, expand $e^{\sum_i t_i X_i} = e^{\frac{1}{2}\sum_{i,j} t_i t_j \mathbb{E} X_i X_j} e^{\sum_i t_i X_i - \frac{1}{2}\sum_{i,j} t_i t_j \mathbb{E} X_i X_j}$ for the finite Gaussian tuple $X_i = \psi(f_i)$ and compare the coefficient $t_1 \cdots t_n$. It gives each partial pairing once, with the remaining factors Wick ordered. Multiply by F, integrate and use the representation of h_k to obtain $\mathbb{E}_\pi \prod_i \psi(f_i) = \int M_n \prod_i f_i$ for product tests. Every integral is absolute by the displayed L^2 bounds. Product-test density gives the unique distribution on B^n.

For a set partition π_0 with blocks A, define

$$M_n^T(x) = \sum_{\pi_0} (-1)^{|\pi_0|-1}(|\pi_0|-1)! \prod_{A\in\pi_0} M_{|A|}(x_A). \tag{33}$$

Every block product is an L^2 tensor product on disjoint coordinates, of norm $\prod_A \|M_{|A|}\|_2$. The finite cumulant partition identity and Fubini identify its product-test pairing with the joint cumulant. Density again gives uniqueness. Real kernels extend complex-linearly, as required; no point field of the unregulated distribution is used. □

Theorem 10.16 (Target properties; reduced lemma CT-R). *(i) ν^S is invariant under the full Euclidean group of $\mathbb{R}^2$ and under $\psi \mapsto -\psi$.*

(ii) For every line L, with Θ_L the reflection in L and Σ_+^L generated by smearings supported in one open half-plane bounded by L: $\int \overline{\Theta_L F}\, F\, d\nu^S \geq 0$ for every $F \in L^2(\Sigma_+^L, \nu^S)$ and every Σ_+^L-measurable F with $\overline{\Theta_L F} F$ integrable.

(iii) For $G, G' \in L^2(\mathcal{S}', \mathcal{C}, \nu^S)$, $\mathbb{E}[G\,(G' \circ A_b)] \to \mathbb{E}G\,\mathbb{E}G'$ as $|b| \to \infty$ (strong mixing). Consequently every $E \in \mathcal{C}$ with $\nu^S(E \triangle A_b^{-1}E) = 0$ for all $b \in \mathbb{R}^2$ has $\nu^S(E) \in \{0, 1\}$ (translation ergodicity), and $\mathbb{E}[\overline{\Theta F}\, \tau_r G] \to \overline{\mathbb{E}F}\, \mathbb{E}G$ for $F, G \in L^2(\Sigma_+)$.

(iv) With a real nonnegative smooth partition of unity $\sum_{k\in\mathbb{Z}^2} \vartheta_k = 1$, $\vartheta_k = \vartheta_0(\cdot - k)$, $\mathcal{A}(f) = \sum_k \|\vartheta_k f\|_2$ is a norm continuous on $\mathcal{S}$, and with $K = \|F_B\|_{L^2(P_m)} \geq 1$ for a square $B \supseteq \operatorname{supp}\vartheta_0$ and $D = 3^{1/4}K^{1/2}/m$: $\mathbb{E}e^{t\psi(f)} \leq K \exp(t^2\mathcal{A}(f)^2/m^2)$, $\|\psi(f)\|_{L^2} \leq D\mathcal{A}(f)$, and $|\mathbb{E}e^{\psi(f)} - \mathbb{E}e^{\psi(g)}| \leq 2DK^{1/2}e^{2/m^2}\mathcal{A}(f-g)$ for $\mathcal{A}(f), \mathcal{A}(g) \leq 1$. Hence condition (c) of C3(i) holds with $|||\cdot||| = \mathcal{A}$.

(v) For every k there is a unique $\operatorname{Schw}_k \in \mathcal{S}'(\mathbb{R}^{2k})$ with $\operatorname{Schw}_k(f_1 \otimes \cdots \otimes f_k) = \int \prod_r \psi(f_r)\, d\nu^S$ for $f_r \in \mathcal{S}(\mathbb{R}^2; \mathbb{C})$ (complex-linear extension); it is symmetric and real on real tests.

(vi) ν^S is not Gaussian; its local moment and truncated kernels lie in $L^2(B^n)$ for bounded B; and some even truncated Schwinger distribution of order at least four is nonzero on real tests supported in pairwise disjoint time slabs.

(vii) By C3(i), applied with (i), (ii), (iv) and (iii): the moments of ν^S are the Schwinger functions of a unique relativistic quantum field theory in $1+1$ dimensions satisfying all the Wightman axioms (temperedness, Poincaré invariance, positivity, spectrum condition, locality, clustering), with a unique physical vacuum; its field fails Gaussian factorization.

Proof. All parts of this proof use only C1–C3 and C6, with the source-defined target identity. The proofs of Lemmas 10.1, 10.5, 10.13, 10.14 and 10.15 have supplied the needed classical objects directly. Their use here does not import the separate common-space Gaussian/Wick foundations.

(i), invariance. This is Lemma 10.13, proved without translated-density norms, local kernels, mixing or reconstruction.

(ii), full measurable reflection positivity. Under the measure reading C1(i)(b_M), the L^2 time-reflection inequality for v transfers directly by positive dilation: T_s commutes with Θ and its pullback preserves the positive-half-plane algebra. Under the weaker reading C1(i)(b_P), positive dilation first transfers exactly the inequality for polynomials whose individual monomials have strictly ordered positive-time supports. Reflection is an $L^2(\nu^S)$ isometry by (i).

Let B be bounded open and let $V_n(G)$ be the polynomial associated to a finite sum G of compact smooth product tests on $(B \cap H_+)^n$. The local $2n$-point moment kernel of Lemma 10.15 gives

$$\|V_n(G)\|^2_{L^2(\nu^S)} = M_{2n}(\overline{G} \otimes G) \leq \|M_{2n}\|_{L^2(B^{2n})} \|G\|^2_{L^2(B^n)}. \tag{34}$$

Thus this map extends boundedly from product sums to $L^2((B \cap H_+)^n)$. Products of real smooth tests with pairwise disjoint positive-time supports span a dense subspace there. Indeed the union of equal-time hyperplanes has Lebesgue measure zero; its complement has a countable cover by relatively compact product boxes with disjoint positive time projections. A function orthogonal to all smooth products on those boxes vanishes almost everywhere on each box, hence on the whole

complement. For each such product, reorder the commuting field factors into increasing time order. It is then exactly a monomial admitted in (b_P). The bounded map above therefore approximates every compact positive-half-plane polynomial in $L^2(\nu^S)$ by admitted polynomials.

Every finite vector of compact evaluations has finite exponential moments in all real directions by (28). The polynomial density theorem of C6 makes its polynomials dense in its L^2 space. For every positive-half-plane Schwartz test, compact spatial cutoffs converge in $\mathcal{S}$, keep support in the open half-plane, and converge in evaluation pointwise on $\mathcal{S}'$. Thus compact evaluations generate the stated Σ_+. Countable dependence supplies a countable family for each observable; conditional-expectation approximation for increasing finite families and truncation now make the compact polynomials dense in $L^2(\Sigma_+, \nu^S)$. If $F_j \to F$ there, reflection isometry and Cauchy–Schwarz give

$$\left|\mathbb{E}\overline{\Theta F_j} F_j - \mathbb{E}\overline{\Theta F} F\right| \le \|F_j - F\|_2(\|F_j\|_2 + \|F\|_2).$$

Positivity therefore holds for the full L^2 class under both readings. For a measurable F whose reflected product is integrable, use $F_j = F\mathbf{1}_{\{|F|\le j\}}$: the reflected products converge pointwise and are dominated by $|\Theta F|\,|F|$. Bounded positivity and dominated convergence prove the additional integrable-form clause. For any line, conjugate the time-line inequality by a Euclidean isometry using (i). This concerns the target; the finite neural stages retain only their two stated central coordinate lines.

(iii), Euclidean mixing and ergodicity before reconstruction. At mass M use the unique Gibbs extension μ of v supplied by C2. Let $D \in F_0$ be its increasing square family and $K_D F(\eta) = E^{\eta}_{D^c} F$. For an integrable local polynomial F, integrability follows from C1(ii) and Gaussian moments, and the Gibbs identity gives $\mu(K_D F) = \mu(F)$. If H is bounded and Σ_{D^c}-measurable, the kernel's point-fixing property applied to H and $|H|^2$ shows that H equals the constant $H(\eta)$ under each kernel probability (apply to real and imaginary parts). Hence $K_D(HF) = HK_D F$ wherever the integral of $|F|$ is finite. The Gibbs identity for $|F|$ gives that finiteness μ-a.e.; its identity for HF yields

$$\mu(HF) = \mu(HK_D F). \tag{35}$$

For a real local polynomial F, C2(iii) with both Gibbs measures equal to μ and Jensen imply

$$\|K_D F - \mu F\|_1 \le \iint |K_D F(\eta) - K_D F(\eta')|\,\mu(d\eta)\mu(d\eta') \longrightarrow 0. \tag{36}$$

Complex polynomials follow by real and imaginary parts. Thus

$$|\mu(FH) - \mu F\,\mu H| \le \|H\|_\infty \|K_D F - \mu F\|_1.$$

For a fixed bounded compact cylinder G, $H = G \circ A_b$ is Σ_{D^c}-measurable for all sufficiently large $|b|$: its finitely many translated compact supports lie outside $\bar{D}$, hence in every open neighborhood used in the closed-set outside algebra. This condition holds uniformly over the direction of b. First choose D sufficiently far along F_0, then let $|b|$ grow. Translation invariance of v gives $\mu(G \circ A_b) = \mu G$, so mixing holds for each such F, G. Local polynomials are dense in $L^2(v)$ by the local exponential moments, finite-dimensional polynomial density and increasing finite-coordinate conditional expectations. Compact evaluations generate $\mathcal{C}$ by Lemma 10.12. Bounded compact cylinders are also dense by the same conditional expectations and truncation. Translations preserve $L^2(v)$ norms, so Cauchy–Schwarz extends the limit to all complex $G, G' \in L^2(v)$, in every direction.

The identity $A_b T_s = T_s A_{sb}$ transfers this strong mixing to ν^S, since $s > 0$ implies $|sb| \to \infty$. For an invariant event E as in the statement, take both observables to be $\mathbf{1}_E$. The mixing product

equals $\nu^S(E)$ for every b and tends to $\nu^S(E)^2$; its probability is therefore zero or one. The half-plane clustering formula follows by taking the first observable $\overline{\Theta F}$ and translating the second in positive time; (i) gives $\mathbb{E}\Theta F = \mathbb{E}F$. No reconstruction theorem, tube uniqueness, spectrum condition, unique-vacuum assertion, FS Theorem 4.4 or OS-energy identification has entered this argument. Condition C3(i)(d) is established at this point.

(iv), the exact norm and exponential regularity. Fix a real nonnegative $\vartheta_0 \in C_c^\infty$ with $\sum_{k\in\mathbb{Z}^2} \vartheta_0(\cdot - k) = 1$, put $\vartheta_k = \vartheta_0(\cdot - k)$ and $\mathcal{A}(f) = \sum_k \|\vartheta_k f\|_2$. The partition identity gives a norm. For each integer $N > 2$, bounded support of ϑ_0 gives $\|\vartheta_k f\|_2 \le c_N(1+|k|)^{-N}\|f\|_{(N)}$; summation proves finiteness and Schwartz continuity. Bounded overlap of the translated supports, the product rule and arbitrarily higher Schwartz decay show that the partition sums converge to f in every Schwartz seminorm. Choose a bounded open square $B \supset \operatorname{supp}\vartheta_0$ and set $K = \|F_B\|_2 \ge \mathbb{E}F_B = 1$, $D = 3^{1/4}K^{1/2}/m$. The translated-density norm equality has already been proved after invariance. For real g supported in $B + k$, the local density and free Gaussian formulas give

$$\mathbb{E}e^{t\psi(g)} \le Ke^{t^2\|g\|_2^2/m^2}, \qquad \mathbb{E}\psi(g)^2 \le K(\mathbb{E}_{P_m}\psi(g)^4)^{1/2} \le \sqrt{3}Km^{-2}\|g\|_2^2. \tag{37}$$

Minkowski bounds each finite partition sum in L^2 by $D\sum_k \|\vartheta_k f\|_2$. The same estimate for its tails makes it Cauchy in L^2; its pointwise Schwartz-evaluation limit is $\psi(f)$. Thus $\|\psi(f)\|_2 \le D\mathcal{A}(f)$ for every real Schwartz f.

Let $a_k = \|\vartheta_k f\|_2$ and, when $\mathcal{A}(f) > 0$, put $\alpha_k = a_k/\mathcal{A}(f)$ for the nonzero terms. For any finite set J, generalized Hölder with weights α_k (including a factor one with the unused weight) and (37) imply

$$\begin{aligned}\mathbb{E}\exp\left(t\sum_{k\in J}\psi(\vartheta_k f)\right) &\le K^{\sum_J \alpha_k}\exp\left(t^2m^{-2}\sum_J a_k^2/\alpha_k\right)\\ &\le K\exp(t^2\mathcal{A}(f)^2/m^2).\end{aligned}$$

Fatou passes this to the pointwise full sum. The zero-norm case is immediate. For $\mathcal{A}(f), \mathcal{A}(g) \le 1$, use $|e^x - e^y| \le |x-y|(e^x + e^y)$ and Cauchy–Schwarz, with $\|e^{\psi(f)}\|_2 \le K^{1/2}e^{2/m^2}$, to obtain exactly

$$|\mathbb{E}e^{\psi(f)} - \mathbb{E}e^{\psi(g)}| \le 2DK^{1/2}e^{2/m^2}\mathcal{A}(f-g). \tag{38}$$

The unit ball has exponential bound Ke^{1/m^2} and this Lipschitz constant, so C3(i)(c) holds. For later use, $\mathbb{E}e^{|\psi(f)|/\mathcal{A}(f)} \le 2Ke^{1/m^2} =: K'$ when $f \ne 0$, whence $\|\psi(f)\|_k \le (k!K')^{1/k}\mathcal{A}(f)$ for integers $k \ge 1$.

(v), tempered Schwinger distributions. For a k-linear form M on $\mathcal{S}(\mathbb{R}^2;\mathbb{C})^k$ with $|\mathsf{M}(f_1,\dots,f_k)| \le C\prod_i \|f_i\|_{(N)}$, there is a unique tempered kernel. Here are the required reduction details. For $k = 1$ it is the definition. Inductively, at a fixed last test ϕ, apply the $(k-1)$-variable assertion to obtain $\mathsf{L}(\phi)$. Kernel uniqueness makes L linear. For a tensor sum Ψ_j, $\phi \mapsto \mathsf{L}(\phi)(\Psi_j)$ is continuous. Approximate an arbitrary Schwartz Ψ by these sums; the values converge pointwise because each $\mathsf{L}(\phi)$ is a distribution. Banach–Steinhaus on the Fréchet space $\mathcal{S}(\mathbb{R}^2)$ makes the limiting functional continuous. C3(iii), with $N = 2(k-1)$ and $n = 2$, now gives the joint tempered kernel. Product-test density gives uniqueness. Apply this to $\mathsf{M}_k(f_1,\dots,f_k) = \mathbb{E}\prod_i \psi(f_i)$. Hölder and (iv) give $|\mathsf{M}_k| \le k!K'\prod_i \mathcal{A}(f_i)$ for real tests, and $\mathcal{A}(f) \le C_{\mathcal{A}}\|f\|_{(3)}$. Splitting complex tests into real and imaginary parts gives the required complex multilinear bound. Kernel uniqueness transfers symmetry and reality of the moments to $\mathrm{Schw}_k \in \mathcal{S}'_{\mathbb{C}}(\mathbb{R}^{2k})$.

(vi), a local quartic conditional density excludes Gaussianity. Take an open centered axis-parallel square B and nonzero real $h \in C_c^\infty(B)$. Put

$$k = (-\Delta + m^2)h, \quad q = \int hk = \int (|\nabla h|^2 + m^2 h^2) > 0, \quad X = \psi(k)/q, \quad \eta = \psi - Xh.$$

Under P_m, $\operatorname{Var} X = 1/q$ and $\operatorname{Cov}(X, \psi(f)) = q^{-1} \int hf$, so $\operatorname{Cov}(X, \eta(f)) = 0$ for every real Schwartz f. Gaussianity and the monotone-class theorem make X independent of the full residual evaluation algebra. Its law is $\gamma_q(dx) = (q/(2\pi))^{1/2} e^{-qx^2/2} dx$.

The boundary factor g_B in (27) has a residual-measurable version. For BC1 choose a boundary neighborhood disjoint from $\operatorname{supp} h$; each generating evaluation there is unchanged by replacing ψ by η. For BC2, boundary-supported $f \in H^{-1}$ has $\langle f, h \rangle = 0$. If real Schwartz $f_j \to f$ in H^{-1}, then $\langle f_j, h \rangle \to 0$ by $h \in H^1$, so the defining $L^2(P_m)$ limit of $\psi(f_j)$ is also the limit of $\eta(f_j)$. Gaussian completion changes versions only on null sets. This covers each stated boundary algebra separately.

For the fixed mollifier, $\psi_\delta = \eta_\delta + X h_\delta$, $h_\delta = \rho_\delta * h$. Let $H_j(x; 1/q)$ be the monic Hermite polynomials for γ_q. At each positive mollifier scale,

$$J_\delta = u \int_B :\psi_\delta^4: = \sum_{j=0}^{4} A_{j,\delta}(\eta) H_j(X; 1/q).$$

Its residual coefficients are square integrable by finite Gaussian moments and Minkowski on bounded B. Independence and Hermite orthogonality give

$$\|J_\delta - J_\epsilon\|_2^2 = \sum_{j=0}^{4} j! q^{-j} \|A_{j,\delta} - A_{j,\epsilon}\|_2^2. \tag{39}$$

The direct L^2 Wick limit therefore gives residual coefficients A_j with

$$U^S_{m,u,B} = \sum_{j=0}^{4} A_j(\eta) H_j(X; 1/q), \qquad A_4 = u \int_B h^4 =: c > 0. \tag{40}$$

This is an actual degree-four polynomial in the scalar X, with deterministic leading coefficient c; it is obtained without multiplying unregulated distributions.

Suppose ν^S were Gaussian. The local measurable density F_B depends on a countable family $\psi(f_j)$, $f_j \in C_c^\infty(B; \mathbb{R})$; include k in this list. Write $Y = (\eta(f_j))_j$ and $F_B = b(X, Y)$. Both X and each Y_j are local linear evaluations. Under P_m, X is independent of Y, and the target law of (X, Y) has density $b(x, y)$ relative to $\gamma_q(dx) P_Y(dy)$, by the local Radon–Nikodym identity. The full residual variables used by $A_0, \dots, A_4, g_B, Y$ depend on a countable residual family too. They need not all be measurable for the smaller local family Y. Independence of X from that full residual family and Fubini give, for almost every full residual configuration v_0,

$$\log b(x, Y(v_0)) = -\sum_{j=0}^{4} A_j(v_0) H_j(x; 1/q) - \alpha_\infty |sB| + \log g_B(v_0) \tag{41}$$

for Lebesgue-a.e. x. This uses the square factorization, not a global target density. In particular $b > 0$ almost everywhere. The integral $\int b(x, y) \gamma_q(dx)$ is positive and finite for P_Y-a.e. y by positivity and $\mathbb{E} F_B = 1$; the target marginal of Y is equivalent to P_Y. The conditional target density of X at such y is proportional to $b(x, y) e^{-qx^2/2}$.

Under the assumed Gaussian target, every finite vector $(X, Y_1, \dots, Y_j)$ is Gaussian. Finite Gaussian regression gives conditional characteristic functions $e^{itm_j - t^2 v_j/2}$, where $m_j = \mathbb{E}_{\nu^S}[X \mid Y_1, \dots, Y_j]$ converges in L^2 by the increasing-filtration theorem and the deterministic $v_j \geq 0$ decrease. Applying the same conditional-expectation theorem to e^{itX} identifies the conditional characteristic function given Y as $e^{it\mathbb{E}[X|Y] - t^2 v/2}$, $v = \lim v_j \geq 0$, first for rational t and then by continuity. Thus it is a Gaussian conditional law, possibly degenerate. Its exceptional Y set is null for P_Y by equivalence. Intersect with the full-measure residual set in (41). A positive Lebesgue density excludes zero variance. For positive variance its log density is quadratic, whereas (41) plus $-qx^2/2$ has leading coefficient $-c < 0$ in degree four. Equality a.e. would make the polynomials identical, which is impossible. This proves non-Gaussianity while preserving the local/full-residual distinction and the fixed boundary trace.

The local moment and truncated L^2 kernels for every bounded open B and every order are Lemmas 10.14 and 10.15. Field-sign invariance annihilates every odd joint cumulant. If all even compact cumulants of order at least four vanished as well, the moment-cumulant formula would leave only pair partitions. Local exponential moments and moment determinacy would make every compact linear evaluation centered Gaussian, hence every finite compact vector jointly Gaussian. For any real Schwartz test take compact cutoffs converging in $\mathcal{S}$; their evaluations converge pointwise on $\mathcal{S}'$, hence weakly in law. Centered Gaussian laws are closed under such weak convergence: unbounded variances would force a discontinuous limiting characteristic function at zero; a bounded-variance subsequence has a Gaussian limit. Apply this to each real linear combination to obtain Gaussian Schwartz vectors, contradicting the preceding argument. This passage needs no global neural Schwartz estimate.

Consequently some even $n \geq 4$ has a nonzero compact joint cumulant. Choose bounded open B containing its supports. Its real kernel $M_n^T \in L^2(B^n)$ is nonzero. The union of hyperplanes $x_{i,2} = x_{j,2}$ is Lebesgue null. The complement has a countable cover by relatively compact product boxes $U_1 \times \cdots \times U_n$ whose time projections lie in pairwise disjoint bounded open intervals. At least one box has nonzero kernel restriction; otherwise the squared integral would vanish on that countable cover. Density of real smooth product tests in its L^2 space then gives individual $f_i \in C_c^\infty(U_i; \mathbb{R})$ with

$$S_n^T(f_1, \dots, f_n) = \int M_n^T(x) \prod_i f_i(x_i)\, dx \neq 0. \tag{42}$$

Thus the witness is a real tensor product in disjoint time slabs at the same even order $n \geq 4$, rather than only a general joint test or a distribution on coincident times.

(vii), reconstruction and its precise non-Gaussian consequence. For ν^S on $(\mathcal{S}', \mathcal{C})$, condition (a) of C3(i) is (i), condition (b) is the full L^2 time-line assertion of (ii) under either positivity reading, condition (c) is the exact norm and estimates of (iv), and condition (d) was proved by Euclidean mixing in (iii). Apply C3(i) only now. It gives the stated unique scalar relativistic theory in $1+1$ dimensions, the listed Wightman axioms and unique physical vacuum; (v) identifies its tempered symmetric Schwinger representatives.

Only after this application use C3(ii),(ii'). On a common ordered tube, form the truncated holomorphic n-point function by the finite cumulant partition sum of the holomorphic moment functions, preserving the order within each block. Every factor is holomorphic there and uses disjoint variables from the other block factors. Its distribution boundary is the truncated Wightman distribution: the block products are tensor products, so no undefined multiplication of coincident-variable distributions is involved. At pairwise distinct Euclidean arguments its values are the truncated Schwinger function by C3(ii) and the same partition identity. If its Wightman boundary

vanished, C3(ii') would force the holomorphic function to vanish and hence its analytic continuation at those Euclidean points. Permute the disjoint time slabs of (42) into one strict order; symmetry of the Schwinger kernels preserves that nonzero pairing. Vanishing analytic values would make the pairing zero, a contradiction. The truncated Wightman distribution at this same even order $n \geq 4$ is therefore nonzero. In particular the field fails Gaussian factorization at some order at least three, exactly the claimed consequence. No scattering or quantitative mass-gap conclusion follows from this argument. □

Remark 10.17 (Order of the argument). Ergodicity, condition (d), is established in (iii) by a Euclidean mixing argument from the Gibbs identity and the boundary-condition independence of C2, before [8, Thm. 1.0] is applied. The spectrum condition, clustering in the Wightman sense and uniqueness of the vacuum are consequences of the reconstruction, not inputs to (d). In particular neither a unique-vacuum statement nor an ergodic form of the reconstruction theorem is used to prove ergodicity. A historical route through [8, Thm. 4.4] and an identification of the Osterwalder–Schrader energy with a Wightman energy is recorded, with its first unsupported step, in Appendix G; it is not used. The main theorem uses only part (v) of Theorem 10.16, in clause (D). The word "interacting" means the non-Gaussian quantum field of (vii); no scattering or quantitative mass-gap statement is made.

11 Local moments and the common sequence

To turn the iterated limits into one sequence serving all later observables of Theorem 4.1, each arrow is first upgraded to carry all moments of the local seminorms (Proposition 11.6), using uniform moment bounds and Lemma 11.5. Proposition 11.7 then chooses at stage n parameters that make each of four discrepancies smaller than $2^{-n}/4$, and the triangle inequality gives (15). Since each compact smearing is controlled by one seminorm through (16), clause (B) follows (Proposition 11.10).

The moment-transfer and selection arguments use the field-space, Gaussian, Wick, fixed-volume and continuum-volume results proved above, including Theorem 10.10. Their provenance is recorded in Appendix D.

Throughout, $S_j = \sum_{k\in\mathbb{Z}^2} \lambda_{j,k}^{-2}$, which is finite because the shell $|k|_\infty = r \geq 1$ has $8r$ points with $\lambda_{j,k} \geq 1 + \pi^2 r^2/(j+1)^2$; thus $B_j = \|\chi_j\|_2^2 S_j/|\mathbb{T}_j|$ in Theorem 7.10(e). Put

$$K_j = \|\chi_j\|_2/2, \qquad \tau_j = m^{-2}B_j, \qquad \sigma_j = \frac{\|\chi_j\|_1^2 S_j}{4|\mathbb{T}_j|}, \qquad L_p = \sum_{q=1}^{p} q^{2p}, \qquad (2p-1)!! = \prod_{i=1}^{p}(2i-1).$$

11.1 Moment bounds for the reference fields

Lemma 11.1 (Hilbert sums)**.** *Let $(a_k)_{k\in\mathbb{Z}^2}$ be complex random variables, $w_k \geq 0$ and $p \geq 1$. Then $\mathbb{E}(\sum_k w_k|a_k|^2)^p \leq (\sum_k w_k\|a_k\|_{2p}^2)^p$. In particular, for a Borel random element Ξ of X, with $a_k = \Xi(\chi_j\overline{e_{j,k}})$, $\mathbb{E}p_j(\Xi)^{2p} \leq (\sum_k \lambda_{j,k}^{-2}\|\Xi(\chi_j\overline{e_{j,k}})\|_{2p}^2)^p$.*

Proof. Minkowski's inequality in L^p for finite partial sums, $\||a_k|^2\|_p = \|a_k\|_{2p}^2$, and monotone convergence. □

Corollary 11.2 (Neural reference moments)**.** *For $b \in \{\mathrm{prod}, \mathrm{Mon}\}$, a regulator a, $N \geq 1$, $j \geq 1$ and $p \in \mathbb{N}$:*

$$\mathbb{E}_b p_j(\iota F_{N,a}^b)^{2p} \leq L_p K_j^{2p}. \tag{43}$$

The separate bound $\mathbb{E}_b p_j(\iota F^b_{N,a})^{2p} \le L_p \sigma_j^p$ also holds. Both are uniform over every finite N and every regulator $a > 0$; K_j and $\sigma_j^{1/2}$ are distinct constants.

Proof. For a fixed spatial point z, $X_i = g^b_{i,a}(z)$ are independent, centered and bounded by $1/2$. In the $2p$-slot expansion of $\mathbb{E}|N^{-1/2}\sum_i X_i|^{2p}$, every singleton label has zero contribution. A surviving tuple has $q \le p$ distinct labels; there are at most $N^q q^{2p}$ tuples with q labels, each term bounded by 2^{-2p} in absolute value. Since $N^{q-p} \le 1$,

$$\mathbb{E}_b |F^b_{N,a}(z)|^{2p} \le L_p / 4^p. \tag{44}$$

Joint measurability and $|F_N| \le \sqrt{N}/2$ make all spatial integrals finite for fixed N. Integral Minkowski in L^p, including $p = 1$, gives

$$\left\| \int \chi_j(z)^2 |F_N(z)|^2 dz \right\|_{L^p} \le \int \chi_j(z)^2 (\mathbb{E}|F_N(z)|^{2p})^{1/p} dz \le L_p^{1/p} \|\chi_j\|_2^2 / 4.$$

Parseval and $\lambda_{j,k} \ge 1$ give $p_j(\iota F_N)^2 \le \|\chi_j F_N\|_2^2$. Raising to power p proves (43) with the original K_j exactly. This uses only positive-regulator pointwise neurons; no point field at $a = 0$ appears.

For the separate σ_j bound, let $f \in L^1(\mathbb{R}^2; \mathbb{C})$, $Y_i = \int g_i f$ and $c = \|f\|_1/2$. The Y_i are independent, identically distributed, centered (Fubini and $\mathbb{E}g = 0$), and $|Y_i| \le c$ because $|g_i| = 1/2$; put $s^2 = \mathbb{E}|Y_1|^2 \le c^2$. Expanding $\mathbb{E}|\sum_{i \le N} Y_i|^{2p} = \mathbb{E}(\sum_i Y_i)^p (\sum_i \overline{Y_i})^p$ over index tuples, a term vanishes unless each index occurring in it occurs at least twice. A term with exactly q distinct indices, with multiplicities $n_i \ge 2$ summing to $2p$, is bounded by $\prod_i \mathbb{E}|Y_i|^{n_i} \le \prod_i c^{n_i - 2} s^2 = c^{2p-2q} s^{2q}$; there are at most $\binom{N}{q} q^{2p} \le N^q q^{2p}$ such tuples and $q \le p$. Hence $\mathbb{E}|N^{-1/2}\sum_i Y_i|^{2p} \le \sum_{q \le p} q^{2p} N^{q-p} c^{2p-2q} s^{2q} \le L_p c^{2p}$. For $f = \chi_j \overline{e_{j,k}}$ we have $(\iota F_N)(f) = N^{-1/2}\sum_i Y_i$ and $\|f\|_1 \le \|\chi_j\|_1 |\mathbb{T}_j|^{-1/2}$, so $\|(\iota F_N)(f)\|_{2p}^2 \le L_p^{1/p} \|\chi_j\|_1^2 / (4|\mathbb{T}_j|)$; Lemma 11.1 concludes. □

Lemma 11.3 (Gaussian localized trace)**.** *Let Ξ be a Borel random element of X such that for every complex $f \in C_c^\infty$ the real and imaginary parts of $\Xi(f)$ are centered and jointly Gaussian with $\mathbb{E}|\Xi(f)|^2 \le m^{-2}\|f\|_2^2$. Then $\mathbb{E}p_j(\Xi)^{2p} \le (2p-1)!!\, \tau_j^p$ for all $j, p \ge 1$. This applies to Φ_a under $\mathbb{P}_0$ for $0 \le a < 1/m$, to $\iota\gamma'$ under $G'_{\Omega,a}$ and to $\iota\gamma_\mu$ under G_μ.*

Proof. For $T = X' + iY'$ with X', Y' real centered Gaussian, $\|T\|_{2p}^2 = \|X'^2 + Y'^2\|_p \le \|X'\|_{2p}^2 + \|Y'\|_{2p}^2 = ((2p-1)!!)^{1/p} \mathbb{E}|T|^2$. With $\mathbb{E}|\Xi(\chi_j \overline{e_{j,k}})|^2 \le m^{-2}\|\chi_j\|_2^2/|\mathbb{T}_j|$, Lemma 11.1 gives the bound. For Φ_a the hypothesis is Theorem 7.10(a) with Theorem 7.8. For a continuous centered Gaussian field γ with covariance kernel C and complex $f \in C_c^\infty$, $\int \gamma f$ is an L^2-limit of Riemann sums, so its real and imaginary parts are jointly Gaussian, and $\mathbb{E}|\int \gamma f|^2 = \langle f, Cf\rangle \le m^{-2}\|f\|_2^2$ when $0 \le C \le C_a$ as operators (Theorems 7.5 and 7.2(d)); this covers $C'_{\Omega,a}$ and C_μ^{end}. □

Lemma 11.4 (Density bounds)**.** *Fix a regulator a, a physical square Λ and $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. For Borel $G : X \to [0, \infty]$:*

$$\mathbb{E}_{P^b_{N,\varepsilon,\Omega,a;\Lambda}} G \le \frac{M_{\mathrm{full}}}{Z^b_N} \mathbb{E}_b G(\iota F^b_{N,a}) \qquad (\varepsilon > 0,\ N \ge 1). \tag{45}$$

In particular, before any eventual-width denominator replacement, every finite N, $j, p \in \mathbb{N}$, both architectures and every fixed regulator satisfy

$$P^b_{N,\varepsilon,\Omega,a;\Lambda}(p_j^{2p}) \le \frac{M_{\mathrm{full}} L_p K_j^{2p}}{Z^b_N} \qquad (\varepsilon > 0). \tag{46}$$

Only after these exact finite-width bounds, the eventual-width and other density bounds are

$$\mathbb{E}_{P^b_{N,\varepsilon,\Omega,a;\Lambda}} G \le \frac{2M_{\mathrm{full}}}{Z_{\mathrm{full}}}\, \mathbb{E}_b\, G(\iota F^b_{N,a}) \quad (\varepsilon > 0,\ N \ge N_0), \tag{47}$$

$$\mathbb{E}_{\iota_* \mathrm{Res}_{\varepsilon,\Omega,a;\Lambda}} G \le M_{\mathrm{res}}\, e^{3u|\Lambda|c_a^2 + 3\varepsilon|\Omega\setminus\Lambda|c_a^2}\, \mathbb{E}_{G'_{\Omega,a}} G(\iota\gamma') \quad (\varepsilon \ge 0), \tag{48}$$

$$\mathbb{E}_{\nu_{a',\Lambda}} G \le \Xi_2(\Lambda)^{1/2} \big(\mathbb{E}_0 G(\Phi_{a'})^2\big)^{1/2} \quad (0 \le a' < 1/m), \tag{49}$$

where M_{full} *is given by* (18), $M_{\mathrm{res}} = e^{6u|\Lambda|c_a^2}$, *and* N_0 *depends on* $(b, \varepsilon, \Omega, a, \Lambda)$ *and is such that* $Z^b_N \ge Z_{\mathrm{full}}/2$ *for* $N \ge N_0$. *At* $\varepsilon = 0$ *the factor in* (48) *does not depend on* Ω.

Proof. (45) is the exact normalized expectation bound from $0 < e^{-V} \le M_{\mathrm{full}}$ and $Z^b_N > 0$, valid at every finite width. Combining it with (43) gives (46). Only afterwards, (47) uses $Z^b_N \to Z_{\mathrm{full}} > 0$ (Proposition 9.2). (48): the density of Res_ε with respect to $G'_{\Omega,a}$ is $e^{-W-\varepsilon J}/z_\varepsilon \le M_{\mathrm{res}}/z_\varepsilon$, and z_ε is bounded below in Proposition 9.3. (49): $Z_{a',\Lambda} \ge 1$ and Cauchy–Schwarz with Theorem 8.2(e) for $p = 2$. □

Lemma 11.5 (Unbounded moment transfer)**.** *Let* $\rho_k \Rightarrow \rho$ *be Borel probabilities on* X, $j, r \ge 1$, *and* $\sup_k \rho_k(p_j^{2r}) < \infty$. *Then* $\rho(p_j^{2r}) \le \liminf_k \rho_k(p_j^{2r})$, *and* $\rho_k(H) \to \rho(H)$ *for every continuous* $H : X \to \mathbb{C}$ *with* $|H| \le c(1 + p_j^r)$.

Proof. For $T > 0$, $\min(p_j^{2r}, T)$ is bounded and continuous; let $T \to \infty$ by monotone convergence. With $H_T = H \min(1, T/|H|)$, bounded and continuous, $|\rho_k(H) - \rho_k(H_T)| \le \rho_k(|H|^2)/T \le 2c^2(1 + \rho_k(p_j^{2r}))/T$, and the same for ρ; let $k \to \infty$, then $T \to \infty$. □

11.2 Moments along the arrows

Proposition 11.6 (Moments along the arrows)**.** *Fix* $j, r \ge 1$, *a physical square* Λ *and, where they occur,* b, *a regulator* a, $\varepsilon > 0$ *and* $\Omega = [-R, R]^2 \supseteq \bar{\Lambda}$. *Every law below has finite moments of every power of* p_j, *and:*

(i) width: $P^b_{N,\varepsilon,\Omega,a;\Lambda}(p_j^r) \to \nu^v_{\varepsilon,\Omega,a;\Lambda}(p_j^r)$ *as* $N \to \infty$;

(ii) stabilizer: $\nu^v_{\varepsilon,\Omega,a;\Lambda}(p_j^r) \to \nu^{(2)}_{\Omega,a;\Lambda}(p_j^r)$ *as* $\varepsilon \downarrow 0$;

(iii) auxiliary square: $\nu^{(2)}_{\Omega,a;\Lambda}(p_j^r) \to \nu_{a,\Lambda}(p_j^r)$ *as* $R \to \infty$, *and* $\sup_R \nu^{(2)}_{\Omega,a;\Lambda}(p_j^{2r}) \le M_{\mathrm{res}} e^{3u|\Lambda|c_a^2}(2r-1)!!\, \tau_j^r$;

(iv) regulator: $\nu_{a,\Lambda}(p_j^r) \to \nu_{0,\Lambda}(p_j^r)$ *as* $a \downarrow 0$, *and* $\sup_{0 \le a < 1/m} \nu_{a,\Lambda}(p_j^{2r}) \le \Xi_2(\Lambda)^{1/2}((4r-1)!!)^{1/2} \tau_j^r$;

(v) physical volume: $\nu_n(p_j^r) \to \nu(p_j^r)$, *with* $\sup_n \nu_n(p_j^{2r}) < \infty$.

Consequently, for every fixed n, dist_n *between consecutive laws of each arrow tends to* 0 *along the arrow.*

Proof. (i) By (47) and Corollary 11.2, $\sup_{N \ge N_0} P^b_N(p_j^{2r}) \le 2M_{\mathrm{full}} L_r K_j^{2r}/Z_{\mathrm{full}}$ by the exact finite-width inequality (46); for $N < N_0$ each law has $p_j(\iota F_N) \le \|\chi_j\|_2 \sqrt{N}/2$. Apply Lemma 11.5 with Proposition 9.2. (ii) By Proposition 9.3, $\nu^v_\varepsilon = \iota_* \mathrm{Res}_\varepsilon$; the densities $e^{-W-\varepsilon J}/z_\varepsilon$ converge pointwise to e^{-W}/z_0 and are bounded by $M_{\mathrm{res}} e^{3u|\Lambda|c_a^2 + 3|\Omega\setminus\Lambda|c_a^2}$ for $\varepsilon \le 1$ (Proposition 9.4), and

$p_j(\iota\gamma')^r \in L^1(G'_{\Omega,a})$ by Lemma 11.3; dominated convergence applies. (iii) (48) at $\varepsilon = 0$, Lemma 11.3, Proposition 9.5 and Lemma 11.5. (iv) (49) with $G = p_j^{2r}$, Lemma 11.3 for $\mathbb{E}_0 p_j(\Phi_a)^{4r}$, Proposition 9.6 and Lemma 11.5. (v) Since $\lambda_{j,k} \geq 1$, $p_j(v) \leq \|\chi_j v\|_{H^{-3/2}(\mathbb{T}_j)}$, whose moments are bounded uniformly in n by Theorem 10.10(iii); $\nu_n \Rightarrow \nu$ there, and Lemma 11.5 applies. The last sentence combines (i)–(v) with the convergences in β of Propositions 9.2, 9.4 (variation distance dominates β), 9.5, 9.6 and Theorem 10.10(iii) with Proposition 5.1(v). □

11.3 The common sequence

Proposition 11.7 (Selection of the common sequence)**.** *Put $\eta_n = 2^{-n}/4$ and $\Omega_k = [-k,k]^2$. Define recursively, for $n = 1, 2, \ldots$ (conditions involving index $n-1$ only for $n > 1$):*

(1) $a_n = a(k)$ for the least $k \in \mathbb{N}$ with $a(k) < \min(1/n, 1/(2m))$, $a(k) < a_{n-1}/2$ and $\mathrm{dist}_n(\nu_{a(k),\Lambda_n}, \nu_n) < \eta_n$;

(2) $R_n = R(k) = k$ for the least $k \in \mathbb{N}$ with $k \geq \max(n, n\ell)$, $k > R_{n-1}$ and

$$\mathrm{dist}_n(\nu^{(2)}_{\Omega_k,a_n;\Lambda_n}, \nu_{a_n,\Lambda_n}) < \eta_n;$$

(3) $\varepsilon_n = \varepsilon(k) = 1/k$ for the least $k \in \mathbb{N}$ with $1/k < 1/n$, $1/k < \varepsilon_{n-1}/2$ and

$$\mathrm{dist}_n(\nu^{v}_{1/k,\Omega_n,a_n;\Lambda_n}, \nu^{(2)}_{\Omega_n,a_n;\Lambda_n}) < \eta_n,$$

where $\Omega_n = \Omega_{R_n}$;

(4) $N_n = N(k) = k$ for the least $k \in \mathbb{N}$ with $k > n$, $k > N_{n-1}$ and

$$\mathrm{dist}_n(P^{b}_{k,\varepsilon_n,\Omega_n,a_n;\Lambda_n}, \nu^{v}_{\varepsilon_n,\Omega_n,a_n;\Lambda_n}) < \eta_n$$

for both b.

Each least index exists, and the resulting sequence satisfies all conditions and (15) *of Theorem 4.1(S). It is determined by m, ℓ, u, the localization system and the enumerations* (13)*.*

Proof. In each step the order constraints hold for all large k, and by Proposition 11.6 the discrepancy tends to 0 as $k \to \infty$ (in (1) because $a(k) \downarrow 0$, in (2) along the increasing squares Ω_k, in (3) because $1/k \downarrow 0$, and in (4) for each of the two architectures); so some k satisfies all conditions and the least one is well defined. In (2), $k \geq n\ell$ gives $\bar{\Lambda}_n \subseteq \Omega_k$, so every law in (2)–(4) is defined, and $a_n < 1/(2m)$ is a regulator. Every law entering dist_n has finite moments of p_j^r (Proposition 11.6). Since dist_n satisfies the triangle inequality, $\mathrm{dist}_n(P^b_n, \nu_n) < 4\eta_n = 2^{-n}$. The laws involved are determined by m, u, ℓ, the neuron definitions and the parameters, and dist_n by the localization system; no other object enters. □

Remark 11.8 (The selection is not an algorithm)*.* The least-index rule makes the sequence a well-defined deterministic function of the data listed in Proposition 11.7. It is not an algorithm: the predicates involve exact values of β and of moments of non-explicit laws, and no procedure deciding them, no bound on the selected indices and no end-to-end rate is given. Selecting the four parameters in the order $a_n, R_n, \varepsilon_n, N_n$ follows the order of the iterated limits of Section 9; no interchange of limits is used or claimed.

11.4 Whole-field convergence, moments and later tests

Proposition 11.9 (Clauses (0) and (A)). *For both architectures every P_n^b is a Borel probability on X with positive finite normalizer and $p_j(v) \le \|\chi_j\|_2\sqrt{N_n}/2$ on its support; $\beta(P_n^b,\nu)\to 0$; and for all $j,r\ge 1$, $P_n^b(p_j^r)\to\nu(p_j^r)$ and $M_{j,r}=\sup_{\rho\in\mathcal{L}}\int p_j^r\,d\rho<\infty$.*

Proof. Lemma 11.11 below and Theorem 6.5(a) give the Borel law and the normalizer bounds; Parseval gives $p_j(\iota F)\le\|\chi_j F\|_{L^2}$. Next, $\beta(P_n^b,\nu)\le \mathrm{dist}_n(P_n^b,\nu_n)+\beta(\nu_n,\nu)<2^{-n}+\beta(\nu_n,\nu)\to 0$ by Theorem 10.10(iii) and Proposition 5.1(v). For $n\ge\max(j,r)$, $|P_n^b(p_j^r)-\nu_n(p_j^r)|<2^{-n}$, and $\nu_n(p_j^r)\to\nu(p_j^r)<\infty$ by Proposition 11.6(v). A convergent sequence of finite numbers is bounded, and the finitely many stages with $n<\max(j,r)$ have finite moments. □

Proposition 11.10 (Clause (B)). *Let real $f_1,\dots,f_k\in C_c^\infty(\mathbb{R}^2)$ and a real or complex polynomial Q of degree d be given after the sequence. Then the expectations in Theorem 4.1(B) are absolutely finite, uniformly bounded, and converge; the laws of $(v(f_1),\dots,v(f_k))$ converge weakly.*

Proof. The evaluations are continuous on X (Proposition 5.1(iii)), so the continuous-mapping theorem gives weak convergence of the vectors, independently of Q. If Q is constant, the assertions involving Q follow directly from probability normalization. Otherwise its degree $d\ge 1$. Choose j with all supports in $(-j,j)^2$. By (16), $|v(f_i)|\le p_j(v)\|f_i\|_{H^2(\mathbb{T}_j)}$, so

$$|Q(v(f_1),\dots,v(f_k))|\le c_Q(1+p_j(v)^d)$$

and, for $q>0$, $|Q|^q\le c_{Q,q}(1+p_j^{\lceil dq\rceil})$; Proposition 11.9 bounds these uniformly over $\mathcal{L}$. Lemma 11.5 with $H=Q(v(f_1),\dots,v(f_k))$ and $r=d$ gives the convergence of expectations. □

11.5 Finite-stage reflection positivity

Lemma 11.11 (The stages satisfy the positivity hypotheses). *For every n, $\Lambda_n=(-n\ell,n\ell)^2$ is a physical square in $\Omega_n=[-R_n,R_n]^2$ (Definition 2.1); both $\bar\Lambda_n$ and Ω_n are centered closed axis-parallel squares invariant under θ_1 and θ_2; and $P_n^b=P_{N_n}^{v_V}$ with the integrand v_V of (7) for $(\varepsilon_n,\Omega_n,a_n;\Lambda_n)$, which agrees off the null set $\partial\Lambda_n\times\mathbb{R}$ with the integrand of Corollary 6.6 for the rectangles $\bar\Lambda_n=[-n\ell,n\ell]^2$ and Ω_n.*

Proof. $R_n\ge n\ell$ by Proposition 11.7; $\varepsilon_n>0$ and a_n is a regulator. The remaining assertions are the manuscript-tilt part of Corollary 6.6. □

Proposition 11.12 (Clause (E1)). *For $\theta\in\{\theta_1,\theta_2\}$, every n and both architectures, P_n^b satisfies (11) for every bounded complex Σ_+^θ-measurable F.*

Proof. Lemma 11.11 and Corollary 6.6, which gives Theorem 6.5(b) across both central lines; the integrand is θ-invariant because both the inner square and the outer square are. □

12 Joint compact-test correlations

Correlation functions involve products of field values at several points, which have no pointwise meaning for a distribution. This section uses instead the tensor power $v^{\otimes k}$, a distribution in k plane variables that equals $\prod_r v(f_r)$ on product tests; it is built from localized Fourier coefficients and

bounded by $p_j(v)^k$ times a norm of the test (Proposition 12.3). Averaging over a law gives the correlation distributions, the moment bounds of Section 11 give their convergence, and the same objects form the polynomial positivity expressions of clause (E2) (Proposition 12.9).

For $\kappa = (\kappa_1, \dots, \kappa_k) \in (\mathbb{Z}^2)^k$ put $e_{j,\kappa}(z_1, \dots, z_k) = \prod_r e_{j,\kappa_r}(z_r)$ and $\lambda_{j,\kappa} = \prod_r \lambda_{j,\kappa_r}$.

Lemma 12.1 (Fourier facts)**.** *Let $F \in \mathcal{D}(((-j,j)^2)^k)$, extended by zero to a smooth periodic function on $\mathbb{T}_j^k$, and $F_\kappa = \int_{\mathcal{C}_j^k} \overline{e_{j,\kappa}} F$. Then the Fourier series $\sum_\kappa F_\kappa e_{j,\kappa}$ converges to F in $C^\infty(\mathbb{T}_j^k)$,*

$$\sum_\kappa \lambda_{j,\kappa}^2 |F_\kappa|^2 = A_{j,k}(F)^2, \qquad A_{j,k}(F) \le 3^k |\mathbb{T}_j|^{k/2} \|F\|_{C^{2k}}. \tag{50}$$

Proof. $\prod_r (1 - \Delta_{z_r}) e_{j,\kappa} = \lambda_{j,\kappa} e_{j,\kappa}$, so Parseval on $\mathbb{T}_j^k$ gives the identity; repeated integration by parts makes F_κ decay faster than every power of $|\kappa|$, which gives convergence with all derivatives. Expanding the product of the operators $1 - \partial_1^2 - \partial_2^2$ gives 3^k terms, each a derivative of order at most $2k$, and $\|\cdot\|_{L^2(\mathbb{T}_j^k)} \le |\mathbb{T}_j|^{k/2} \|\cdot\|_\infty$. □

Lemma 12.2 (Support-preserving product density)**.** *Let $U \subseteq \mathbb{R}^2$ be open and $F \in \mathcal{D}(U^k)$. There are j, a real $\zeta \in \mathcal{D}(U \cap (-j,j)^2)$ and finite sums F_l of products $f_1 \otimes \cdots \otimes f_k$ with $f_r \in \mathcal{D}(U)$ and $\operatorname{supp} f_r \subseteq \operatorname{supp} \zeta$, such that $F_l \to F$ in C^M for every M, all supports lie in $(\operatorname{supp} \zeta)^k$, and $A_{j',k}(F_l - F) \to 0$ for every $j' \ge j$.*

Proof. Let $K_0 \subset U$ be the union of the coordinate projections of $\operatorname{supp} F$, choose j with $K_0 \subset (-j,j)^2$ and a real $\zeta \in \mathcal{D}(U \cap (-j,j)^2)$ with $\zeta = 1$ near K_0. Then $F = \zeta^{\otimes k} F$, and with the Fourier partial sums $S_l F$ on $\mathbb{T}_j^k$, $F_l = \zeta^{\otimes k} S_l F = \sum_{|\kappa|_\infty \le l} F_\kappa \bigotimes_r (\zeta e_{j,\kappa_r})$ converges to F in every C^M by Lemma 12.1 and the Leibniz rule; (50) gives the last assertion. □

Proposition 12.3 (Tensor evaluations; clause (C1))**.** *For $v \in X$ and $F \in \mathcal{D}(((-j,j)^2)^k)$ put $v^{\otimes k}(F) = \sum_\kappa F_\kappa \prod_r v(\chi_j e_{j,\kappa_r})$. The series converges absolutely, does not depend on j, and defines the unique $v^{\otimes k} \in \mathcal{D}'((\mathbb{R}^2)^k)$ with $v^{\otimes k}(f_1 \otimes \cdots \otimes f_k) = \prod_r v(f_r)$; it is symmetric, satisfies $\overline{v^{\otimes k}(F)} = v^{\otimes k}(\overline{F})$, $v \mapsto v^{\otimes k}(F)$ is continuous on X, and $|v^{\otimes k}(F)| \le p_j(v)^k A_{j,k}(F) \le 3^k |\mathbb{T}_j|^{k/2} p_j(v)^k \|F\|_{C^{2k}}$.*

Proof. Cauchy–Schwarz with the weights $\lambda_{j,\kappa}^{\pm 1}$, (50), $v(\chi_j e_{j,\kappa_r}) = v(\chi_j \overline{e_{j,-\kappa_r}})$ and $\lambda_{j,-\kappa_r} = \lambda_{j,\kappa_r}$ give absolute convergence and the bound. For $F = f_1 \otimes \cdots \otimes f_k$ the coefficients factor and the series is $\prod_r \sum_{\kappa_r} (f_r)_{\kappa_r} v(\chi_j e_{j,\kappa_r}) = \prod_r v(f_r)$ by the expansion in the proof of (16); this holds for every admissible j. For general F, Lemma 12.2 gives product sums $F_l \to F$ with $A_{j,k}(F_l - F) \to 0$ and $A_{j',k}(F_l - F) \to 0$, so the values for j and $j' \ge j$ agree. The bound shows continuity on each $\mathcal{D}_K$, so $v^{\otimes k} \in \mathcal{D}'$; uniqueness follows from the density of product sums. Symmetry and the reality relation hold on product tests (for the latter because v is real) and extend by density. For $v, w \in X$, telescoping gives $|v^{\otimes k}(F) - w^{\otimes k}(F)| \le k A_{j,k}(F)\, p_j(v - w) \max(p_j(v), p_j(w))^{k-1}$, and $p_j(v - w) \le 2^j d(v, w)$ when $d(v, w) < 2^{-j}$. □

Remark 12.4 (Meaning of the tensor evaluation). $v^{\otimes k}$ is the k-fold tensor power of the distribution v; for background on tensor products of distributions see [22], cited for orientation only. Every property used here is proved in Proposition 12.3. The evaluation is defined for every $v \in X$, with no exceptional set.

Lemma 12.5 (Complexification, reflection and products)**.** *Let $v \in X$, $\theta \in \{\theta_1, \theta_2\}$, $G \in \mathcal{D}((\mathbb{R}^2)^r)$ and $H \in \mathcal{D}((\mathbb{R}^2)^s)$. Then (a) $(\theta v)^{\otimes r}(G) = v^{\otimes r}(G \circ \theta^{\times r})$, where $(G \circ \theta^{\times r})(z_1, \dots, z_r) = G(\theta z_1, \dots, \theta z_r)$; (b) $v^{\otimes r}(G)\, v^{\otimes s}(H) = v^{\otimes (r+s)}(G \otimes H)$; (c) if $G \in \mathcal{D}((H_+^\theta)^r)$, then $v \mapsto v^{\otimes r}(G)$ is Σ_+^θ-measurable; (d) for $G \in \mathcal{D}((H_+^\theta)^r)$, $\overline{(\theta v)^{\otimes r}(G)} = v^{\otimes r}(\theta_r \overline{G})$ with $(\theta_r \overline{G})(z) = \overline{G(\theta z_1, \dots, \theta z_r)}$.*

Proof. On product tests, (a) is $(\theta v)(f) = v(f \circ \theta)$ (Proposition 5.1(vii), extended complex-linearly) and (b) is multiplicativity. Both sides of (a) are continuous in G for $A_{j,r}$ with supports in a fixed compact, since $p_j(\theta v) \le C_j p_{j+1}(v)$ and θ preserves $\mathbb{T}_j$ and the Laplacian; for (b), $A_{j,r+s}(G \otimes H) = A_{j,r}(G)A_{j,s}(H)$. Lemma 12.2 concludes. For (c), the approximants of Lemma 12.2 with $U = H^\theta_+$ give $v^{\otimes r}(G)$ as a pointwise limit of finite sums of products $\prod_i v(f_i)$ with $f_i \in \mathcal{D}(H^\theta_+)$, whose real and imaginary parts are real tests in $C^\infty_c(H^\theta_+)$. (d) combines (a) with the reality relation of Proposition 12.3. □

Proposition 12.6 (Correlation distributions; clause (C2))**.** *For $\rho \in \mathcal{L}$ and $F \in \mathcal{D}((\mathbb{R}^2)^k)$, $\mathcal{S}^\rho_k(F) = \int v^{\otimes k}(F)\,\rho(dv)$ is absolutely defined, $\mathcal{S}^\rho_k \in \mathcal{D}'((\mathbb{R}^2)^k)$ is the unique distribution with $\mathcal{S}^\rho_k(f_1 \otimes \cdots \otimes f_k) = \int \prod_r v(f_r)\,\rho(dv)$, and $\sup_{\rho\in\mathcal{L}} |\mathcal{S}^\rho_k(F)| \le M_{j,k}A_{j,k}(F)$ for $\operatorname{supp} F \subset ((-j,j)^2)^k$.*

Proof. $v \mapsto v^{\otimes k}(F)$ is continuous, hence Borel, with $|v^{\otimes k}(F)| \le p_j^k A_{j,k}(F)$, and $M_{j,k} < \infty$ (Proposition 11.9). Linearity is clear; the bound and (50) give continuity on each $\mathcal{D}_K$; uniqueness is the density of product sums (Lemma 12.2). □

Proposition 12.7 (Kernel convergence; clause (C3))**.** *For every k and $F \in \mathcal{D}((\mathbb{R}^2)^k)$, $\mathcal{S}^b_{n,k}(F) \to \mathcal{S}_{\infty,k}(F)$ for both architectures.*

Proof. $H(v) = v^{\otimes k}(F)$ is continuous with $|H| \le A_{j,k}(F)p_j^k$; $P^b_n \Rightarrow \nu$ and $\sup_n P^b_n(p_j^{2k}) < \infty$ (Proposition 11.9); apply Lemma 11.5. □

Proposition 12.8 (Target identification; clause (D))**.** *For every k, $\mathcal{S}_{\infty,k} = \mathrm{Schw}_k|_{\mathcal{D}((\mathbb{R}^2)^k)}$, with Schw_k the tempered Schwinger distribution of ν^S (Theorem 10.16(v)).*

Proof. For $f_r \in \mathcal{D}(\mathbb{R}^2)$, $\nu = \mathcal{R}_*\nu^S$ and $\mathcal{R}(\psi)(f) = \psi(f)$ (Theorem 10.10(iii), extended complex-linearly) give $\mathcal{S}_{\infty,k}(f_1 \otimes \cdots \otimes f_k) = \int \prod_r \psi(f_r)\,d\nu^S = \mathrm{Schw}_k(f_1 \otimes \cdots \otimes f_k)$. The restriction of Schw_k to $\mathcal{D}$ is a distribution because $\mathcal{D} \hookrightarrow \mathcal{S}$ is continuous; two distributions that agree on product tests agree (Lemma 12.2). □

Proposition 12.9 (Polynomial reflection positivity; clause (E2))**.** *Let $\theta \in \{\theta_1, \theta_2\}$, $J \ge 0$, $f_0 \in \mathbb{C}$, $f_r \in \mathcal{D}((H^\theta_+)^r)$ and $\rho \in \mathcal{L}$. Then $\mathcal{O}_f, \mathcal{O}_f \circ \theta \in L^2(\rho)$ with $\|\mathcal{O}_f\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^J A_{j,r}(f_r)M^{1/2}_{j,2r}$ and $\|\mathcal{O}_f \circ \theta\|_{L^2(\rho)} \le |f_0| + \sum_{r=1}^J C_j^r A_{j,r}(f_r)M^{1/2}_{j+1,2r}$ for j with all supports in the corresponding cubes; every term of $\Pi_\rho(f)$ is absolutely finite; $\Pi_\rho(f) = \int \overline{\mathcal{O}_f(\theta v)}\,\mathcal{O}_f(v)\,\rho(dv)$; $\Pi_{P^b_n}(f) \ge 0$; and $\Pi_{P^b_n}(f) \to \Pi_\nu(f) \ge 0$.*

Proof. The L^2 bounds follow from Proposition 12.3, $p_j(\theta v) \le C_j p_{j+1}(v)$ and Minkowski. By Lemma 12.5(b),(d), $\overline{(\theta v)^{\otimes r}(f_r)}\,v^{\otimes s}(f_s) = v^{\otimes(r+s)}((\theta_r \overline{f_r}) \otimes f_s)$ for every v; integrating gives the terms of (12), which are absolutely finite by Cauchy–Schwarz, and summing gives the integral formula. For $\rho = P^b_n$, $\mathcal{O}_f$ is Σ^θ_+-measurable (Lemma 12.5(c)), and the clipped observables $\mathrm{clip}_T\mathcal{O}_f = \mathcal{O}_f \min(1, T/|\mathcal{O}_f|)$ are bounded and Σ^θ_+-measurable, with $(\mathrm{clip}_T\mathcal{O}_f)(\theta v) = \mathrm{clip}_T(\mathcal{O}_f(\theta v))$. Proposition 11.12 gives $\int \overline{(\mathrm{clip}_T\mathcal{O}_f)(\theta v)}(\mathrm{clip}_T\mathcal{O}_f)(v)\,P^b_n(dv) \ge 0$; as $T \to \infty$ the integrands converge pointwise and are dominated by the integrable $|\mathcal{O}_f(\theta v)||\mathcal{O}_f(v)|$, so $\Pi_{P^b_n}(f) \ge 0$. Each term converges by Proposition 12.7, so $\Pi_{P^b_n}(f) \to \Pi_\nu(f)$, which is therefore real and nonnegative. □

Remark 12.10 (What the correlation statements mean). Convergence in (C3) is convergence of values on each fixed compactly supported smooth joint test, that is, in $\sigma(\mathcal{D}'((\mathbb{R}^2)^k), \mathcal{D}((\mathbb{R}^2)^k))$; no Schwartz or other noncompact test is used at the neural stages, and no strong-dual convergence is

claimed. The bounds $M_{j,k}$ hold along the selected sequence and at ν. The positivity of $\Pi_\nu(f)$ is obtained as a limit; it is consistent with, but not derived from, Theorem 10.16(ii).

13 Proof of the main theorem

Lemma 13.1 (One set of limiting objects)**.** *The law ν in clauses (0), (A)–(C) and (E) is $\nu = \mathcal{R}_*\nu^S$ of Theorem 10.10(iii), with ν^S the target of Theorem 10.8. Any Borel probability on X whose finite-dimensional laws on real compact smooth tests are those of ν^S equals ν. The law ν_n in* (15) *is ν_{0,Λ_n} of Definition 2.10(d),(e).*

Proof. The cylinder events of compact smooth evaluations form a $\cap$-stable generator of $\mathcal{B}(X)$ (Proposition 5.1(iii)); apply Dynkin's theorem. □

Proof of Theorem 4.1. The classical inputs enter only through Theorems 10.8, 10.10 and 10.16 and Lemma 10.14; the foundations of Sections 5–8 are proved in this paper.

Clause	Proved by
(S)	Proposition 11.7, from Propositions 9.2–9.6, 11.6 and Theorem 10.10
(0)	Proposition 11.9, Lemma 11.11, Theorem 6.5(a), Theorem 10.10(iii)
(A)	Proposition 11.9
(B)	Proposition 11.10
(C1)–(C3)	Propositions 12.3, 12.6, 12.7
(D)	Proposition 12.8, Theorem 10.16(v)
(E1)	Proposition 11.12 (Corollary 6.6)
(E2)	Proposition 12.9

Every clause uses the one sequence of Proposition 11.7 with no subsequence, and the objects quantified in (A)–(E) are chosen after it (Remark 4.3). Theorem 4.2 is Theorem 10.16. □

Remark 13.2 (No circularity). The order of proof is: field spaces (Section 5); neurons and their positivity, Gaussian references and the Wick theory (Sections 6–8, Appendices A–C), none of which uses an interacting law; the fixed-volume chain (Section 9); the continuum lemmas (Section 10), where Theorem 10.8 uses C1, C2 and the Wick identification, Theorem 10.10 uses C4, C5 and Theorem 10.8, and the restricted proof of Theorem 10.16 uses C1–C3, C6, the direct classical Wick lemma and the source portion of Theorem 10.8, without its common-space bridge; then Sections 11–12. Ergodicity of the target is proved in Theorem 10.16(iii) before [8, Thm. 1.0] is applied in (vii). No statement uses a later one.

14 Contribution and relation to prior work

14.1 Contribution

The quadratic term enters through a Gaussian identity. After the width limit, the quadratic tilt of the Gaussian endpoint reference over Ω is again a Gaussian law, with covariance $C'_{\Omega,a}$; as Ω grows, $C'_{\Omega,a}$ converges locally uniformly to C_a, whose Fourier multiplier is $(m^2 + |p|^2 + a^2 p_1^2 p_2^2)^{-1}$ by the mass-shift identity $1/\hat{C}_\mu^{\mathrm{end}} - \kappa_a^2 = 1/\hat{C}_a$. The regulator a therefore controls the anisotropy of this Gaussian reference covariance, and the anisotropy disappears as $a \to 0$. The quartic interaction is

separate from this identity: the finite-width laws and the interacting laws are non-Gaussian, and their covariances are not given by it.

For both architectures we prove the following. First, at fixed physical volume and for every $m, u > 0$, the laws converge successively in the width, the stabilizer, the auxiliary volume and the regulator to the free-boundary Wick-quartic law on the physical square, together with all positive integer powers of the local seminorms. Second, for $0 < u/m^2 < g_*$, one deterministic sequence of all parameters, fixed before any observable is chosen and never passed to a subsequence, converges weakly on $H^{-2}_{\mathrm{loc}}(\mathbb{R}^2)$ to the H^{-2}_{loc}-valued version of the classical infinite-volume weak-coupling Wick-quartic measure; the same sequence gives convergence, with uniform moment bounds, of every polynomial in finitely many compactly supported smooth smearings, and convergence of every compact joint-test correlation distribution. Third, every stage of this sequence is reflection positive across the two central coordinate lines $z_1 = 0$ and $z_2 = 0$, both for bounded half-plane observables and for compact joint-test polynomial forms. We prove no reflection-positivity statement for other lines, and we claim neither Euclidean invariance nor clustering of the finite stages.

These statements rest on attributed classical inputs and on the foundations proved here: the endpoint neurons and their finite-width central reflection positivity, the Gaussian reference constructions, and the fixed-volume Wick estimates. The infinite-volume target, its Osterwalder–Schrader properties and its reconstruction are classical results used with attribution.

14.2 Relation to prior work

Frank [7] gives a constructive neural-network approach to the finite-volume Wick-ordered ϕ^4_2 measure at every coupling. His networks are quenched random-feature shells with Gaussian coefficients and a smooth spatial cutoff; for almost every draw of the features, along a prescribed nested width schedule, the joint laws of finitely many Schwartz smearings converge in either order of the width and ultraviolet limits, with explicit width thresholds and width-uniform ultraviolet tail bounds. His coefficient $\lambda/4!$ corresponds to our u. Our construction differs in its architectures, in using annealed rather than quenched laws, in its sharp-square interaction with a stabilizer and a mass tilt, in its finite-stage reflection positivity, and in including a physical-volume limit, which we obtain by combining the fixed-volume neural limits with classical infinite-volume results. Frank's construction stops at finite volume and leaves open whether neural constructions reach infinite volume themselves; we do not claim that an analogous classical thermodynamic step could not be attached to it. At fixed volume, his width-uniform estimates imply, by a standard uniform-integrability argument (Remark 14.1), convergence of all polynomial moments of finitely many smeared fields for his laws, so we make no novelty claim based on moment convergence. Whether his estimates also yield fixed-volume convergence of joint compact-test kernels or of the field on H^{-2}_{loc} requires estimates that we have not established, chiefly a bound on localized smeared fields summed over all frequency shells uniformly in the localized Fourier mode, followed by tightness and an H^{-2}_{loc}-valued realization of his infinite-cutoff field. We base no novelty claim on the kernel or topology labels either.

The annealed reweighting is the parameter-density deformation $P = P_G\, e^{-\lambda \int \mathcal{O}}$ of Demirtas et al. [3]; Zhang [21] writes the same normalized reweighting of a free ensemble. In the parts of the former work that we consulted, the deformation is formal and at infinite width. We realize it at finite width for the two architectures above, with finite positive normalization and with the limits stated above.

The reflection-positivity argument uses conditional factorization. For product neurons, conditioning on the complete boundary-data tuple of all N neurons makes the two half-plane sides

independent with equal conditional laws. For Mondrian neurons, the square identity is first proved in symmetric finite boxes using the extended trace, including positions, auxiliary times and colors (Corollary A.12). Whole-plane half-plane positivity then follows by L^2 approximation of bounded observables (Proposition A.15). In both cases, a reflection-invariant additive potential is absorbed into the side observables, preserving positivity (Lemma A.14 and Theorem 6.5). This instantiates the conditional-factorization principle underlying the one-dimensional parameter-splitting mechanism of Ferko and Halverson [4]. They also show that symmetric Markov processes are reflection positive, a connection they attribute to Nelson and others, and that reflection positivity survives the linear combinations and pointwise neural maps they consider, for example of independent inputs and with random maps drawn independently of the input. Reflection positivity of Markov fields and lattice models is classical [9]. We instantiate the mechanism in two dimensions for these architectures, across the two central coordinate lines.

Harvey [13] shows that, for $d \geq 2$, a pointwise ensemble whose two-point function is Euclidean invariant, $S_2(gx, gy) = S_2(x, y)$ for all translations, rotations and reflections g, and continuous, and which is two-point reflection positive across one hyperplane, has a constant two-point function; he presents this as an answer, for that finite-variance class, to the question of reflection positivity in $d \geq 2$ raised in [6]. The invariance hypothesis concerns the two-point function, not the full law. Our finite stages have bounded values. We do not claim that the two-point functions of our normalized interacting finite-stage laws are Euclidean invariant, and we do not determine whether they are; we therefore assert neither that the theorem applies to these stages nor that it does not, and the statements we prove do not conflict with it. He notes that products of coordinate cosine networks have the product kernel $\prod_i e^{-m|x_i - y_i|}$ with two-point reflection positivity across coordinate hyperplanes, giving up exact rotation invariance. Our construction does not use rotation invariance either: the endpoint covariance $e^{-\mu|z|_1}/4$ of the neurons is not rotation invariant. This is a property of that reference covariance; it is not a statement about the two-point function of the interacting finite-stage laws. What we add at finite width is reflection positivity of the full normalized law for bounded observables across the two central coordinate lines, rather than two-point positivity alone.

Ferko and Mutchler [5] prove obstructions for specified finite-width classes: finite expansions in fixed deterministic features, including distributional neurons, cannot reproduce a nontrivial Källén–Lehmann covariance; ensembles with bounded mollified coincident variance, such as networks with locally bounded pointwise second moments, are incompatible with a nontrivial Källén–Lehmann representation, which temperedness, Euclidean invariance and reflection positivity would furnish; and finite single-layer ridge networks cannot reproduce all correlation functions of an Osterwalder–Schrader field with a nontrivial Källén–Lehmann measure, a result they state is specific to single-layer networks. They also stress that a parameter count alone cannot define finiteness, since a single Borel parameter can encode any field law. Our finite stages have bounded values and are not claimed to be Euclidean invariant, so they do not conflict with the variance obstruction, and the ridge theorem does not describe product-telegraph or Mondrian neurons. Ferko, Halverson and Mutchler [6] show that every law on $\mathcal{S}'(\mathbb{R}^d)$ admits a neural-network representation with countably many parameters, or formally with a single one; we therefore do not claim representability of the target as a contribution.

For fields built from neurons see Halverson, Maiti and Stoner [12] and Halverson [11]. Zhang [21] and Ageev and Ageeva [1] study adjacent architecture and cost questions that are not addressed here.

Remark 14.1 (Fixed-volume moments for the laws of [7]). The moment sentence above rests on the following short derivation, which uses only estimates stated in [7]; the conclusion is not stated there. Let X be the pairing of the cutoff field of [7] with a Schwartz function f. The normalizer there is

at least one [7, Lemma 2.8], so Cauchy–Schwarz gives $\mathbb{E}_\mu|X|^q \le (\mathbb{E}_\gamma e^{-2V})^{1/2}(\mathbb{E}_\gamma|X|^{2q})^{1/2}$, where γ is the Gaussian prior. The first factor is the normalizer at doubled coupling, which [7] bounds uniformly in the shell cutoff and the width index on the almost-sure feature event. Under the prior, X is centered Gaussian with variance bounded, for every feature realization, by the deterministic bound on the shell pairing coefficients [7, (6.38)]. Gaussian moments therefore bound $\sup \mathbb{E}|X|^q$ over the width index and the cutoff for every q, likewise for finite tuples, and Fatou passes the bounds to the limit laws. The weak convergence statements of [7, Lemma 7.5, Thm. 6.6(c), Thm. 7.6, Cor. 7.8], together with uniform integrability, then give convergence of every polynomial moment along each iterated limit. We do not claim the corresponding statements for joint compact-test kernels or for the topology of H^{-2}_{loc}; they would require the estimates named above.

15 Limitations

The construction is existential. It provides no effective end-to-end parameter schedule, computational cost, sampling guarantee or numerical weak-coupling threshold: the parameter sequence is shown to exist but is not computed. The fixed-volume Wick step includes explicit $\sqrt{a}$ bounds whose constants depend on the volume; these are fixed-volume upper bounds, not practical convergence rates.

The infinite-volume limit, the target, its Euclidean invariance, clustering and reflection positivity across every line, and its $1 + 1$-dimensional Wightman reconstruction are classical results used with attribution. No new interacting theory and no statement in other dimensions is claimed. The classical inputs C1–C5 are used in the formulations of Section 3; their proofs were not audited, the bodies of [16, 17] and [20] and the antecedents of the review in [10, pp. 316–319] were not available, and the page-level access record is in Appendix D.

Finite-stage reflection positivity is proved across the two central coordinate lines, for the centered axis-parallel squares used here and, more generally, for every bounded auxiliary region and every measurable potential, bounded on bounded field ranges, that are invariant under both central reflections. Nothing is claimed for other lines. For a rotated physical square nothing is claimed beyond this invariant case; in particular, no failure of reflection positivity is asserted for any rotated square. The finite stages are not claimed to be Euclidean invariant, and no finite-stage clustering statement is made.

The laws are annealed; nothing is claimed about quenched or fixed-feature laws.

Neural convergence is proved for compactly supported test functions only. No global Schwartz or strong-dual convergence is claimed.

The reflection-positivity mechanism, the density-deformation mechanism and the endpoint covariance kernel all have prior instances in the cited works.

The Wick interaction at $a = 0$ is constructed on one common Gaussian space as an L^2 limit in the fourth chaos, for a fixed physical square; its exceptional null sets are fixed per regulator value and per test, or common to countable families. Nothing is asserted about other representations of that variable, about uncountable families of regulators, or about volume-uniform versions of the Wick estimates beyond $Z_{a,\Lambda} \ge 1$.

The review status of this exact version, including that of Theorem 4.1, is recorded in its companion decision record, not in this text. Appendix D identifies the components that are retained, modified, re-expressed or newly written here, and the current-text review that the project's review policy requires for each. All review recorded for the antecedents was internal model review; none of it is independent expert review.

A Proofs for the process neurons

This appendix proves Theorems 6.1–6.5 and Corollaries 6.6, 6.7. Besides C6 it uses: (K1) Dynkin's π–λ theorem and the functional monotone-class theorem; (K2) Doob–Dynkin factorization, and the fact that a function measurable for the σ-algebra generated by a family of maps depends on countably many of them; (K3) if Y is Polish and continuous real $l_1, l_2, \dots$ separate its points, then $y \mapsto (l_k(y))_k$ is a continuous injection into $\mathbb{R}^{\mathbb{N}}$, hence (Lusin–Souslin) a Borel isomorphism onto its Borel image, so $\sigma(l_k : k \geq 1) = \mathcal{B}(Y)$; (K4) finite counting measures on a Polish space form a standard Borel space whose σ-algebra is generated by $\nu \mapsto \nu(A)$, and $\nu \mapsto \int f\, d\nu$ is Borel for bounded Borel f; (K5) Kolmogorov extension for sequences of standard Borel spaces; (K6) the Poisson facts of C6(h). For $\theta \in \{\theta_1, \theta_2\}$ write $H_+ = H_+^\theta$ and $Q_n = Q_n^\theta = H_+ \cap (-n, n)^2$, and $r_N = \sqrt{N}/2$.

A.1 Measurability

Lemma A.1 (Partition spaces)**.** *Let C be a box with nonempty interior. (i) $\mathrm{Part}^{\mathrm{col}}(C)$ is a Borel subset of the standard Borel space of finite counting measures on $\mathbb{R}^4 \times \{-1, 1\}$, hence standard Borel; restriction res_B, the union map $\cup(P_0, P_1) = P_0 \cup P_1$ (for daughters of a cut), and images under translations and θ_1, θ_2 are Borel. (ii) For open $U \subseteq \operatorname{int} C$, the field map $P \mapsto \mathrm{fld}(P)|_U$, $\mathrm{fld}(P) = \sum_{(R,\xi)\in P} (\xi/2)\mathbf{1}_{R^{\mathrm{ll}}}$, is Borel into $L^1(U)$. (iii) $(z, P) \mapsto \mathrm{fld}(P)(z)$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{B}(\mathrm{Part}^{\mathrm{col}}(C))$-measurable.*

Proof. (i) On the Borel set of counting measures with exactly k atoms of mass one, the atoms can be listed lexicographically by a Borel map into $(\mathbb{R}^4 \times \{\pm 1\})^k$. The conditions "each atom codes a box with nonempty interior contained in C, interiors pairwise disjoint, areas summing to $|C|$" are finitely many Borel conditions on the list, and they characterize the k-cell partitions: the union of the boxes is closed, and a missing point of C would have a missing relative neighbourhood of positive area. Restriction, union and the maps act coordinatewise on the list by Borel operations (maxima and minima of corners, discarding pieces of zero width or height). (ii) For bounded Borel f on U, $P \mapsto \int_U f\mathrm{fld}(P) = \int h_f\, dP$ with $h_f(a, b, c, d, \xi) = (\xi/2) \int_{U\cap[a,b]\times[c,d]} f$, Borel by (K4); the indicators of rational boxes with closure in U give continuous functionals on $L^1(U)$ that separate points (Lebesgue differentiation), so (K3) applies. (iii) The integrand $(z, a, b, c, d, \xi) \mapsto (\xi/2)\mathbf{1}\{a \leq z_1 < b,\ c \leq z_2 < d\}$ is Borel, and integrals of jointly Borel bounded functions against counting measures are jointly measurable (K1). □

Lemma A.2 (Random fields and parameter maps)**.** *Let $r \geq 0$, $U \subseteq \mathbb{R}^2$ open and $\mathrm{clip}_r(x) = \max(-r, \min(r, x))$. (i) If $F : \mathbb{R}^2 \times \Gamma \to \mathbb{R}$ is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable with $|F| \leq r$, then $\omega \mapsto [F(\cdot, \omega)|_U]$ is measurable into $\mathrm{Ball}_r(U)$, and for $U = \mathbb{R}^2$ into E. (ii) Two such fields with the same finite-dimensional distributions have the same law of $F|_U$. (iii) Let $(Y_{\mathrm{sp}}, \mathcal{Y})$ be a measurable space and $G : \mathbb{R}^2 \times Y_{\mathrm{sp}} \to \mathbb{R}$ be $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{Y}$-measurable. Then $\Gamma_G^r(y) = [\mathrm{clip}_r(G(\cdot, y))|_U]$ is defined for every y, takes values in $\mathrm{Ball}_r(U)$ and is measurable. If $Y : \Gamma \to Y_{\mathrm{sp}}$ is measurable and $F(z, \omega) = G(z, Y(\omega))$ for almost every $z \in U$ for every outcome of an almost sure event, then $[F|_U] = \Gamma_G^r(Y)$ almost surely, and the law of $[F|_U]$ is the image of the law of Y. If $|G| \leq r$, the clip changes nothing.*

Proof. (i) For $s_0 \in L^1_{\mathrm{loc}}(U)$ and compact $K \subset U$, $\omega \mapsto \int_K |F(z, \omega) - s_0(z)|dz$ is measurable by Tonelli; $L^1_{\mathrm{loc}}(U)$ is separable metric, so open sets are countable unions of balls; the same argument works in E. (ii) Indicators of rational boxes with closure in U generate the Borel σ-algebra (K3), finite cylinders form a π-system (K1), and the law of a bounded vector $(\int f_{k_1} F, \dots, \int f_{k_m} F)$ is determined by its mixed moments $\int \cdots \int \prod_j f_{k_j}(z_j) \mathbb{E} \prod_j F(z_j)\, dz$. (iii) Each section of G is

Borel and clip_r is continuous, so the clipped section is Borel and bounded by r. For a Borel representative s_0 and compact K, $(z,y) \mapsto \mathbf{1}_K(z)|\mathrm{clip}_r(G(z,y)) - s_0(z)|$ is product measurable, so $y \mapsto \int_K |\mathrm{clip}_r G(z,y) - s_0(z)|dz$ is measurable (the measurability part of Tonelli needs σ-finiteness only in z: the sets D whose sections satisfy that $y \mapsto |D_y \cap K_m|$ is measurable form a Dynkin system containing rectangles); the metric of $L^1_{\mathrm{loc}}(U)$ and separability give measurability of Γ^r_G. If $F = G(\cdot, Y)$ almost everywhere on U and $|F| \le r$, then $\mathrm{clip}_r(G(\cdot,Y)) = F$ almost everywhere on U. Without the clip the map need not be defined: $G(z,1) = |z|^{-2}$ is not locally integrable at 0. □

Lemma A.3 (Integrated potentials of bounded fields)**.** *Let $U \subseteq \mathbb{R}^2$ be open, $A \subseteq U$ bounded and Lebesgue measurable, $r \ge 0$, and v an $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{B}(\mathbb{R})$-measurable function bounded on $A \times [-r,r]$. Then $\Psi_v(s) = \int_A v(z, s(z))\, dz$ is well defined and Borel on $\mathrm{Ball}_r(U)$.*

Proof. For a representative with values in $[-r,r]$, $z \mapsto (z, s(z))$ is $(\mathcal{L}, \mathcal{L} \otimes \mathcal{B})$-measurable, so $z \mapsto h(z, s(z))$ is Lebesgue measurable, and representatives differ on null sets. For $h = \mathbf{1}_{A'}(z)\varphi(x)$ with φ continuous, the map is continuous on $\mathrm{Ball}_r(U)$: if $s_k \to s$ in $L^1_{\mathrm{loc}}(U)$ within Ball_r and $K \subset U$ is compact with $|A \setminus K| < \delta$, then $\int_A |s_k - s| \le \int_K |s_k - s| + 2r\delta$, so $s_k \to s$ in measure on A and the integrals converge by uniform continuity of φ on $[-r,r]$ (the bound r handles sets A touching ∂U). The class of bounded h for which the map is Borel is a vector space closed under bounded monotone limits and contains a multiplicative generating class; (K1) applies to $v\mathbf{1}_{A\times[-r,r]}$. □

Lemma A.4 (Restrictions and point recovery)**.** *Fix θ and $r \ge 0$. (i) $r_n : \mathrm{Ball}_r(H_+) \to L^1(Q_n)$, $s \mapsto s|_{Q_n}$, is well defined and continuous, and $r_m = \rho_{n,m} \circ r_n$ for $m \le n$ with the continuous restriction $\rho_{n,m}$. (ii) The relative Borel σ-algebra of $\mathrm{Ball}_r(H_+)$ equals $\sigma(r_n : n \ge 1)$, and $\sigma(r_1, \dots, r_n) = \sigma(r_n)$ increases with n. (iii) For $z \in H_+$ with discs $D_j(z)$ of radius $1/j$, $L_z(s) = \limsup_j |D_j(z)|^{-1} \int_{D_j(z)} s$ is Borel on $\mathrm{Ball}_r(H_+)$ with values in $[-r,r]$, and $L_z(s) = c$ if s has a representative equal to c near z.*

Proof. (i) $|s| \le r$ and $|Q_n| < \infty$; if $s_j \to s$ in $L^1_{\mathrm{loc}}(H_+)$ within Ball_r, choose compact $K \subset Q_n$ with $|Q_n \setminus K| < \delta$. (Without the bound the map is undefined: $1/z_1$ is locally integrable on $\{z_1 > 0\}$ but not on $(0,n) \times (-n,n)$.) (ii) Each compact $K \subset H_+$ lies in some Q_n, and $s \mapsto \|s - s_0\|_{L^1(K)}$ is a continuous function of $r_n(s)$; relatively open sets are countable unions of such balls. (iii) For large j the closed disc lies in H_+, the averages are continuous, and a lim sup of countably many Borel maps is Borel. □

A.2 The canonical box Mondrian

For each finite binary tree τ (a finite prefix-closed set of words in which every non-leaf has both children), the event that the born nodes form τ is defined by finitely many Borel inequalities in (C, s) and the ingredients of the nodes of τ, and on it the output is a Borel function of finitely many variables; on the complement of the union of these countably many events (a null set, by the next lemma) set the output to $\{(C, \xi_\emptyset)\}$. For a box $C = [a,b] \times [c,d]$ with $p(C) > 0$, let κ_C be the law of the line which is vertical with probability $(b-a)/p(C)$ at a uniform position in (a,b) and horizontal with probability $(d-c)/p(C)$ at a uniform position in (c,d); for a line λ meeting $\mathrm{int}\, C$, $C_0(\lambda)$ and $C_1(\lambda)$ are the closed left/lower and right/upper pieces.

Lemma A.5 (Nonexplosion)**.** *Almost surely finitely many nodes are born; the mean number of born nodes is at most $2e^{sp(C)} - 1$; a fixed $z \in \mathrm{int}\, C$ is almost surely interior to a cell; and the event that z lies on a cell boundary is Borel in $\mathrm{Part}^{\mathrm{col}}(C)$.*

Proof. A daughter never has larger half-perimeter than its parent, so a born node w at depth k with strict ancestors $w_0, \dots, w_{k-1}$ has $s > b_w = \sum_j E_{w_j}/p(R_{w_j}) \geq \sum_j E_{w_j}/p(C)$. For a fixed word of length k the ancestors' clocks are independent $\mathrm{Exp}(1)$, so with $c = sp(C)$ the mean number of born nodes is at most $\sum_k 2^k \mathbb{P}(\Gamma_k \leq c) = \sum_k 2^k \mathbb{P}(\mathrm{Poi}(c) \geq k) = \sum_j \mathbb{P}(\mathrm{Poi}(c) = j)(2^{j+1} - 1) = 2e^c - 1$, Γ_k a sum of k independent $\mathrm{Exp}(1)$. For a fixed node the cut position is $a + X_w(b-a)$ or $c + X_w(d-c)$ with X_w uniform and independent of the node's box, so it passes through z with probability 0; cell boundaries inside $\operatorname{int} C$ lie on ancestors' cut lines; there are countably many nodes. The boundary event is Borel by (K4). □

Definition A.6 (First-step recursion)**.** Let I be an index set and, for $i \in I$, C_i a box with nonempty interior, $s_i \geq 0$ and Q_i a probability on $\mathrm{Part}^{\mathrm{col}}(C_i)$. A *daughter assignment* gives, for every i, $t \in [0, s_i)$, line λ meeting $\operatorname{int} C_i$ and $\epsilon \in \{0, 1\}$, a probability $Q_{i,t,\lambda,\epsilon}$ on $\mathrm{Part}^{\mathrm{col}}(C_\epsilon(\lambda))$ which is either Q_j for some $j \in I$ with $(C_j, s_j) = (C_\epsilon(\lambda), s_i - t)$ or $M^{\mathrm{col}}(C_\epsilon(\lambda), s_i - t)$, such that $(t, \lambda) \mapsto [Q_{i,t,\lambda,0} \otimes Q_{i,t,\lambda,1}](\cup^{-1} A)$ is Borel for Borel A. The family satisfies the *first-step recursion* if for every i and Borel A,

$$Q_i(A) = e^{-s_i p(C_i)} \mathbb{P}_\xi(\{(C_i, \xi)\} \in A) + \int_0^{s_i} p(C_i) e^{-p(C_i)t} \int \kappa_{C_i}(d\lambda)\, [Q_{i,t,\lambda,0} \otimes Q_{i,t,\lambda,1}](\cup^{-1} A)\, dt,$$

with ξ a uniform sign.

Lemma A.7 (First-step identity and characterization)**.** *(i) The laws $M^{\mathrm{col}}(C, s)$ satisfy the first-step recursion with daughter laws $M^{\mathrm{col}}(C_\epsilon(\lambda), s - t)$. (ii) Every family satisfying the first-step recursion with a daughter assignment has $Q_i = M^{\mathrm{col}}(C_i, s_i)$ for all i. (iii) For T a translation, θ_1 or θ_2, $T_* M^{\mathrm{col}}(T^{-1}C, s) = M^{\mathrm{col}}(C, s)$.*

Proof. (i) The root ingredients are independent of the two subfamilies indexed by $0w$ and $1w$, which are canonical families; $d_\emptyset = E_\emptyset/p(C) \sim \mathrm{Exp}(p(C))$ and $(D_\emptyset, X_\emptyset)$ produce $\lambda \sim \kappa_C$ independent of $d_\emptyset$. On $\{d_\emptyset \geq s\}$ the output is $\{(C, \xi_\emptyset)\}$; on $\{d_\emptyset < s\}$ the subtree rooted at ϵ, with births shifted by $d_\emptyset$, is the construction on $C_\epsilon(\lambda)$ with budget $s - d_\emptyset$; Fubini gives the formula, and joint Borel measurability of the output gives the measurability. (ii) Write $Q^{(k)}(A) = Q(A \cap \mathrm{Part}^{\mathrm{col}}_k)$ for the k-cell part. Cells of the two daughters have disjoint interiors, so the k-cell part of the integrand involves only $Q^{(k_0)} \otimes Q^{(k_1)}$ with $k_0 + k_1 = k$, $k_\epsilon \geq 1$. Induct on k simultaneously for all i: for $k = 1$ only the first term contributes; for $k \geq 2$ every daughter law agrees with M^{col} on fewer than k cells by hypothesis, so $Q_i^{(k)}$ equals the k-cell part of the right side of (i). Both laws are carried by finite partitions (Lemma A.5). (iii) The family $T_* M^{\mathrm{col}}(T^{-1}C, s)$, indexed by all (C, s), satisfies the recursion: T commutes with $\cup$, preserves half-perimeters, maps $\kappa_{T^{-1}C}$ onto κ_C and daughters onto daughters, possibly exchanged, which does not change $[Q_0 \otimes Q_1](\cup^{-1} A)$. Apply (ii). □

Lemma A.8 (Competing clocks, conditional form)**.** *Let $A_1, \dots, A_k$ be independent, $A_j \sim \mathrm{Exp}(\alpha_j)$, and Z a random element independent of them; $\alpha = \sum \alpha_j$, $\tau = \min A_j$, $J = \operatorname{argmin} A_j$. Then $\mathbb{P}(J = j) = \alpha_j/\alpha$, and conditionally on $\{J = j\}$ the random elements τ, $(A_l - \tau)_{l \neq j}$ and Z are independent, with $\tau \sim \mathrm{Exp}(\alpha)$, $A_l - \tau \sim \mathrm{Exp}(\alpha_l)$ and Z distributed as unconditionally.*

Proof. $\mathbb{P}(J = j, \tau \in dt, A_l - \tau > r_l\ (l \neq j), Z \in G) = \alpha_j e^{-\alpha_j t} \prod_{l \neq j} e^{-\alpha_l (t + r_l)} \mathbb{P}(Z \in G)\, dt$. The residual vector is indexed by $l \neq J$ and is in general not independent of J: with $\alpha_1 = 1$, $\alpha_2 = 2$, the losing residual R has $\mathbb{P}(R > 1 \mid J = 1) = e^{-2}$ and $\mathbb{P}(R > 1 \mid J = 2) = e^{-1}$. □

A.3 The core-exterior construction

Definition A.9 (Core-exterior construction)**.** Let $K \subseteq C$ be boxes (K possibly degenerate), C with nonempty interior, $s \geq 0$. Ingredients, all independent: a canonical family $h = (E_w, D_w, X_w, \xi_w)_{w \in \mathrm{Tr}}$ (the *core*) and, for every $w \in \mathrm{Tr}$, side $\sigma \in \{L, R, D, U\}$ and $k \geq 1$, a triple $(E^\sigma_{w,k}, X^\sigma_{w,k}, G^\sigma_{w,k})$ with $E \sim \mathrm{Exp}(1)$, $X \sim U(0,1)$ and G a canonical family. Run the canonical construction on K with h (core boxes K_w, births b_w, deaths d_w, split lines); $e_w = \min(d_w, s)$. Each born core node carries an *enclosing box* $C_w(t) = [a_w(t), b_w(t)] \times [c_w(t), d_w(t)] \supseteq K_w = [a'_w, b'_w] \times [c'_w, d'_w]$, $t \in [b_w, e_w]$, with *gaps* $g^L = a'_w - a_w$, $g^R = b_w - b'_w$, $g^D = c'_w - c_w$, $g^U = d_w - d'_w$; initially $C_\emptyset(0) = C$. On $[b_w, e_w)$ side σ jumps at the times $t^\sigma_{w,k} = t^\sigma_{w,k-1} + E^\sigma_{w,k}/g^\sigma(t^\sigma_{w,k-1})$, $t^\sigma_{w,0} = b_w$ (no jumps while the gap is 0). At a jump time $t < e_w$ the gap g becomes $X^\sigma_{w,k} g$, and the removed slab (the *detached box*, of full current extent in the other coordinate) receives $\Pi^{\mathrm{can}}(\mathrm{slab}, s-t)$ built from $G^\sigma_{w,k}$: on the slab itself for $\sigma = R, U$, and on its image under θ_1 (for L) or θ_2 (for D), mapped back by the same reflection. If $d_w < s$, the core split line of K_w cuts $C_w(d_w)$ and each child receives the piece containing its core box (gap 0 on the side of the cut). The output is $\Pi^{\mathrm{ce}}(C, K, s) = \{(C_w(e_w), \xi_w) : w \text{ a core leaf}\} \cup (\text{all detached colored partitions})$, with law $\mathcal{Q}(C, K, s)$.

Trace data. For $C = [-L, L] \times J_0$ ($L > 0$, J_0 a compact interval of positive length) and $K = \{0\} \times J_0$, all core splits are horizontal, $K_w = \{0\} \times J_w$ and $g^D = g^U = 0$. Write $Rr = (E^R_{w,k}, X^R_{w,k}, G^R_{w,k})_{w,k}$ and Ll likewise. The *straddling cells* are the enclosing boxes, and the trace data are $\mathrm{tr}(h) = ((J_w, b_w, d_w)_{w \text{ born}}, (\xi_w)_{w \text{ core leaf}})$: positions and auxiliary times of the horizontal cuts of straddling cells, and colors of the final straddling cells. For a side family Y (independent triples with the laws above) define widths $x_w(t)$: $x_\emptyset(0) = L$, children inherit $x_w(d_w)$, and on $[b_w, e_w)$ the width jumps from x to $X^Y_{w,k} x$ at times $t^Y_{w,k} = t^Y_{w,k-1} + E^Y_{w,k}/x$; for each jump with $t^Y_{w,k} < e_w$ put $S^Y_{w,k} = [X^Y_{w,k} x, x] \times J_w$, and

$$\Psi(h, Y) = \{([0, x_w(e_w)] \times J_w, \xi_w) : w \text{ core leaf}\} \cup \bigcup_{t^Y_{w,k} < e_w} \Pi^{\mathrm{can}}(S^Y_{w,k}, s - t^Y_{w,k}; G^Y_{w,k}),$$

which depends on h only through $\mathrm{tr}(h)$. The same definitions with coordinates exchanged apply to $C = I_0 \times [-L, L]$, $K = I_0 \times \{0\}$.

Lemma A.10 (The core-exterior construction)**.** *(i) Almost surely the construction has finitely many events, its output is a finite colored partition of C and a Borel function of (C, K, s) and the ingredients; for every born w and $t \in [b_w, e_w]$, $K_w \subseteq C_w(t)$, $C_w(t) \cap K = K_w$, $C_w(t)$ has nonempty interior, and every detached box is disjoint from K. (ii) The family $\mathcal{Q}(C, K, s)$ satisfies the first-step recursion, with daughter laws $\mathcal{Q}(C_\epsilon(\lambda), K \cap C_\epsilon(\lambda), s - t)$ for a line meeting the relative interior of K in its cut coordinate, $\mathcal{Q}(C_\epsilon(\lambda), K, s - t)$ for the daughter containing K and $M^{\mathrm{col}}(C_\epsilon(\lambda), s - t)$ for the other daughter when the line lies in a gap, and M^{col} for the null set of lines through ∂K. (iii) $\mathcal{Q}(C, K, s) = M^{\mathrm{col}}(C, s)$.*

Proof. (i) The core is the canonical construction on K. A gap never exceeds $p(C)$, so inter-jump times are at least $E/p(C)$ and the number of jumps of one gap process in a window of length at most s is dominated by a $\mathrm{Poi}(sp(C))$ variable; each detached output is finite (Lemma A.5); there are countably many items. Measurability: decompose by the finite core tree, the finite numbers and order of side events and the finite detached trees. Invariants by induction: a jump on side R removes the slab $[b'_w + Xg, b'_w + g] \times [c_w, d_w] \subset \{z_1 > b'_w\}$, which misses K because $C_w \cap K = K_w \subseteq \{z_1 \leq b'_w\}$; a core split of K_w by $\{z_1 = x\}$ gives $C_{w0} \cap K = K_w \cap \{z_1 \leq x\} = K_{w0}$; other sides and horizontal splits are analogous; core splits are strictly inside K_w's extent and

gaps are multiplied by $X \in (0,1)$. (ii) The root clocks are the core clock $A_K = E_\emptyset/p(K)$ (if $p(K) > 0$) and the gap clocks $A_\sigma = E^\sigma_{\emptyset,1}/g^\sigma(0)$ (if positive), independent of each other and of all remaining ingredients Z; the rates sum to $p(K) + \sum_\sigma g^\sigma = p(C)$. By Lemma A.8, $\tau \sim \mathrm{Exp}(p(C))$, $\mathbb{P}(J = \text{core}) = p(K)/p(C)$, $\mathbb{P}(J = \sigma) = g^\sigma(0)/p(C)$. The first line is a function of J and one uniform position variable (a component of Z): a core line has law κ_K, a gap line on side R is uniform on (b', b), and so on; hence (τ, λ) has law $p(C)e^{-p(C)t}dt\,\kappa_C(d\lambda)$, and off a null set J is a function of λ. If $\tau \geq s$ the output is $\{(C, \xi_\emptyset)\}$. On a core event at $\tau < s$, the children's futures are the construction on $(C_\epsilon(\lambda), K_\epsilon, s - \tau)$ driven by distinct components of Z, conditionally independent of (τ, λ) with the ingredient law. On a gap event on side σ, the slab receives a canonical output of law $M^{\mathrm{col}}(S, s - \tau)$ (Lemma A.7(iii) for L, D), and the other daughter C' carries the continuation driven by the family Y' in which the core clock is replaced by $p(K)(A_K - \tau)$, side σ by its shifted sequence with the new gap, and each other side σ' by $g^{\sigma'}(0)(A_{\sigma'} - \tau)$; pathwise every later event of the original construction is the corresponding event of the construction on $(C', K, s - \tau)$ with ingredients Y', shifted by τ, and by Lemma A.8 the scaled residuals are fresh Exp(1) variables independent of τ and of Z. Integrating over (τ, λ) gives the recursion. (iii) Apply Lemma A.7(ii) to the family indexed by all triples. □

Corollary A.11 (Restriction and two points)**.** *(i) If $K = B$ has nonempty interior,*

$$\mathrm{res}_B(\Pi^{\mathrm{ce}}(C, B, s)) = \Pi^{\mathrm{can}}(B, s; h),$$

and $(\mathrm{res}_B)_* M^{\mathrm{col}}(C, s) = M^{\mathrm{col}}(B, s)$. (ii) For $z \neq z'$ in C and K the box they span, z and z' lie in one cell of $\Pi^{\mathrm{ce}}(C, K, s)$ if and only if the core root does not split before s; hence for $M^{\mathrm{col}}(C, s)$ the probability that some cell contains both is $e^{-s|z-z'|_1}$.

Proof. (i) $C_w(e_w) \cap B = K_w$ for core leaves and detached cells miss B (Lemma A.10(i)); apply Lemma A.10(iii). (ii) If the root does not split, $z, z' \in K \subseteq C_\emptyset(s)$; if it splits, the line lies strictly between the two points in one coordinate, and every final cell lies in a detached slab (disjoint from K) or in one closed side of the line. $p(K) = |z - z'|_1$, and the event is Borel. □

Corollary A.12 (Trace decomposition and box positivity)**.** *Let $C = [-L, L] \times J_0$ and $K = \{0\} \times J_0$. (i) $g^D = g^U = 0$ for all nodes and times; the right and left gaps evolve as the widths x_w of Definition A.9 driven by Rr and by Ll respectively. (ii) $\mathrm{res}_{[0,L]\times J_0}(\Pi^{\mathrm{ce}}) = \Psi(h, Rr)$ and $\theta_1 \mathrm{res}_{[-L,0]\times J_0}(\Pi^{\mathrm{ce}}) = \Psi(h, Ll)$. (iii) h, Rr, Ll are independent, $Rr \overset{d}{=} Ll$, and $\Psi(h, Y) = \hat{\Psi}(\mathrm{tr}(h), Y)$ for a Borel $\hat{\Psi}$. (iv) For N independent copies and bounded complex Borel Φ on $\mathrm{Part}^{\mathrm{col}}([0, L] \times J_0)^N$, $\mathbb{E}[\overline{\Phi((\Psi(h^i, Ll^i))_i)}\Phi((\Psi(h^i, Rr^i))_i)] = \mathbb{E}|\bar{\Phi}(h^1, \ldots, h^N)|^2$ with $\bar{\Phi} = \int \Phi((\Psi(h^i, y^i))_i) \prod_i P_{Rr}(dy^i)$, a function of the trace data. (v) For independent $\Pi^1, \ldots, \Pi^N$ with law $M^{\mathrm{col}}(C, s)$, $T^i_+ = \mathrm{res}_{[0,L]\times J_0}(\Pi^i)$ and $T^i_- = \theta_1 \mathrm{res}_{[-L,0]\times J_0}(\Pi^i)$, $\mathbb{E}[\overline{\Phi(T^1_-, \ldots, T^N_-)}\Phi(T^1_+, \ldots, T^N_+)] \geq 0$; the same holds across θ_2 for $C = I_0 \times [-L, L]$.*

Proof. (i) The vertical extents of C and K coincide initially, core lines are horizontal (K has zero width) and cut C_w and K_w at the same height, and side jumps L, R do not change vertical extents; at a horizontal straddling cut both daughters inherit the current widths and then use their own triples. (ii) Final straddling cells are $[-l_w(e_w), r_w(e_w)] \times J_w$ and restrict to $[0, r_w(e_w)] \times J_w$; R-slabs $[Xr, r] \times J_w$ carry $\Pi^{\mathrm{can}}(\cdot; G^R)$ and L-slabs carry $\theta_1 \Pi^{\mathrm{can}}([Xl, l] \times J_w; G^L)$; cells of one half-box meet the other in at most a segment and are discarded by the restriction. (iii) The ingredients are independent by definition, and Ψ uses h only through the born-node data and the leaf colors; measurability as in Lemma A.10(i). (iv) By Fubini, $\mathbb{E}[\Phi((\Psi(h^i, Rr^i))_i) \mid h] = \bar{\Phi}(h)$ and $\mathbb{E}[\overline{\Phi((\Psi(h^i, Ll^i))_i)} \mid h] = \overline{\bar{\Phi}(h)}$,

and the two are conditionally independent given h. (v) By Lemma A.10(iii) the law of $(T^i_+, T^i_-)_i$ is that of N independent copies of $(\Psi(h^i, Rr^i), \Psi(h^i, Ll^i))$; apply (iv). Only the complete trace data are used; the finite-box hand checks of the endpoint record exhibit loss of conditional independence when either the auxiliary times or the positions are dropped from the conditioning. □

Proof of Theorems 6.2 and 6.3. Theorem 6.2: (a) Lemma A.5 and the measurability paragraph before it; (b) Lemma A.7(i),(ii); (c) Lemma A.7(iii); (d) Corollary A.11(i); (e) Corollary A.11(ii) and Lemma A.5. Theorem 6.3: Lemma A.10(iii), Corollary A.11(i) and Corollary A.12. □

A.4 The whole-plane Mondrian neuron

Proof of Theorem 6.4(a). The bonding maps $\pi_n = \mathrm{res}_{B_n}$ are Borel with $(\pi_n)_* M^{\mathrm{col}}(B_{n+1}, \mu) = M^{\mathrm{col}}(B_n, \mu)$ (Corollary A.11(i)); drawing $P_m \sim M^{\mathrm{col}}(B_m, \mu)$ and setting $P_n = \pi_n \circ \cdots \circ \pi_{m-1}(P_m)$ gives consistent finite-dimensional laws, and (K5) gives a probability on $\prod_n \mathrm{Part}^{\mathrm{col}}(B_n)$, under which the Borel set S of consistent sequences has full measure. For any consistent tessellation $(P_1, \ldots, P_m)$ is almost surely the image of P_m under the bonding maps, so the law is unique (K1). Restriction is transitive ($(R \cap B_n) \cap B = R \cap B$, and a nonempty interior of $R \cap B$ forces one of $R \cap B_n$), so $\mathrm{res}_B(P_m) = \mathrm{res}_B(P_n)$ on S, with law $M^{\mathrm{col}}(B, \mu)$. For $T = \theta_j$, $(TP)_n = TP_n$ is consistent because $TB_n = B_n$, with marginals $M^{\mathrm{col}}(B_n, \mu)$ (Lemma A.7(iii)); for a translation by x and $m = n + \lceil |x|_\infty \rceil$, $(TP)_n = \mathrm{res}_{B_n}(TP_m)$ does not depend on the admissible m, is consistent, and has law $M^{\mathrm{col}}(B_n, \mu)$; uniqueness gives $TP \overset{d}{=} P$. The last sentence of (a) is part (A) of the next proof. □

Proof of Theorem 6.4(b),(c). (A) Let $P \in \mathrm{Part}^{\mathrm{col}}(D)$, $D = [a_D, b_D] \times [c_D, d_D]$ with nonempty interior, and $z \in [a_D, b_D) \times [c_D, d_D)$. The points $z + e(1,1)$, $0 < e < e_0$, lie in $\mathrm{int}\, D$ and avoid the finitely many cell boundaries (an axis-parallel segment meets the open diagonal ray in at most one point), so they lie in the interior of one cell R, and then $z \in R^{\mathrm{ll}}$; conversely $z \in R^{\mathrm{ll}}$ forces $z + e(1,1) \in \mathrm{int}\, R$ for small e. So the lower-left representatives partition $[a_D, b_D) \times [c_D, d_D)$. (B) On the consistency event, for $z \in [-n, n)^2$: $R_n(z) = R_{n+1}(z) \cap B_n$, $\xi_n(z) = \xi_{n+1}(z)$ and $R_n(z)^{\mathrm{ll}} \subseteq R_{n+1}(z)^{\mathrm{ll}}$, since z lies in the representative of $R_{n+1}(z) \cap B_n$, a cell of P_n with the same color; uniqueness in (A) gives the claim. (C) For every outcome $g(z) = \sum_n \mathbf{1}\{n_z = n\} \int hh(z, r)\, P_n(dr)$ with $hh(z, (a, b, c, d, \xi)) = (\xi/2)\mathbf{1}\{a \le z_1 < b, c \le z_2 < d\}$, exactly one atom contributing by (A); so g is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable (Lemma A.1(iii)) with $|g| = 1/2$. (D) If $R(z) \not\subseteq z + (-m, m)^2$, some $R_n(z)$ contains one of $z \pm me_1$, $z \pm me_2$; the events that some cell of P_n contains two fixed points increase in n and have probability $e^{-\mu|y-y'|_1}$ (Corollary A.11(ii)); so $\mathbb{P}(R(z) \not\subseteq z + (-m, m)^2) \le 4e^{-\mu m}$ and almost surely every $R(q)$, $q \in \mathbb{Q}^2$, is bounded. Every z has a rational q in the interior of $R_{n_z}(z)^{\mathrm{ll}}$, and then $R(q) = R(z)$, $H(q) = H(z)$. (E) If $R(z) \subseteq B_k$, $k \ge n_z$, then $R_n(z) = R_{n+1}(z)$ for $n \ge k$, so $R(z) = R_k(z)$ and $H(z) = R(z)^{\mathrm{ll}}$. If $w \in H(z) \cap H(z')$, then $R_n(z) = R_n(z') = R_n(w)$ for large n; so distinct representatives are disjoint and cover $\mathbb{R}^2$; $R(z) = \mathrm{cl}\, H(z)$ and $\mathrm{int}\, R(z) = \mathrm{int}\, H(z)$; a representative meeting B_k contains a representative of a cell of P_{k+1}, so finitely many meet B_k; $g = \xi(z)/2$ is constant on $H(z)$. (F) Under $M^{\mathrm{col}}(B_n, \mu)$ the colors are independent uniform given the uncolored partition, so $\mathbb{E}g(z) = 0$ and $\mathbb{E}g(z)g(z') = \frac{1}{4}\mathbb{P}(R_n(z) = R_n(z')) = \frac{1}{4}e^{-\mu|z-z'|_1}$, using that z, z' are almost surely interior to cells (Lemma A.5). □

Proof of Theorem 6.4(d). On the consistency event, if z is interior to a cell of P_{n_z+1}, that cell is the only one containing z, its color is $\xi(z)$, and every admissible convention gives $g'(z) = g(z)$; the exceptional points lie on countably many segments. Let $Y_{\mathrm{sp}} = \prod_n \mathrm{Part}^{\mathrm{col}}(B_n)$ and $G(z, (P'_n)_n) = \sum_n \mathbf{1}\{n_z = n\} \int hh(z, r)\, P'_n(dr)$. Then G is product measurable and $|G| = 1/2$ on all of $\mathbb{R}^2 \times Y_{\mathrm{sp}}$,

consistent or not, because the representatives of any element of $\mathrm{Part}^{\mathrm{col}}(B_{n_z})$ partition $[-n_z, n_z)^2$ (part (A) above uses no consistency). So $\Gamma_G^{1/2}$ of Lemma A.2(iii) is a total measurable map with $[g(\cdot,\omega)|_U] = \Gamma_G^{1/2}((P_n(\omega))_n)$ for every outcome, and the law of $g|_U$ is the image of the unique law of $(P_n)_n$. An admissible g' is clipped to $F' = \mathrm{clip}_{1/2}(g')$, which equals g' on the consistency event and g almost everywhere there, so $g'|_U$ has the same law. For $T = \theta_j$, the field of TP equals $g \circ \theta_j$ off a null set on the consistency event and is $G(\cdot, (TP_n)_n)$, so $g \circ \theta_j \overset{d}{=} g$ because $TP \overset{d}{=} P$; translations are the same with $(TP)_n = \mathrm{res}_{B_n}(TP_m)$. □

A.5 The product neuron

Proof of Theorem 6.1. (a) The Poisson process $\mathcal{N}$ is almost surely locally finite, has no atom at 0 and has unbounded atoms in both directions; then σ is piecewise constant and càdlàg, and $|\sigma| = 1$. $\mathbb{E}\sigma(t) = 0$; for $t < t'$, $\sigma(t)\sigma(t') = (-1)^{\mathcal{N}((t,t'])}$ in each sign configuration, so $\mathbb{E}\sigma(t)\sigma(t') = e^{-2(\mu/2)|t-t'|}$ (K6); given $\sigma(t_1)$, parities on disjoint later intervals are independent, which gives the Markov chain with transition probabilities $(1 + e^{-\mu u})/2$.

(b) A right-continuous process is jointly measurable ($\sigma(t) = \lim_k \sigma(2^{-k}\lceil 2^k t\rceil)$), so g is $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable, $|g| = 1/2$, $\mathbb{E}g = 0$ and $\mathbb{E}g(z)g(z') = \frac{1}{4}e^{-\mu|z_1 - z_1'|}e^{-\mu|z_2 - z_2'|}$; g is constant on the boxes $[t_i, t_{i+1}) \times [y_j, y_{j+1})$ of the jump points.

(c) Across θ_1 let $b = (\epsilon_\sigma, \tau)$, $M_+ = \mathcal{N}|_{(0,\infty)}$ and M_- the image of $\mathcal{N}|_{(-\infty,0)}$ under $t \mapsto -t$. These are random elements of the space N of $\{0, 1, \ldots, \infty\}$-valued measures on $(0,\infty)$ (with the σ-algebra of the maps $m \mapsto m(A)$); by (K6) they are independent with the same law P_{M} and independent of b. Let $D_\pm$ be the càdlàg functions $\mathbb{R} \to \{-1, 1\}$ with the evaluation σ-algebra; $(t, x) \mapsto x(t)$ is product measurable by right continuity. Put $Y_{\mathrm{prod}} = \{-1, 1\} \times D_\pm \times \mathsf{N}$, $\mathrm{Good}_+ = \{m : m((0,k]) < \infty \ \forall k\}$, $\mathrm{par}(m, t) = (-1)^{m((0,t])}$ on Good_+ and 1 otherwise, and $G_{\mathrm{prod}}(z, (e, x, m)) = \frac{1}{2}e\,\mathrm{par}(m, z_1)x(z_2)$ for $z_1 > 0$, 0 otherwise. Then G_{prod} is product measurable with $|G_{\mathrm{prod}}| \le 1/2$ everywhere, and Lemma A.2(iii) gives a total measurable $\Psi_{\mathrm{prod}} : Y_{\mathrm{prod}} \to \mathrm{Ball}_{1/2}(H_+)$. On the almost sure event of (a), $s_+ = G_{\mathrm{prod}}(\cdot, (\epsilon_\sigma, \tau, M_+))$ on H_+, and $s_-(z) = \frac{1}{2}\epsilon_\sigma(-1)^{\mathcal{N}((-z_1,0])}\tau(z_2)$ differs from $G_{\mathrm{prod}}(z, (\epsilon_\sigma, \tau, M_-))$ only on countably many vertical lines; so $[s_\pm] = \Psi_{\mathrm{prod}}(b, M_\pm)$ almost surely, and Fubini gives $\mathbb{E}[f(s_+)f'(s_-) \mid b] = \bar{f}(b)\bar{f}'(b)$ with $\bar{f}(b) = \int f(\Psi_{\mathrm{prod}}(b, m))P_{\mathsf{M}}(dm)$. Across θ_2 exchange σ and τ.

(d) For independent neurons, $(b_i)_i$, $(M_+^i)_i$, $(M_-^i)_i$ are independent families, the last two independent and identically distributed with the same law, and $S_\pm = N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(b_i, M_\pm^i)$ almost surely; every value of $N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(y_i)$ lies in $\mathrm{Ball}_{\sqrt{N}/2}(H_+)$. The same Fubini computation with the complete tuple $(b_1, \ldots, b_N)$ gives conditional independence and equal conditional laws. With only b_1 given and $N = 2$, neuron 2 is shared by both sides: $\mathrm{Cov}(F_2(t, 0), F_2(-t, 0) \mid b_1) = e^{-2\mu t}/8 > 0$ for $t > 0$, and point values are almost surely Borel functions of the sides (proof of Theorem 6.5(c) below), so the sides are not conditionally independent given b_1 alone. □

A.6 Finite-width annealed reflection positivity

Fix b, $N \ge 1$, θ, Ω and v as in Theorem 6.5; $|F_N| \le r_N$ everywhere.

Lemma A.13 (Projections)**.** *If $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \cdots$ are σ-algebras, $\mathcal{G}_\infty = \sigma(\bigcup_n \mathcal{G}_n)$ and $Y \in L^2(\mathcal{G}_\infty)$, then $\mathbb{E}[Y \mid \mathcal{G}_n] \to Y$ in L^2.*

Proof. The conditional expectations are orthogonal projections onto increasing closed subspaces V_n;

the events $A \in \mathcal{G}_\infty$ with $\mathbf{1}_A$ in the closure of $\bigcup V_n$ form a Dynkin system containing the π-system $\bigcup \mathcal{G}_n$, so the closure is $L^2(\mathcal{G}_\infty)$; for $W \in V_m$ with $\|Y-W\| < e$ and $n \geq m$, $\|Y-\mathbb{E}[Y \mid \mathcal{G}_n]\| \leq \|Y-W\|$. □

Lemma A.14 (Weight factorization and reduction to the sides)**.** *Let $\Psi_+(s) = \int_{\Omega\cap H_+} v(z, s(z))\, dz$ on $\mathrm{Ball}_{r_N}(H_+)$ (Borel by Lemma A.3). Then $U = \Psi_+(S_+) + \Psi_+(S_-)$ for every outcome. For bounded complex Σ_+^θ-measurable F, with the representation of Proposition 5.1(x), $\Phi_F(s) = g((\int f_k s)_k)$ is bounded Borel on $\mathrm{Ball}_{r_N}(H_+)$, $F(\iota F_N) = \Phi_F(S_+)$, $F(\theta\iota F_N) = \Phi_F(S_-)$, and $\int_X \overline{F(\theta v)} F(v)\, P_N^v(dv) = Z^{-1}\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)]$ with $\Phi = \Phi_F e^{-\Psi_+}$.*

Proof. The line ∂H_+ is null; the substitution $z' = \theta z$, $\theta(\Omega\cap H_+) = \Omega\cap H_-$ and $v(\theta z', x) = v(z', x)$ give $\Psi_+(S_-) = \int_{\Omega\cap H_-} v(z', F_N(z'))\, dz'$. For the second part, $\int f_k F_N = \int f_k S_+$ and $\int f_k (F_N \circ \theta) = \int f_k S_-$ because $\mathrm{supp}\, f_k \subset H_+$, and $\theta\iota F_N = \iota(F_N \circ \theta)$ (Proposition 5.1(vii)). □

Proposition A.15 (Unweighted side positivity)**.** *$(S_+, S_-) \overset{d}{=} (S_-, S_+)$, and $\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)] \geq 0$ for every bounded complex Borel Φ on $\mathrm{Ball}_{r_N}(H_+)$.*

Proof. Product neurons. By Theorem 6.1(c),(d), $S_\pm = N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(b_i, M_\pm^i)$ with independent families and equal laws of the last two, which gives the symmetry, and Fubini gives $\mathbb{E}[\overline{\Phi(S_-)}\Phi(S_+)] = \mathbb{E}|\bar\Phi(b_1, \dots, b_N)|^2$ with $\bar\Phi(b) = \int \Phi(N^{-1/2}\sum_i \Psi_{\mathrm{prod}}(b_i, y_i)) \prod_i P_{\mathsf{M}}(dy_i)$.

Mondrian neurons. Step 1: $g_i \circ \theta \overset{d}{=} g_i$ (Theorem 6.4(d)), so $F_N \circ \theta \overset{d}{=} F_N$, and the continuous map $w \mapsto (w|_{H_+}, (w \circ \theta)|_{H_+})$ gives the symmetry. Step 2: take $\theta = \theta_1$, $B_n^+ = [0, n] \times [-n, n]$, $Q_n = (0, n) \times (-n, n)$. On the consistency events, off the null cell boundaries, $g_i = \mathrm{fld}(\mathrm{res}_{B_n^+}(P_n^i))$ and $g_i \circ \theta_1 = \mathrm{fld}(\theta_1 \mathrm{res}_{[-n,0]\times[-n,n]}(P_n^i))$ on Q_n; so $r_n(S_\pm) = \Lambda_n((T_{\pm,n}^i)_i)$ with the Borel map $\Lambda_n((P'^i)_i) = N^{-1/2}\sum_i \mathrm{fld}(P'^i)|_{Q_n}$, and Corollary A.12(v) with $C = B_n$ gives $\mathbb{E}[\overline{\Phi_n(r_n S_-)}\Phi_n(r_n S_+)] \geq 0$ for bounded Borel Φ_n on $L^1(Q_n)$. Step 3: by Lemma A.4(ii) the σ-algebras $\mathcal{G}_n = \sigma(r_n(S_+))$ increase and generate $\sigma(S_+)$; with $Y = \Phi(S_+)$ and $\mathbb{E}[Y \mid \mathcal{G}_n] = \Phi_n(r_n S_+)$ (K2; truncated to $|\Phi_n| \leq \sup|\Phi|$), Lemma A.13 gives $\Phi_n(r_n S_+) \to \Phi(S_+)$ in L^2, Step 1 gives the same for the minus side, and Cauchy–Schwarz passes the inequality to the limit. □

Proof of Theorem 6.5. (a) $(z, \omega) \mapsto v(z, F_N(z, \omega))$ is $\mathcal{L}(\mathbb{R}^2) \otimes \mathcal{A}$-measurable, so U is a random variable with $|U| \leq |\Omega| M_N$ and the stated bounds on Z (no bound $Z \geq 1$ is claimed). F_N is a Borel random element of $\mathrm{Ball}_{r_N}(\mathbb{R}^2)$ and of E (Lemma A.2), and on Ball_{r_N} the inclusion into E is continuous ($|s_k - s|^4 \leq (2r_N)^3 |s_k - s|$), so ιF_N is Borel and P_N^v is a Borel probability. $U = \Psi_v(F_N)$ (Lemma A.3), so P_N^v depends only on the law of F_N, which is the image of the N-fold product of one neuron's law under the continuous sum map; that law is determined by the finite-dimensional laws for product neurons (Lemma A.2(ii)) and by Theorem 6.4(d) for Mondrian neurons. Admissible conventions change F_N only on a null set of z, on the almost sure intersection of the consistency events, so U and ιF_N are unchanged almost surely. (b) Lemma A.14 gives $e^{-U}\overline{\Phi(S_-)}\Phi(S_+) = \overline{\Phi'(S_-)}\Phi'(S_+)$ with $\Phi' = \Phi e^{-\Psi_+}$; apply Proposition A.15, and for X-observables Lemma A.14 and $Z > 0$. (c) The first assertion is the computation in (b). For each fixed z, almost surely F_N is constant near z (for Mondrian neurons z is interior to a cell of every P_n^i with $z \in \mathrm{int}\, B_n$; for product neurons z_1, z_2 are not atoms of the locally finite Poisson processes). Hence, almost surely for all k, $L_{z_k}(S_+) = F_N(z_k)$ and $L_{z_k}(S_-) = F_N(\theta z_k)$ (Lemma A.4(iii)), so $Y = \Phi(S_+)$ and $Y^\theta = \Phi(S_-)$ with the bounded Borel $\Phi = \tilde{Y} \circ (L_{z_k})_k$. If Φ' comes from another representation, $\Phi(S_+) = \Phi'(S_+)$ almost surely, and $S_- \overset{d}{=} S_+$ gives $\Phi(S_-) = \Phi'(S_-)$ almost surely. □

Proof of Corollaries 6.6 and 6.7. Corollary 6.6: each coordinate interval of Λ and Ω is symmetric about 0, so both rectangles are invariant under θ_1 and θ_2; v is Borel, bounded on $\Omega \times [-r, r]$ by $\max_{|x|\le r}(|q_{\text{in}}| + |q_{\text{out}}| + \kappa^2 x^2/2)$ and invariant under both reflections on Ω; Theorem 6.5 applies to each θ separately. For a physical square, $\bar{\Lambda} = [-l, l]^2$ and $\Omega = [-R, R]^2$ are such rectangles; off $\partial\Lambda$, $\mathbf{1}_\Lambda = \mathbf{1}_{\bar{\Lambda}}$ and $\mathbf{1}_{\Omega\setminus\Lambda} = \mathbf{1}_{\Omega\setminus\bar{\Lambda}}$, so v_V agrees with the displayed v off the null set $\partial\Lambda \times \mathbb{R}$, and for every outcome $\int_\Omega v_V(z, F_N(z))\, dz = V_{\varepsilon,\Omega,\underline{a};\Lambda}(F_N)$, which is also the U of the displayed v. Axis-parallelism is used exactly in the identification $\bar{\Lambda} = [-l, l]^2$; for the rotated square of Remark 6.8(i) no conclusion, positive or negative, is drawn. The hypotheses "Ω centered" and "Λ centered" are both used: an outer rectangle that is invariant under θ_1 only does not satisfy the hypothesis for θ_2, and no claim is made for it. Corollary 6.7: the hypotheses of Theorem 6.5 hold for each θ. □

Remark A.16 (What the endpoint results do not contain). The conditional identities of Corollary A.12 are proved on the extended probability space of the core-exterior construction in a box; the auxiliary times are not functions of the final tessellation, and no whole-plane auxiliary-time history is constructed. Not needed and not proved: the Poisson law of trace cut points; reflection positivity across shifted or rotated lines; any Euclidean invariance of tilted laws; reflection positivity of the limits in N, ε, Ω or a by these arguments; and reflection positivity, or its failure, for tilts built on rotated squares that are not invariant under both reflections. The measurability hypothesis is used exactly in Lemma A.3 and in part (a) of Theorem 6.5; the broader reading of Remark 6.8(iii) fails.

B Proofs for the Gaussian reference fields

Besides C6(c)–(g) the following elementary Gaussian facts are used: an L^2-limit of centered jointly Gaussian vectors is centered jointly Gaussian; uncorrelated jointly Gaussian families are independent; for $\xi \sim N(0, \lambda)$, $\lambda > 0$, $t\lambda < 1$: $\mathbb{E}e^{t\xi^2/2} = (1 - t\lambda)^{-1/2}$, and the law with density proportional to $e^{tx^2/2}$ relative to $N(0, \lambda)$ is $N(0, \lambda/(1 - t\lambda))$; a centered Gaussian of variance v has $\mathbb{E}|Y|^{2n} = (2n - 1)!!v^n$; and *regression*: if $(Y, \xi_1, \dots, \xi_N)$ is centered jointly Gaussian, $Y \in \mathbb{R}^n$, with independent ξ_j of variances $\lambda_j > 0$, then $Y - \sum_j \mathbb{E}[Y\xi_j]\xi_j/\lambda_j$ is independent of (ξ_j) with covariance $\operatorname{Cov} Y - \sum_j \mathbb{E}[Y\xi_j]\mathbb{E}[Y\xi_j]^T/\lambda_j$. The path space $C(\mathbb{R}^2)$ carries the metric of uniform convergence on compacts; it is separable, its Borel σ-algebra is generated by the evaluations at rational points, and two Borel probabilities on it with the same finite-dimensional laws at rational points coincide.

Lemma B.1 (Spectral point fields)**.** *Let ν be an even σ-finite Borel measure on $\mathbb{R}^2$, $H_\nu = \{h \in L^2(\nu; \mathbb{C}) : h(-p) = \overline{h(p)}\}$ with $\langle g, h\rangle = \int \bar{g}h\, d\nu$, W an isonormal process on H_ν, $\sigma \ge 0$ even and Borel with $S_\sigma = \int \sigma^2 d\nu < \infty$, and $X(z) = W(e_z\sigma)$.*

(S0) H_ν is a separable real Hilbert space.

(S1) X is centered Gaussian with $\mathbb{E}[X(z)X(z')] = k(z - z')$, $k(x) = \int \cos(p \cdot x)\sigma^2 d\nu$, real, even, continuous, $|k| \le k(0) = S_\sigma$; and $\mathbb{E}[X(z)W(h)] = \int e^{ip\cdot z}\sigma h\, d\nu$.

(S2) $\mathbb{E}|X(z + h) - X(z)|^2 = \int 2(1 - \cos(p \cdot h))\sigma^2 d\nu = 2(k(0) - k(h))$.

(S3) If there are constants $M > 0$ and $\vartheta > 0$ with $2(k(0) - k(h)) \le M|h|^\vartheta$ for all $h \in \mathbb{R}^2$, there is a $\mathcal{B}(\mathbb{R}^2) \otimes \mathcal{F}$-measurable $\tilde{X}$ with continuous paths for every outcome, $\mathbb{P}(\tilde{X}(z) = X(z)) = 1$ for every z, which is a Borel $C(\mathbb{R}^2)$-valued random element whose law is the unique Borel law with centered Gaussian coordinates of covariance $k(z - z')$.

(S4) Under (S3), for real $g \in L^1(\mathbb{R}^2)$: $\hat{g}\sigma \in H_\nu$, $\int |g||\tilde{X}| < \infty$ almost surely, and $\int g\tilde{X}\,dz = W(\hat{g}\sigma)$ almost surely.

Proof. (S0) $p \mapsto -p$ preserves ν, so $\langle g, h\rangle$ is real; H_ν is closed in $L^2(\nu)$ (an a.e.-convergent subsequence preserves the symmetry) and separable. (S1) $e_z\sigma \in H_\nu$; isometry gives $\mathbb{E}[X(z)X(z')] = \int e^{ip\cdot(z-z')}\sigma^2 d\nu$, whose sine part vanishes by evenness; continuity by dominated convergence; all $X(z)$ lie in the range of W. (S2) $|e^{-it}-1|^2 = 2(1-\cos t)$. (S3) Increments are Gaussian with $\mathbb{E}|X(z+h)-X(z)|^{2n} \le (2n-1)!!M^n|h|^{n\vartheta}$; for an integer $n > 2/\vartheta$ (which exists because $\vartheta > 0$) the continuity criterion C6(d) applies with $\alpha = 2n$, $\beta = n\vartheta - 2 > 0$. The applications below use $(M,\vartheta) = (\mu/\sqrt{2}, 1)$, $(M'_{a,1/2}, 1/2)$ and $(1/(4\pi), 2)$, all with $M, \vartheta > 0$. Joint measurability: the dyadic approximations $\tilde{X}(2^{-l}\lfloor 2^l z\rfloor, \omega)$ are product measurable and converge everywhere by continuity. Borel measurability and the law follow from the path-space facts. (S4) $|\hat{g}| \le \|g\|_1$ and $\hat{g}(-p) = \overline{\hat{g}(p)}$; $\mathbb{E}\int |g||\tilde{X}| \le \|g\|_1 S_\sigma^{1/2}$ and $\mathbb{E}(\int |g||\tilde{X}|)^2 \le \|g\|_1^2 S_\sigma$, which justify Fubini in: $\mathbb{E}I^2 = \iint g(z)g(z')k(z-z') = \int |\hat{g}|^2\sigma^2 d\nu$, $\mathbb{E}[IW(\hat{g}\sigma)] = \int g(z)\mathbb{E}[X(z)W(\hat{g}\sigma)]dz = \int |\hat{g}|^2\sigma^2 d\nu$ and $\mathbb{E}W(\hat{g}\sigma)^2 = \int |\hat{g}|^2\sigma^2 d\nu$, with $I = \int g\tilde{X}$; hence $\mathbb{E}|I - W(\hat{g}\sigma)|^2 = 0$. □

Proof of Theorem 7.1. Take $\nu = (2\pi)^{-2}\,dp$ and $\sigma = (\hat{C}_\mu^{\mathrm{end}})^{1/2}$. In one variable $\int e^{-iwt}e^{-\mu|t|}dt = 2\mu/(\mu^2+w^2)$, so $\hat{C}_\mu^{\mathrm{end}}$ is as in (1), $\int \hat{C}_\mu^{\mathrm{end}}\,dp = \pi^2$, $S_\sigma = 1/4$, and by Fourier inversion $k = C_\mu^{\mathrm{end}}$. Then $2(k(0)-k(h)) = (1-e^{-\mu|h|_1})/2 \le \mu|h|_1/2 \le \mu|h|/\sqrt{2}$, and Lemma B.1(S3) with $\vartheta = 1$ gives γ_μ and the law G_μ. □

Proof of Theorem 7.2. (a) $\hat{C}_a \le \hat{C}_0 \le m^{-2}$; since $a^2m^2 < 1$, termwise $m^2(1+a^2w^2)(1+a^2q^2) \le m^2+w^2+q^2+a^2w^2q^2$, which gives the product bound and $\int \hat{C}_a \le m^{-2}\pi^2/a^2$; $\int_{\mathbb{R}} dw/(A+Bw^2) = \pi/\sqrt{AB}$ with $A = m^2+q^2$, $B = 1+a^2q^2$ gives (2), whose integrand is at most $1/m$ and at most $1/(aq^2)$. (b) $1/\hat{C}_\mu^{\mathrm{end}} = (\mu^2+w^2)(\mu^2+q^2)/\mu^2 = \mu^2+w^2+q^2+a^2w^2q^2$, and $\hat{C}_\mu^{\mathrm{end}} \le \mu^{-2}$ gives $\kappa_a^2\hat{C}_\mu^{\mathrm{end}} \le r_a < 1$, so the geometric series converges pointwise. (c) $s_0 = C_\mu^{\mathrm{end}}$ is continuous, positive, $\sup s_0 = 1/4$, $\|s_0\|_1 = \mu^{-2}$; inductively $s_{k+1} = \kappa_a^2 C_\mu^{\mathrm{end}} * s_k$ has $\sup s_{k+1} \le r_a \sup s_k$, $\|s_{k+1}\|_1 = r_a\|s_k\|_1$ (Tonelli) and is continuous (continuity of translation in L^1); $\hat{s}_k = \kappa_a^{2k}(\hat{C}_\mu^{\mathrm{end}})^{k+1}$. The sum $S = \sum s_k$ converges uniformly and in L^1; $\hat{S} = \hat{C}_a$ by (b), and inversion gives $S = C_a$. Hence $C_a \ge C_\mu^{\mathrm{end}} > 0$, $|C_a| \le C_a(0) = c_a$, $\|C_a\|_1 = \hat{C}_a(0) = m^{-2}$, and $\|C_a\|_2^2 = (2\pi)^{-2}\int \hat{C}_a^2 \le (2\pi)^{-2}\int (m^2+|p|^2)^{-2}dp = 1/(4\pi m^2)$. (d) Convolution by C_μ^{end} is the multiplier $\hat{C}_\mu^{\mathrm{end}}$ with norm $\hat{C}_\mu^{\mathrm{end}}(0) = \mu^{-2}$; Young's inequality and the uniform-and-L^1 convergence give $C_a = \sum_k \kappa_a^{2k}(C_\mu^{\mathrm{end}})^{k+1}$ in operator norm, $= C_\mu^{\mathrm{end}}(I - \kappa_a^2 C_\mu^{\mathrm{end}})^{-1}$; the quadratic form is $(2\pi)^{-2}\int |\hat{f}|^2\hat{C}_a \in [0, m^{-2}\|f\|_2^2]$. (e) $2(c_a - C_a(h)) = (2\pi)^{-2}\int 2(1-\cos(p\cdot h))\hat{C}_a\,dp$. Use $1-\cos t \le \min(2,|t|)$, $\min(2,|\alpha+\beta|) \le \min(2,|\alpha|)+\min(2,|\beta|)$, the product bound of (a) and $\int (1+a^2q^2)^{-1}dq = \pi/a$: the right side is at most $(2\pi)^{-2}2m^{-2}(\pi/a)[I_a(h_1)+I_a(h_2)]$ with $I_a(s) = \int \min(2,|w||s|)(1+a^2w^2)^{-1}dw$. Splitting at $|w| = 2/|s|$, $I_a(s) \le (|s|/a^2)[\log(1+4a^2/s^2)+2]$, and the prefactor is $(2\pi m^2a^3)^{-1}$; this is $\varpi_a(h)^2$, for every h. (f) For $y \ge 0$ and $\beta \in (0,1]$, $\log(1+y) \le y^\beta/\beta$ (the difference has nonnegative derivative). With $\beta = (1-\alpha)/2$, $|h_i|\log(1+4a^2/h_i^2) \le 2(2a)^{1-\alpha}|h_i|^\alpha/(1-\alpha)$; for $|h| \le 1$, $2|h_i| \le 2|h_i|^\alpha$ and $\sum_i |h_i|^\alpha \le 2|h|^\alpha$, giving $M_{a,\alpha}|h|^\alpha$; for $|h| \ge 1$, $C_a(h) \ge 0$ gives $2(c_a - C_a(h)) \le 2c_a|h|^\alpha$. □

Proof of Theorem 7.3. Let B be the L^2 inverse transform of $(\hat{C}_\mu^{\mathrm{end}})^{1/2}$: real, even, $\|B\|_2^2 = 1/4$, and $B * B = C_\mu^{\mathrm{end}}$ pointwise. $(Tf)(u) = \int_\Omega B(u-x)f(x)\,dx$ is Hilbert–Schmidt from $L^2(\Omega)$ to $L^2(\mathbb{R}^2)$ with $\|T\|_{\mathrm{HS}}^2 = |\Omega|/4$, and Fubini gives $\langle Tf, Tg\rangle = \langle f, K_\Omega g\rangle$, so $K_\Omega = T^*T$ is positive and trace class with $\operatorname{tr} K_\Omega = |\Omega|/4$. $\langle f, K_\Omega f\rangle = \langle R^*f, C_\mu^{\mathrm{end}}R^*f\rangle \le \mu^{-2}\|f\|_2^2$, with R^* extension by zero. The kernel

is real and symmetric, so the spectral theorem gives a real eigenbasis with eigenvalues in $[0,\mu^{-2}]$ summing to $|\Omega|/4$. $\square$

Proof of Theorem 7.4. Work with $\gamma=\gamma_\mu$; $T=T_{\Omega,a}$ is continuous on $C(\mathbb{R}^2)$. Put $\xi_j=\int_\Omega \gamma e_j$ for the eigenbasis (e_j,λ_j) of $K=K_\Omega$; by Lemma B.1(S4) every finite vector of point values and ξ_j's is almost surely in the range of the isonormal process, hence centered jointly Gaussian, with $\mathbb{E}\xi_j\xi_l=\lambda_j\delta_{jl}$ and $\mathbb{E}[\gamma(z)\xi_j]=\langle g_z,e_j\rangle$, $g_z=C_\mu^{\mathrm{end}}(z-\cdot)|_\Omega$. The ξ_j are independent $N(0,\lambda_j)$ and, by Parseval for every outcome, $T(\gamma)=(\kappa_a^2/2)\sum_j\xi_j^2$. (a) With T_N the partial sums, $e^{T_N}\uparrow e^{T(\gamma)}$ and $\mathbb{E}e^{T_N}=\prod_{j\le N}(1-\kappa_a^2\lambda_j)^{-1/2}$, so $\mathrm{Det}_{\Omega,a}=\prod_j(1-\kappa_a^2\lambda_j)^{-1/2}\ge 1$; with $-\log(1-x)\le x/(1-r_a)$ on $[0,r_a]$, $1-r_a=a^2m^2$ and $\sum\lambda_j=|\Omega|/4$, $\log\mathrm{Det}\le\kappa_a^2|\Omega|/(8a^2m^2)$; monotonicity in Ω is pointwise. (b) $0\le c_k(z,z')\le\kappa_a^{2k}(C_\mu^{\mathrm{end}})^{*(k+1)}(z-z')=s_k(z-z')$; symmetry by relabelling; C_μ^{end} is Lipschitz with constant $\mu/4$ in $|\cdot|_1$, so c_k is Lipschitz; the operator $S_k=C_\mu^{\mathrm{end}}(\kappa_a^2PC_\mu^{\mathrm{end}})^k$, $P=\mathbf{1}_\Omega$, has kernel c_k (Fubini, using $\int c_k(x,z')|f(z')|dz'\le\|f\|_2r_a^k/(2\mu)$); $\|S_k\|\le\mu^{-2}r_a^k$, so $\sum S_k=C_\mu^{\mathrm{end}}(I-\kappa_a^2PC_\mu^{\mathrm{end}})^{-1}$ in norm, and its kernel is $C'=\sum c_k$ (uniform limit). Since $PC_\mu^{\mathrm{end}}=R^*RC_\mu^{\mathrm{end}}$ and $(\kappa_a^2R^*RC_\mu^{\mathrm{end}})^{k+1}=\kappa_a^2R^*(\kappa_a^2K)^kRC_\mu^{\mathrm{end}}$, the restriction form follows. (c) Fix points $z_1,\dots,z_n$ anywhere in the plane, $Y=(\gamma(z_l))_l$ and $b_j=\mathbb{E}[Y\xi_j]$. Regression on the ξ_j with $\lambda_j>0$, $j\le N$, and the Gaussian tilting fact show that under $e^{T_N}\mathbb{P}/\mathbb{E}e^{T_N}$ the vector Y is centered Gaussian with covariance $\mathrm{Cov}\,Y+\kappa_a^2\sum_{j\le N}b_jb_j^T/(1-\kappa_a^2\lambda_j)$, and $\mathbb{E}[e^{it\cdot Y}e^{T_N}]=\mathbb{E}[e^{T_N}]\exp(-\frac12t^T\Sigma_Nt)$. Dominated convergence (majorant $e^{T(\gamma)}$) and absolute convergence of the series give, in the limit, the covariance $C_\mu^{\mathrm{end}}(z_l-z_{l'})+\kappa_a^2\langle g_{z_l},(I-\kappa_a^2K)^{-1}g_{z_{l'}}\rangle$; expanding the resolvent and using the iterated kernels, this is $C'(z_l,z_{l'})$. Hence under $G'_{\Omega,a}$ the coordinates at all points are centered Gaussian with covariance C'; uniqueness is the path-space fact; the change-of-measure identity is the definition of G', extended to integrable F by splitting into positive and negative parts. $\square$

Proof of Theorem 7.5. Pointwise: $c_0=C_\mu^{\mathrm{end}}>0$, $c_k\ge0$ and $c_k\le s_k$ give $C_\mu^{\mathrm{end}}\le C'\le\sum s_k=C_a\le c_a$; each c_k is nondecreasing in Ω. As operators: $C'-C_\mu^{\mathrm{end}}=\kappa_a^2C_\mu^{\mathrm{end}}R^*(I-\kappa_a^2K_\Omega)^{-1}RC_\mu^{\mathrm{end}}\ge0$ because the resolvent has spectrum in $[1,(a^2m^2)^{-1}]$; $C_a-C'=\Delta\ge0$ by Theorem 7.6(b), whose proof uses only Theorem 7.4(b), the pointwise part and Theorem 7.2. $\square$

Proof of Theorem 7.6. (a) For $x=(x_1,\dots,x_n)$, $1-\prod_i\mathbf{1}_\Omega(x_i)=\sum_i[\prod_{l<i}\mathbf{1}_\Omega(x_l)]\mathbf{1}_{\Omega^c}(x_i)$ (first exit), so $s_n(z-z')-c_n(z,z')=\sum_{i=1}^n\kappa_a^2\int_{\Omega^c}c_{i-1}(z,x)s_{n-i}(x-z')\,dx$ by Tonelli; summing, $\Delta(z,z')=\kappa_a^2\int_{\Omega^c}C'(z,x)C_a(x-z')\,dx$, which is finite and symmetric, so also $=\kappa_a^2\int_{\Omega^c}C_a(z-x)C'(x,z')\,dx$; substituting $C_a=C'+\Delta$ gives the third form. As operators, with $A=I-\kappa_a^2C_\mu^{\mathrm{end}}$ and $B=I-\kappa_a^2PC_\mu^{\mathrm{end}}$, $C_a=C_\mu^{\mathrm{end}}A^{-1}$, $C'=C_\mu^{\mathrm{end}}B^{-1}$ and $B-A=\kappa_a^2QC_\mu^{\mathrm{end}}$, so $C_a-C'=C_\mu^{\mathrm{end}}A^{-1}(B-A)B^{-1}=\kappa_a^2C_aQC'$; self-adjointness gives the other forms. (b) Continuity is that of C_a and C'. Operator positivity: $\langle f,\Delta f\rangle=\kappa_a^2\|QC'f\|^2+\kappa_a^4\langle QC'f,C_aQC'f\rangle\ge0$. Kernel positivity: with $F(x)=\mathbf{1}_{\Omega^c}(x)\sum_l c_lC'(z_l,x)\in L^1\cap L^\infty$, $\sum c_lc_{l'}\Delta(z_l,z_{l'})=\kappa_a^2\|F\|_2^2+\kappa_a^4\langle F,C_aF\rangle\ge0$. (c) From the last-exit form and $C'\le C_a$; if Ω contains the open r-neighbourhood of K, then $|z-x|\ge r$ for $z\in K$, $x\in\Omega^c$, and Cauchy–Schwarz gives $\epsilon_a(r)$, which decreases to 0 by dominated convergence since $C_a\in L^2$. (d) Choose r with $\epsilon_a(r)\le\eta$; every bounded Ω containing the r-neighbourhood of K has $\sup_{K\times K}\Delta\le\eta$; for $[-R,R]^2$ this holds once $R\ge r+\sup_K|z|_\infty$. (e) With $q_k=k(z+h,z+h)+k(z,z)-2k(z+h,z)$, $q_{C_a}=q_{C'}+q_\Delta$ with both terms nonnegative, so each is at most $2(c_a-C_a(h))\le M'_{a,1/2}|h|^{1/2}$; Kolmogorov extension gives Gaussian processes with covariances Δ and C', and the continuity criterion with $\mathbb{E}|Z(z)-Z(z')|^{12}\le 11!!(M'_{a,1/2})^6|z-z'|^3$ and the dyadic argument give jointly measurable continuous modifications. $\square$

Proof of Theorems 7.7 and 7.8. μ_0 is even and σ-finite, so $H_{\mathbb{C}}$ is a separable real Hilbert space

carrying an isonormal W (C6(f)). For real $f \in L^2$, $\hat{f}\tau_a$ is conjugate-symmetric with $\int |\hat{f}\tau_a|^2 d\mu_0 = (2\pi)^{-2}\int |\hat{f}|^2 \hat{C}_a \le m^{-2}\|f\|_2^2$; all $\phi_a(f)$ lie in the range of the single W, hence are jointly Gaussian; the covariance is the isometry, with $\tau_a\tau_b\hat{C}_0 = (\hat{C}_a\hat{C}_b)^{1/2}$. For Theorem 7.8: with $D = m^2 + |p|^2$, $0 \le \hat{C}_0 - \hat{C}_a = a^2w^2q^2/(D(D+a^2w^2q^2)) \le a^2w^2q^2/D^2 \le a^2/4$ (as $4w^2q^2 \le D^2$), and $(1-\tau)^2 \le 1-\tau^2$ on $[0,1]$, so $\|\phi_a(f) - \phi(f)\|_2^2 = (2\pi)^{-2}\int |\hat{f}|^2(1-\tau_a)^2\hat{C}_0 \le (2\pi)^{-2}\int |\hat{f}|^2(\hat{C}_0 - \hat{C}_a) \le \frac{a^2}{4}\|f\|_2^2$; complex tests split into real and imaginary parts. □

Proof of Theorem 7.9. Lemma B.1 with $\nu = \mu_0$ and $\sigma = \tau_a$, $\int \tau_a^2 d\mu_0 = c_a$ and $k = C_a$: (S1) gives the covariances, with $C_{ab} = \mathcal{I}[(\hat{C}_a\hat{C}_b)^{1/2}]$ where $\mathcal{I}g(x) = (2\pi)^{-2}\int e^{ip\cdot x}g(p)\,dp$ for $g \in L^1$ (here $(\hat{C}_a\hat{C}_b)^{1/2} \le \frac{1}{2}(\hat{C}_a + \hat{C}_b) \in L^1$); (S2) with Theorem 7.2(e) gives the increment; (S3) with Theorem 7.2(f), $\alpha = 1/2$, gives $\tilde{\phi}_a$; (S4) gives the smearing identity, the L^1 and Plancherel transforms agreeing on $L^1 \cap L^2$. Each modification and each smearing identity has its own null set; countable unions of null sets are null; $\tilde{\phi}_a$ is constructed separately for each a, so no common null set over uncountably many regulators arises, and no measurability in a is asserted. □

Proof of Theorem 7.10. $\sigma = \tau_a\rho$ has $\int \sigma^2 d\mu_0 \le (2\pi)^{-2}\int (m^2 + |p|^2)^{-1}(1 + |p|^2)^{-2}dp < \infty$, and $\mathbb{E}|\psi_a(z+h) - \psi_a(z)|^2 \le |h|^2(2\pi)^{-2}\int |p|^2\hat{C}_0(1+|p|^2)^{-2}dp \le |h|^2/(4\pi)$, so Lemma B.1(S3) with $\vartheta = 2$ gives $\tilde{\psi}_a$, a Borel $C(\mathbb{R}^2)$-valued random element, and $\Phi_a = L\tilde{\psi}_a$ is Borel because L is continuous (Proposition 5.1(viii)). (a) For real $f \in C_c^\infty$ and $g = (1-\Delta)f$, $\hat{g} = (1+|p|^2)\hat{f}$ and, for every outcome, $\Phi_a(f) = \int \tilde{\psi}_a g$; by (S4) this is $W(\hat{g}\tau_a\rho) = W(\hat{f}\tau_a) = \phi_a(f)$ almost surely. (b) For $a > 0$, Φ_a and $\iota\tilde{\phi}_a$ agree on the countable family $\mathcal{D}_0$ of Proposition 5.1(ix) off one null set (by (a) and Theorem 7.9), hence everywhere off it. (c) Same argument with Φ'. (d) Countable unions of null sets. (e) By (a) for the countably many tests $\chi_j\overline{e_{j,k}}$, Tonelli and Theorem 7.8, $\mathbb{E}p_j(\Phi_a)^2 = \sum_k \lambda_{j,k}^{-2}\mathbb{E}|T_a(\chi_j\overline{e_{j,k}})|^2 \le m^{-2}B_j$, and the difference bound is the same with $\frac{a^2}{4}$; $B_j < \infty$ by the shell count. Since $\|\min(1,Y)\|_2 \le \min(1,\|Y\|_2)$ for $Y \ge 0$, Minkowski for the series gives $\|d(\Phi_a,\Phi_0)\|_{L^2} \le h(a)$, and $h(a) \to 0$ by dominated convergence. (f) is (a) with Theorem 7.7. □

Remark B.2 (Scope of the Gaussian proofs). These proofs use no interaction functional, Wick power, weight, normalizer or statement of Theorem 8.2; in particular the construction of Φ_0 does not use the existence of U_0. They establish exactly the Gaussian interface used by Theorem 8.2(j) and Corollary 8.3: the common isonormal space, the continuous modifications for regulators, and the Borel versions Φ_a for $0 \le a < 1/m$ with their compact-test identities. That the interacting laws exist on this space is a consequence of Theorem 8.2, not of this appendix. Nothing is asserted at the endpoint value $a = 1/m$.

C Proofs for the fixed-volume Wick theory

Throughout, Λ is a physical square with $A = |\Lambda|$, $0 < a < 1/m$ unless $a = 0$ is stated, $\|\cdot\|_r$ is the norm of $L^r(\mathbb{P}_0)$ or of $L^r(\mathbb{R}^2, dz)$, $D = m^2 + |p|^2$, $\sigma_a = \hat{C}_0 - \hat{C}_a$, $\hat{C}_{ab} = (\hat{C}_a\hat{C}_b)^{1/2}$ and $C_{ab} = \mathcal{I}\hat{C}_{ab}$. The standard inputs are C6(a)–(c), Gaussian characteristic functions $\mathbb{E}e^{iX} = e^{-\operatorname{Var}X/2}$ and $\mathbb{E}\mathrm{He}_4(N(0,1)) = 0$, Fourier series on tori, and Schur's product theorem.

C.1 Gaussian preliminaries

Lemma C.1 (Wick powers, reflection and the free σ-algebra)**.** *(i) For ξ in the first chaos H_1 with $c = \operatorname{Var}\xi \ge 0$, $:\xi^4: = :\xi^4:_c \in H^{:4:}$ (for $c = 0$, $\xi = 0$ almost surely and $:\xi^4: = 0$), $\mathbb{E}[:\xi^4::\eta^4:] = 4!(\mathbb{E}\xi\eta)^4$ for all $\xi,\eta \in H_1$, and $z \mapsto :\phi_a(z)^4:$ is continuous into $H^{:4:}$. For $\xi \sim N(0,c)$ and $c' \ge 0$,*

$\mathbb{E}[\xi^4 - 6c'\xi^2 + 3c'^2] = 3(c-c')^2$, *which vanishes exactly at the own variance. (ii) Let* T *be a surjective real-linear isometry of* H_1 *and* $\mathcal{F}_H = \sigma(H_1) \vee \mathcal{N}$. *There is a unique map* Θ_T *on* $L^0(\mathcal{F}_H)$ *with* $\Theta_T[g((\xi_j)_j)] = g((T\xi_j)_j)$ *for countable* $(\xi_j) \subset H_1$ *and Borel* g*; it is linear, multiplicative, commutes with Borel functions (in particular with conjugation and* $e^{-\cdot}$*), preserves laws, is an isometry of every* L^r *and commutes with* L^r*-limits and Bochner integrals in* L^2. *For* $\theta \in \{\theta_1, \theta_2\}$, $(T_\theta h)(p) = h(\theta p)$ *is orthogonal on* $H_{\mathbb{C}}$, *and with* $T = WT_\theta W^{-1}$, $\Theta = \Theta_T$ *satisfies* $\Theta\phi_a(f) = \phi_a(f \circ \theta)$, $\Theta\phi_a(z) = \phi_a(\theta z)$ *and* $\Theta{:}\phi_a(z)^4{:} = {:}\phi_a(\theta z)^4{:}$. *(iii)* $\{\phi(f) : f \in C_c^\infty(\mathbb{R}^2;\mathbb{R})\}$ *is dense in* H_1, *and* $\mathcal{F}_H = \mathcal{F}_{\text{free}}$.

Proof. (i) C6(b) for positive variances; if a variance vanishes both sides vanish since $|\mathbb{E}\xi\eta| \le (\operatorname{Var}\xi \operatorname{Var}\eta)^{1/2}$. Hence $\mathbb{E}|{:}\xi^4{:} - {:}\eta^4{:}|^2 = 4!(c_\xi^4 - 2(\mathbb{E}\xi\eta)^4 + c_\eta^4)$, and $z \mapsto e_z\tau_a$ is continuous into $H_{\mathbb{C}}$. (ii) H_1 is separable; every $Y \in L^0(\mathcal{F}_H)$ is $g((\xi_j)_j)$ by Doob–Dynkin; if $g((\xi_j)) = g'((\xi'_j))$ almost surely, the vectors $((\xi_j),(\xi'_j))$ and $((T\xi_j),(T\xi'_j))$ are centered Gaussian with the same covariances, hence have the same law, so the Borel set $\{g \ne g'\}$ is null for both. The listed properties follow by merging countable families. T_θ is orthogonal because Lebesgue measure and $\hat{C}_0$ are θ-invariant; $(f \circ \theta)\hat{} = \hat{f} \circ \theta$, $\tau_a \circ \theta = \tau_a$ and $T_\theta(e_z\tau_a) = e_{\theta z}\tau_a$. (iii) For $h \in H_{\mathbb{C}}$ and $R > 0$, $\hat{C}_0 \ge (m^2+R^2)^{-1}$ on $\{|p| \le R\}$, so $h_R = h\mathbf{1}_{\{|p|\le R\}} \in L^2(dp)$ is Hermitian, $f_R = \mathcal{I}h_R$ is real with $\phi(f_R) = W(h_R)$, and $\|h - h_R\|_{H_{\mathbb{C}}} \to 0$; $\|\phi(f) - \phi(f')\|_2 \le m^{-1}\|f - f'\|_2$ and density of C_c^∞ in L^2 conclude. Each element of H_1 is an almost sure limit of free-field smearings, and $\mathcal{N} \subset \mathcal{F}_{\text{free}}$. □

Lemma C.2 (Kernel facts)**.** *(K1) For* $0 \le b \le a < 1/m$: $0 < \hat{C}_a \le \hat{C}_{ab} \le \hat{C}_b \le \hat{C}_0$, *and* $\hat{C}_b - \hat{C}_{ab}$, $\hat{C}_{ab} - \hat{C}_a$, $\hat{C}_b - \hat{C}_a$ *lie in* $[0, \sigma_a]$. *(K2)* $\sigma_a = a^2w^2q^2/(D(D + a^2w^2q^2)) \in [0, \min(a^2/4, \hat{C}_0)]$. *(K3)* $\mathcal{I}$ *satisfies* $\|\mathcal{I}g\|_2 = (2\pi)^{-1}\|g\|_{L^2(dp)}$ *and* $\|\mathcal{I}g\|_6 \le (2\pi)^{-5/3}\|g\|_{6/5}$*; for bounded Hermitian* k, $K[k] = \mathcal{I}(k\hat{C}_0)$ *is real,* $k \mapsto K[k]$ *is real-linear,* $\|K[k]\|_2 = (2\pi)^{-1}\|k\hat{C}_0\|_{L^2(dp)}$ *and* $\|K[k]\|_6 \le \|k\|_\infty M_6$*; in particular* $\|C_{ab}\|_6 \le M_6$ *for* $a, b > 0$, *and each of* $C_b - C_{ab}$, $C_{ab} - C_a$, $C_b - C_a$ *has* L^2 *norm at most* $\kappa_2 a$ *for* $0 < b \le a$. *(K4)* c_a *is strictly decreasing,* $c_a \le (2\pi)^{-1}\log(\kappa_0/(am))$, *and* $c_a > c_{1/m} = 1/4$. *(K5)* $C_a(t,y) = (2\pi)^{-1}\int e^{iqy}e^{-E_a(q)|t|}(2E_a(q)(1+a^2q^2))^{-1}dq$ *with* $E_a(q) = [(m^2+q^2)/(1+a^2q^2)]^{1/2}$, *and* $\iint h(\theta z)C_a(z-z')h(z')\,dz\,dz' \ge 0$ *for real* $h \in C_c^\infty(H_+^\theta)$. *(K6)* C_{ab} *is bounded and continuous for* $a, b > 0$, $\sup|C_a| = c_a$ *and* $C_a(\theta x) = C_a(-x) = C_a(x)$.

Proof. (K1) With $x = \hat{C}_a \le y = \hat{C}_b$: $\sqrt{xy} \in [x,y]$, $y - \sqrt{xy} \le y - x$, $\sqrt{xy} - x \le y - x \le \sigma_a$. (K2) Expand $1/D - 1/(D + a^2w^2q^2)$ and use $w^2q^2 \le D^2/4$. (K3) For $s > 1$, $\int(m^2 + |p|^2)^{-s}dp = \pi m^{2-2s}/(s-1)$, so $\hat{C}_0 \in L^2 \cap L^{6/5}$ with $\|\hat{C}_0\|_{6/5} = (5\pi m^{-2/5})^{5/6}$; Hausdorff–Young for the unnormalized transform gives the factor $(2\pi)^{-2}(2\pi)^{1/3}$; the Plancherel and Hausdorff–Young extensions agree almost everywhere (truncations converge in both spaces); Hermitian symmetry gives reality. For the differences, $\|\sigma_a\|_{L^2(dp)}^2 \le \int_{|p|\le 1/a}(a^2/4)^2 + \int_{|p|>1/a}|p|^{-4} = 17\pi a^2/16$. (K4) $\int_{\mathbb{R}} dw/(\alpha + \beta w^2) = \pi/\sqrt{\alpha\beta}$; the integrand of (2) is strictly decreasing in a for $q \ne 0$; $\int_0^{1/a} dq/\sqrt{m^2+q^2} = \operatorname{asinh}(1/(am)) \le \log((1+\sqrt{2})/(am))$ and $\int_{1/a}^\infty dq/(aq^2) = 1$; at $a = 1/m$ the integral is $(2\pi)^{-1}\pi/2$ (used only for this comparison; $1/m$ is not a regulator). (K5) $\hat{C}_a = [(1+a^2q^2)(w^2+E_a^2)]^{-1}$ and $(2\pi)^{-1}\int e^{iwt}(w^2+E^2)^{-1}dw = e^{-E|t|}/(2E)$; for $\theta = \theta_1$ and $t, t' > 0$ on $\operatorname{supp} h$, Fubini gives $(2\pi)^{-1}\int dq\,[2E_a(1+a^2q^2)]^{-1}|\int h(t,y)e^{-E_a(q)t}e^{iqy}dt\,dy|^2 \ge 0$; $\hat{C}_a$ is symmetric in (w,q), which gives θ_2. (K6) $\hat{C}_{ab} \ge 0$ is integrable and invariant under $p \mapsto \theta p$ and $p \mapsto -p$. □

C.2 The interaction and its L^2 limit

Proof of Theorem 8.2(a), (c), (d) and the existence of U_0*. Pathwise equals Bochner.* $z \mapsto {:}\tilde{\phi}_a(z,\omega)^4{:}$ is continuous for every outcome, so U_a is a finite pathwise integral, measurable by Fubini. $Y(z) = u{:}\phi_a(z)^4{:}$ is continuous into $H^{:4:}$ with $\mathbb{E}Y(z)^2 = 4!u^2c_a^4$, so the Bochner integral

$B = \int_\Lambda Y$ exists in $H^{:4:}$; for bounded V, $\mathbb{E}[VU_a] = \int_\Lambda \mathbb{E}[VY(z)]dz = \mathbb{E}[VB]$, so $U_a = B$ almost surely. *Mean zero and lower bound.* $\mathbb{E}U_a = 0$ because the Wick constant is the own variance; $x^4 - 6cx^2 + 3c^2 = (x^2 - 3c)^2 - 6c^2$ gives $U_a \geq -6uAc_a^2$ for every outcome, and the same for $U_a^\pm = u\int_{\Lambda\cap H_\pm} :\tilde\phi_a^4:$. *$L^2$ estimate.* For $0 < b \leq a$, Fubini for Bochner integrals and Lemma C.1(i) give $\mathbb{E}|U_a - U_b|^2 = 4!u^2 \iint_{\Lambda^2}[C_a^4 - 2C_{ab}^4 + C_b^4](z - z') \leq 4!u^2A(\|C_a^4 - C_{ab}^4\|_1 + \|C_b^4 - C_{ab}^4\|_1)$; $|X^4 - Y^4| \leq 4|X - Y|(|X|^3 + |Y|^3)$, Cauchy–Schwarz and (K3) give $\|C_a^4 - C_{ab}^4\|_1 \leq 8\kappa_2 M_6^3 a$, so $\mathbb{E}|U_a - U_b|^2 \leq 384u^2A\kappa_2M_6^3a = K_\Lambda^2 a$. *Limit.* Along any sequence of regulators tending to 0 the U_{a_n} are Cauchy; interlacing shows that all limits coincide; letting $b \downarrow 0$ gives $\|U_a - U_0\|_2 \leq K_\Lambda\sqrt{a}$ for every regulator; $H^{:4:}$ is closed, and $\mathbb{E}U_0 = \lim \mathbb{E}U_a = 0$. A Wick constant $c' \neq c_a$ would give the nonzero mean $3uA(c_a - c')^2$. □

Proof of Theorem 8.2(b). Continuity of Q. For bounded Hermitian k put $Q(k) = \iint_{\Lambda^2} K[k](z - z')^4$. If $k_n \to k$ almost everywhere with $\sup \|k_n\|_\infty \leq \kappa$, then $|Q(k_n) - Q(k)| \leq (4A/\pi)\kappa^3 M_6^3 \|(k_n - k)\hat C_0\|_{L^2(dp)} \to 0$ by dominated convergence with majorant $4\kappa^2\hat C_0^2$. *All admissible cutoffs.* Since $\int \hat C_0 = \infty$, Fatou gives $c_\epsilon^\chi = \operatorname{Var}\phi_\epsilon^\chi(z) \to \infty$, so $c_\epsilon^\chi > 0$ for small ϵ (if $c_\epsilon^\chi = 0$, then $\chi_\epsilon = 0$ almost everywhere and $N_\epsilon^\chi = 0$). For admissible χ, ψ, $\mathbb{E}[\phi_\epsilon^\chi(z)\phi_\delta^\psi(z')] = K[\overline{\chi_\epsilon}\psi_\delta](z - z')$; the regulator family τ is admissible with $N_a^\tau = U_a$; so $\mathbb{E}|N_\epsilon^\chi - U_a|^2 = 4!u^2[Q(|\chi_\epsilon|^2) - 2Q(\overline{\chi_\epsilon}\tau_a) + Q(\tau_a^2)]$. Let $a \downarrow 0$ at fixed ϵ (existence of U_0 and continuity of Q), then $\epsilon \downarrow 0$ along any sequence: the limit is $4!u^2[Q(1) - 2Q(1) + Q(1)] = 0$. For a mollifier, $\chi_\epsilon(p) = \overline{\hat\eta(\epsilon p)}$ gives $\phi_\epsilon^\chi(z) = \phi(\eta_\epsilon(z - \cdot))$, with $\sup|\chi_\epsilon| \leq \|\eta\|_1$. *Pairings and uniqueness.* For $a > 0$, $\mathbb{E}[U_a{:}\phi(f_1)\cdots\phi(f_4){:}] = 4!u\int_\Lambda \prod_j (C_{a0} * f_j)(z)\,dz$; since $|1 - \tau_a|\hat C_0 \leq \sigma_a \leq a^2/4$ and $\hat f_j \in \mathcal{S}$, $C_{a0} * f_j \to C_{\text{free}} * f_j$ uniformly, and the formula passes to U_0. If $Y \in H^{:4:}$ has the same pairings, $Y - U_0$ is orthogonal to the Wick monomials of the dense set of Lemma C.1(iii), which span a dense subspace of $H^{:4:}$ (C6(b)); so $Y = U_0$. *Measurability and scope.* $U_0 \in H^{:4:} \subset L^2(\mathcal{F}_H) = L^2(\mathcal{F}_{\text{free}})$. The construction uses only the variables $W(h)$, their own-variance Wick polynomials, Bochner integrals and L^2-limits; it never uses a point variable at $a = 0$ (indeed $e_z \notin L^2(\mu_0)$ because $\int \hat C_0 = \infty$) and forms no pointwise product of the distribution-valued free field; nothing is asserted about other representations of U_0. □

C.3 Nelson tail, weights and normalizers

Proof of the tail bound and of Theorem 8.2(e). By C6(a) with $k = 4$, $\|X\|_r \leq (r - 1)^2\|X\|_2$ for $X \in H^{:4:}$ and real $r \geq 2$. Fix $K \geq K_\sharp$ and put $s = (K/(12uA))^{1/2} \geq 1$ and $a' = (\kappa_0/m)e^{-2\pi s}$. (D1) $\kappa_0 = e(1+\sqrt{2}) < e^2 < e^{2\pi}$, so $a'm < 1$ and a' is a regulator. (D2) $c_{a'} \leq s$ by (K4), so $U_{a'} \geq -6uAs^2 = -K/2$ for every outcome. (D3) For $a' \leq a < 1/m$, $c_a \leq c_{a'}$, so $U_a \geq -K/2$ and $\mathbb{P}_0(U_a \leq -K) = 0$. (D4) For $0 < a < a'$, $\{U_a \leq -K\} \subseteq \{|U_{a'} - U_a| \geq K/2\}$. (D5) $X = U_{a'} - U_a \in H^{:4:}$ has $\|X\|_2 \leq K_\Lambda a'^{1/2}$, so for real $r \geq 2$, $\mathbb{P}_0(|X| \geq K/2) \leq [(2K_\Lambda/K)(\kappa_0/m)^{1/2}(r - 1)^2e^{-\pi s}]^r$. (D6) With $r = 1 + e^{\pi s/2 - 1} \geq 2$, $(r - 1)^2 = e^{\pi s - 2}$ and the bracket is $(2K_\Lambda/K)(\kappa_0/m)^{1/2}e^{-2} \leq e^{-1}$ because $K \geq (2/e)K_\Lambda(\kappa_0/m)^{1/2}$. (D7) $\mathbb{P}_0(U_a \leq -K) \leq e^{-r} \leq \exp(-\exp((\pi/2)(K/(12uA))^{1/2} - 1)) = T(K)$. The coarse regulator depends on K only, so the bound is uniform in a. By the layer-cake formula, $\mathbb{E}_0 e^{-pU_a} = \int p\,e^{pK}\mathbb{P}_0(U_a < -K)\,dK \leq \Xi_p$. $\Xi_p < \infty$: with $c = (\pi/2)(12uA)^{-1/2}$ and $x = c\sqrt{K} - 1$, for $K \geq 4/c^2$ one has $e^x \geq x^4/24 \geq c^4K^2/384$, so the integrand is at most pe^{-pK} for $K \geq \max(K_\sharp, 4/c^2, 768p/c^4)$. At $a = 0$ choose $a_j \downarrow 0$ with $U_{a_j} \to U_0$ almost surely; Fatou gives $\mathbb{E}_0 e^{-pU_0} \leq \Xi_p$ and $\mathbb{P}_0(U_0 < -K) \leq \liminf \mathbb{P}_0(U_{a_j} < -K) \leq T(K)$ (open event only). □

Proof of Theorem 8.2(f), (g). $|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y})$ and Hölder with exponents $(2p, 2p)$ give $\|e^{-U_a} - e^{-U_0}\|_p \leq \|U_a - U_0\|_{2p}\|e^{-U_a} + e^{-U_0}\|_{2p} \leq (2p - 1)^2K_\Lambda\sqrt{a}\cdot 2\Xi_{2p}^{1/(2p)}$. Jensen and $\mathbb{E}U_a = 0$

give $Z_a \geq 1$, and (e) with $p = 1$ gives $Z_a \leq \Xi_1$; the lower bound concerns the physical laws on the common Gaussian space only. □

C.4 Central reflection positivity at fixed volume

Proof of Theorem 8.2(h), (i). Half-plane localization. For $z \in H_+$, the mollified tests $\rho_n(\cdot - z)$ lie in $C_c^\infty(H_+)$ for large n and $\phi_a(\rho_n(\cdot - z)) \to \phi_a(z)$ in L^2, so $:\phi_a(z)^4: \in L^2(\mathcal{A}_+^{\theta,a})$ and the Bochner integral U_a^+ lies in $L^2(\mathcal{A}_+^{\theta,a})$; $U_a = U_a^+ + U_a^-$ (the line is null), and $\Theta U_a^+ = U_a^-$ because Θ commutes with Bochner integrals and Λ is θ-invariant. *Exponentials are dense.* The span of $e^{i\phi_a(f)}$, $f \in C_c^\infty(H_+;\mathbb{R})$, is dense in $L^r(\mathcal{A}_+^{\theta,a})$, $1 \leq r < \infty$: an annihilating $g \in L^{r'}$ gives finite complex measures with vanishing Fourier transforms on every finite cylinder, hence $\mathbb{E}[g\mathbf{1}_S] = 0$ on a generating π-system and, by Dynkin, on $\mathcal{A}_+^{\theta,a}$. *Gaussian positivity.* For $G = \sum_k c_k e^{i\phi_a(f_k)}$, $\mathbb{E}[\overline{\Theta G} G] = \sum_{j,k} \overline{d_j} d_k \exp(M_{jk})$ with $d_k = c_k e^{-\langle f_k, C_a f_k\rangle/2}$ and $M_{jk} = \langle f_j \circ \theta, C_a f_k\rangle$; M is real symmetric and positive semidefinite by (K5), (K6), so $\exp \circ M = \sum_n M^{\circ n}/n!$ is positive semidefinite by Schur's theorem; density extends this to $G \in L^2(\mathcal{A}_+^{\theta,a})$. *(h)* With $e^{-U_a^\pm} \leq e^{6uAc_a^2}$, $G = Fe^{-U_a^+} \in L^2(\mathcal{A}_+^{\theta,a})$ and $\Theta G = (\Theta F)e^{-U_a^-}$, so $\mathbb{E}[\overline{\Theta F} F e^{-U_a}] = \mathbb{E}[\overline{\Theta G} G] \geq 0$. *(i)* For $G = \sum_j c_j e^{i\phi(f_j)}$ and $G_a = \sum_j c_j e^{i\phi_a(f_j)}$, (h) gives $\mathbb{E}[\overline{\Theta G_a} G_a e^{-U_a}] \geq 0$, and with $c = \sum |c_j|$, $|e^{ix} - e^{iy}| \leq |x - y|$, Theorem 7.8, (e) with $p = 2$ and (f) with $p = 1$, the difference from $\mathbb{E}[\overline{\Theta G} G e^{-U_0}]$ is at most $c\,\Xi_2^{1/2}(a \sum_j |c_j| \|f_j\|_2 + 2cK_\Lambda\sqrt{a}) \to 0$. For bounded $\mathcal{A}_+^{\theta,0}$-measurable F, approximate in L^4 by such G_n and use Hölder with exponents $(4, 4, 2)$ and $\|e^{-U_0}\|_2 \leq \Xi_2^{1/2}$. □

Proof of Theorem 8.2(j) and of the identification with Definition 2.10(d). For Borel B, $\mathbf{1}_B(\Phi')$ is measurable and $Z_a \in [1, \Xi_1]$, so the formula defines a Borel probability with $\int H\, d\nu = Z_a^{-1}\mathbb{E}_0[H(\Phi')e^{-U_a}]$ for bounded Borel H. For bounded Σ_+^θ-measurable F, Proposition 5.1(x) gives real $f_j \in C_c^\infty(H_+^\theta)$ and a bounded Borel g with $F(v) = g((v(f_j))_j)$ and $F(\theta v) = g((v(f_j \circ \theta))_j)$. Off one null set, $\Phi'(f) = \phi_a(f)$ for the countably many f_j and $f_j \circ \theta$; there $F(\Phi') = G := g((\phi_a(f_j))_j)$, which is $\mathcal{A}_+^{\theta,a}$-measurable, and $F(\theta\Phi') = g((\phi_a(f_j \circ \theta))_j) = \Theta G$ almost surely by Lemma C.1(ii). So $\int \overline{F(\theta v)} F(v)\, \nu(dv) = Z_a^{-1}\mathbb{E}_0[\overline{\Theta G} G e^{-U_a}] \geq 0$ by (h) for $a > 0$ and by (i) for $a = 0$. On the common space, Theorem 7.10(a) supplies the hypothesis on $\Phi' = \Phi_a$ for every $0 \leq a < 1/m$, including $a = 0$; two continuous modifications of ϕ_a agree everywhere off one null set (rational points and continuity), so the pathwise U_a does not depend on the modification; and any other Borel Φ' with the same compact-test identities equals Φ_a almost surely (Theorem 7.10(c)), hence gives the same law. This is Corollary 8.3. □

Remark C.3 (Proof of Theorem 8.2(k)). No step above uses a restriction on u/m^2, and every constant is computed for the fixed square Λ.

Remark C.4 (Volume dependence and unused routes). $K_\Lambda = u(384\kappa_2 M_6^3)^{1/2} A^{1/2}$, $K_\sharp \geq 12uA$ and $\Xi_p \geq e^{12puA}$: every quantitative bound is for a fixed physical square, and the only volume-uniform statement is $Z \geq 1$. A frequency-split tail argument, which gives another family of regulator-uniform moment bounds, was checked in the antecedent record and is not used. The proofs use the Gaussian input only through Theorems 7.7, 7.9 and 7.10 and Proposition 5.1; the weight, the normalizer and positivity enter only afterwards.

D Provenance, review status and source access

Authorship of this text. The manuscript and corrections A, AC and AC2 were written by Claude (Opus 5.5, Anthropic), with retained earlier Codex argument lineage as listed below, in a project directed by Cody Mitchell. The AC3 integration author is GPT-6.1 Sol (max effort, OpenAI), adopting the Sol-authored SB5 and SB6 proof arguments and SMH numerical repair, expanding their complete derivations and writing the dependency/source separation. Older Astra-line H1/K1 lineage and Opus authorship remain attributed. The authorship map records every adopted or rewritten argument; a reviewer is an author only when its own proof argument is adopted, never merely for its review verdict. Historical A/AC/AC2 and adverse review bytes are preserved; all earlier verdicts remain bound to the versions they read. The accompanying archive includes a public review summary and exact source hashes; the detailed internal review records are retained in the project archive. Every review mentioned below was an internal review by language models; none is an independent expert review, and no record is a formal (machine-checked) proof.

Publication revision. The present revision adds explanatory prose by Claude (Opus 5.5), with title, opening, integration and precision clarifications by GPT-6 Astra under Cody Mitchell's direction. All numbered mathematical statements and proof bodies are retained verbatim from the accepted construction manuscript. The accompanying archive records the exact public source and PDF hashes and summarizes the internal revision checks.

Argument lineage in the construction chain. The fixed-volume arguments of Section 9 retain those of two earlier Codex-authored records (F1 and F2 in the table below): the local-L^4 compact containment and finite-vector identification of the width record in Proposition 9.2, and the residual normalization, covariance-defect coupling, estimates and constants of the fixed-volume record in Propositions 9.3–9.6.

Theorems 10.8, 10.10 and 10.16 have their own proofs; no earlier conditional record is cited for them. Their arguments retain forms from earlier Codex-authored records (LR, V1 and CT in the table below): the dilation and characteristic-function route of Theorem 10.8; the lattice inequality, Paley–Zygmund bound, Fourier moments, tightness and version of Theorem 10.10; and the invariance, regularity, non-Gaussianity, kernel and determining arguments of Theorem 10.16 other than part (iii). The Gaussian-realization transfer (Lemma 10.2) and corrected readings lemma (Lemma 10.6) were newly written in the Opus text. The AC3 integration adopts the Sol-authored SB5/SB6 derivations: the direct classical Wick/normalizer lemma, explicit determining/density/kernel proofs, source-only symmetries, full positivity, Euclidean mixing, exact regularity, and the local conditional-density non-Gaussianity and reconstruction route. The mixing argument also retains the earlier Claude classical-input lineage. These are manuscript argument adoptions, not attribution to the reviewers.

Sections 11 and 12 re-express, for the present text, the moment-transfer, selection and joint-correlation arguments of two earlier project records (H1 and K1 in the table below). In those records the fixed-volume arrows, the Gaussian and Wick estimates and the volume limit were premises; here each input is a theorem proved in the main text or Appendices A–C, and every place where an input is used is cited. The arguments are otherwise of the same form.

Antecedent records. The following records were supplied with their acceptance decisions; an acceptance decision covers only its literal scope, and a lemma appearing in a file is not covered merely by being there.

Record	Author family	Use in this text
Endpoint foundations (E)	Claude (Opus 5.5)	Accepted as endpoint foundations under stated classical field-space premises; re-expressed in Section 6 and Appendix A.
Gaussian foundations (G)	Claude (Opus 5.5)	Accepted for its statements G1–G11, the interaction consequence of G11 only conditionally; its Borel-law lemma for X is outside that scope and is replaced by Proposition 5.1(ii),(iii), proved here. Re-expressed in Section 7 and Appendix B.
Wick theory (W)	Claude (Opus 5.5)	Accepted; re-expressed in Section 8 and Appendix C, including the precise construction statement of Theorem 8.2(b).
Classical inputs (C)	Claude (Opus 5.5)	Accepted formulations of C1–C6, used verbatim in content in Section 3. The application question on the reading of "polynomial function" in C2(iii) was closed by a later reading of the AHKZ supplement in the reconstruction-coverage review (RC), not by the C author.
Dependency route (D)	Claude (Opus 5.5)	Accepted route "candidate B": remove four project premises by new direct proofs and prove the other five; implemented here (Appendix E).
Contribution wording (N)	Claude (Opus 5.5)	Accepted wording, used in Sections 1, 14 and 15, including the derivation of Remark 14.1, which is N-authored; the Codex-family review RN reviewed that derivation and is not one of its authors.
Older records F1 (width), F2 (fixed volume), LR (continuum limit), V1 (volume), CT (target)	Codex	Conditional on the nine project premises; their premise-conditional statements are not cited for any conclusion here. Their argument forms are retained in the current text, which is therefore jointly authored (Codex argument lineage; historical Opus integration, current Sol adoption): F1's local-L^4 compact containment and finite-vector identification in Proposition 9.2; F2's field-space construction in Proposition 5.1 and its residual normalization, covariance-defect coupling, estimates and constants in Propositions 9.3–9.6; LR's dilation and characteristic-function route in Lemma 10.5 and Theorem 10.8; V1's lattice inequality, Paley–Zygmund bound, Fourier moments, tightness and version in Theorem 10.10; CT's invariance, regularity, non-Gaussianity, kernel and determining arguments in Theorem 10.16 other than part (iii) and Lemmas 10.12–10.15.
Older records H1 (moments), K1 (kernels)	Codex (requested model of the Astra line)	Reviewed by Claude (Opus 5.5) reviewers in hostile, source, scope, blind and matching roles. Their arguments are re-expressed, with rewritten proofs of the inputs, in Sections 11–12; their authorship is unchanged by this integration. AC2 replaced the original bounded Hilbert-sum constant by a distinct scalar σ_j bound. SMH recorded that literal mismatch. AC3 adopts SMH's Sol-authored pointwise-scalar/spatial-Minkowski repair to restore the exact original K_j inequality and its finite-width Z_N^b consequence, alongside the separate σ_j estimate. The common sequence and compact K1 arguments are unchanged.

Components and the review they require. The retained Opus/Astra-line components and joint claims keep their actual authorship. The adopted SB5/SB6/SMH arguments and new AC3

integration text add Sol authorship. Their current-version qualifying review must therefore cover each actual author model, with fresh contexts and the model-specific gate required by the activated project policy; an author model cannot qualify its own contribution. The exact allocations, distinct-model fallback status, findings, actual-proof matches and acceptance decision belong in the companion records. Historical family labels and earlier review requirements are preserved there; this paragraph does not promote a historical verdict or claim independent expert review. Theorems 4.1 and 4.2 require coverage of every component chain.

Source access. The authors of this text (tasks A, AC, AC2 and AC3) read no original classical source. AC3 read only the supplied frozen manuscript, statement, companion maps, proof modules and internal review records. The page-level access below is that recorded by the classical-input and reconstruction-coverage reviews; "image" means a page image was viewed, "text" means only supplied text of the page was read. Access recorded by later reviews of version A is recorded in their own records and is not credited to the authors of this text.

Source	Page images viewed	Supplied text only
[8]	pp. 495–498, 503, 512–514, 517, 518	pp. 494, 509–511
[2]	pp. 684, 685 (with a detail of eq. (1)), 686–690	p. 691 and a supplement
[10]	pp. 310–312, 316, 324	pp. 317–320
[19]	p. 151	p. 148
[14]	—	excerpts

Further page images of a source of Fröhlich were viewed for an alternative route that is not used here; that source is not cited. Not available: the bodies of [16, 17], [20], the antecedents of the review in [10, pp. 316–319], and the originals behind the Markov and lattice positivity attributions of Section 14. The neural-network literature of Section 14 was read by the contribution analysts, partly in excerpts (see the notes in the bibliography); the authors of this text did not read it.

E Disposition of the earlier premises

An earlier conditional assembly of this construction assumed C1–C6 together with nine project premises P1–P9. Their status here is as follows; C1–C6 are the classical inputs of Section 3, in their final accepted formulations.

Premise	Status	Where
P1 endpoint neurons	proved	Theorems 6.1–6.4; Appendix A
P2 central reflection positivity, all finite parameters (with the general invariant-potential form)	proved, without the earlier degree and mass hypotheses	Theorem 6.5, Corollaries 6.6, 6.7
P3 width moments at fixed parameters	removed; not used	local seminorm moments along the width arrow are proved in Proposition 11.6(i)
P4 finite-region Gaussian mass tilt and increment bound	proved	Theorems 7.2–7.6, 7.9; the endpoint value $a = 1/m$ is excluded
P5 common Gaussian space	proved, with Borel versions at $a = 0$	Theorems 7.7–7.10

P6 fixed-volume Wick integrability	proved, with the precise construction statement	Theorem 8.2
P7 regulator products	removed; not used	regulator arrow proved directly (Propositions 9.6, 11.6(iv))
P8 stagewise finite-vector bridge	removed by new proofs	Section 9 (arrows on X) and Theorem 10.8 (classical continuum limit); the earlier transfer claim is not used
P9 exact-cell midpoint convergence	removed; not used	no midpoint law occurs
C1–C5	classical, attributed	Section 3
C6	standard tools	Section 3

F Notation

Symbol	Meaning
m, ℓ, u; a, $\mu = 1/a$	mass, length, quartic coefficient (no factorial); regulator and endpoint rate
$\kappa_a^2 = a^{-2} - m^2$, $r_a = 1 - a^2 m^2$	mass-tilt coefficient; contraction constant
C_μ^{end}, C_a, C_{free}, $C'_{\Omega,a}$, $\Delta_{\Omega,a}$	endpoint, regulated, free, tilted covariances; $C_a - C'_{\Omega,a}$
$c_a = C_a(0)$	Wick constant: variance of the regulated Gaussian physical reference field (covariance C_a); not the endpoint variance $1/4$, and not asserted to be a variance of any finite-width or interacting law
Λ, $\Lambda_n = (-n\ell, n\ell)^2$, $\Omega = [-R, R]^2$	physical squares (axis-parallel, centered); auxiliary square
$F^b_{N,a}$, $b \in \{\mathrm{prod}, \mathrm{Mon}\}$	width-N neural field
$V = W + \varepsilon J - T$, v_V	full potential and its integrand
$P^b_{N,\varepsilon,\Omega,a;\Lambda}$, ν^v, Res, $\nu^{(2)}$, $\nu_{a,\Lambda}$, ν_n	neural, full-tilt, residual, zero-stabilizer, physical and continuum laws
$\nu_{m,u,\Lambda}$, ν^S, ν	classical finite-volume law, target on $\mathcal{S}'$, its X-valued version
E, X, p_j, d, β, dist_n	field spaces, local seminorms, metric, bounded-Lipschitz distance, selection discrepancy
$(\Gamma_0, \mathcal{F}_0, \mathbb{P}_0, W)$, ϕ_a, $\tilde{\phi}_a$, Φ_a	common Gaussian space, smeared and point fields, continuous modification, Borel version
$U_{a,\Lambda}$, $Z_{a,\Lambda}$, K_Λ, Ξ_p	Wick interaction, normalizer, fixed-volume Wick constants
$K_j = \|\chi_j\|_2/2$, σ_j	distinct neural reference-moment constants; original K_j bound restored in AC3
θ_1, θ_2, H^θ_+, Σ^θ_+	central reflections, half-planes, half-plane σ-algebras
$v^{\otimes k}$, $\mathcal{S}^\rho_k$, $A_{j,k}$, Schw_k	tensor evaluation, correlation distributions, coefficient norm, Schwinger distribution

G Historical records and paths not used

The historical ergodicity route. An earlier target record obtained ergodicity of the target through [8, Thm. 4.4] and an identification of the Osterwalder–Schrader semigroup with a Wightman energy, using the spectrum condition and uniqueness of the vacuum in the course of proving condition

(d) of C3(i). Its first unsupported step was that identification. The route is not used: condition (d) is proved in Theorem 10.16(iii) by Euclidean mixing from [2, (7)–(8), (13)], before [8, Thm. 1.0] is applied.

Replaced sentences of the earlier assembly. (i) A compressed sentence in the setting described the mass-shift identity as if it applied to the neural laws; it is replaced by the stage order of Section 2.2, where the identity concerns Gaussian reference covariances only, the pure quadratic tilt on a finite auxiliary square has covariance $C'_{\Omega,a}$, and C_a appears only in the auxiliary-volume limit. (ii) A universal sentence stating that the interaction is never a pointwise fourth power is replaced by the precise construction statement of Theorem 8.2(b) and Remark 2.9. (iii) The description of the zero-regulator interaction as "the conventional Wick integral" is replaced by the same construction statement and by Proposition 10.4.

Geometry. An earlier formulation required only that the physical square be centered; a rotated centered square then breaks the invariance hypothesis of Theorem 6.5 (Remark 6.8(i)). The definition is restricted to axis-parallel squares, which contain the main sequence unchanged; no failure of positivity is asserted for any rotated square, and the general invariant-potential clause is unchanged. A proposed counterexample with an outer rectangle invariant under only one reflection lies outside the hypotheses, which require both rectangles to be centered.

Historical SMH comparison and the AC3 additions. SMH's comparison of AC2 with original H1 returned MISMATCH: the original K_j bounds (2.2), (2.4) had been replaced by different constants, and the generic centered-square wording had been narrowed to axis-parallel squares. That exact historical verdict remains unchanged. AC3 writes the separate proposed numerical repair in (43) and (46); its current-version review and historical statement match remain companion obligations. The geometry disposition above remains: the original generic orientation wording is not recovered, the actual Λ_n and two central-line conclusions are unchanged, and no new rotated-square claim is made. OB4's earlier PROVED WEAKER verdict and OB4-003/OB4-010 remain historical. The source-only Wick/normalizer, sign, kernel and conditional density proofs now written in this text address their mathematical issues; the author does not close those findings or accept the integrated result.

Corrections of version A. This version corrects the following statements of the earlier version A, whose bytes are preserved unchanged. (i) The readings lemma admitted convergence of a characteristic function at a single frequency; that alternative is false (Remark 10.7) and is replaced by convergence at every real frequency, which the proof of Theorem 10.8 already supplies. (ii) A remark asserted that the full Gaussian normalizer grows with the auxiliary square; that assertion is withdrawn (Remark 9.7); monotonicity of $\mathrm{Det}_{\Omega,a}$ and the residual bounds are unchanged. (iii) The empty-set convention $\sup \emptyset = 0$ is restored for M_N, and the early-volume cutoffs of Theorem 10.10 are chosen at least 2. (iv) The positive constants M, ϑ are restored in the continuity hypothesis of Lemma B.1. (v) The Wick variance c_a is identified throughout as the variance of the regulated Gaussian physical reference field. (vi) Absolute "no rate" wording is narrowed to the absence of an effective end-to-end rate or schedule, with the fixed-volume $\sqrt{a}$ bounds kept. (vii) Authorship lineage is recorded per component (Appendix D). (viii) An undefined macro and several overfull displays are corrected without change of any formula.

Withdrawn formulations in the antecedents. An unclipped parameter-map clause (false as a total map; replaced by Lemma A.2(iii)); an unconditional independence assertion for competing clocks (replaced by Lemma A.8); the reading "Lebesgue measurable on $\mathbb{R}^3$" of the potential hypothesis (fails; Remark 6.8(iii)); a sentence extending the increment bound to $a = 1/m$ (not part of the proved statements and not used); the stagewise transfer and midpoint premises P8, P9 (not used); and a frequency-split Nelson route (checked, not used).

References

[1] Dmitry S. Ageev and Yulia A. Ageeva. Neural Network Quantum Field Theory from Transformer Architectures, 2026. URL `https://arxiv.org/abs/2602.10209v1`. arXiv:2602.10209v1 as acquired. The primary class (cs.LG) is taken from Zhang v1 ref. [17] and was not observed on the primary page. Adjacent architecture comparison only.

[2] S. Albeverio, R. Høegh-Krohn, and B. Zegarlinski. Uniqueness of Gibbs States for General $P(\phi)_2$-Weak Coupling Models by Cluster Expansion. *Communications in Mathematical Physics*, 121:683–697, 1989.

[3] Mehmet Demirtas, James Halverson, Anindita Maiti, Matthew D. Schwartz, and Keegan Stoner. Neural Network Field Theories: Non-Gaussianity, Actions, and Locality, 2023. URL `https://arxiv.org/abs/2307.03223v2`. arXiv:2307.03223v2, latest revision 2023-12-13 per the acquisition record. Supplied only as an excerpt (source lines 1-140, 673-944, 1165-1599). Published version cited by Frank v2 [3], Ferko–Mutchler v1 [17], Harvey v2, Universality v1 [6] and Zhang v1 [3] as Mach. Learn.: Sci. Technol. 5 (2024) 015002; DOI 10.1088/2632-2153/ad17d3 as printed in Ageev–Ageeva v1 ref. (7). Not verified.

[4] Christian Ferko and James Halverson. Quantum Mechanics and Neural Networks, 2025. URL `https://arxiv.org/abs/2504.05462v1`. arXiv:2504.05462v1 as acquired (HTML; NOVELTY-ADJACENT-ACQUISITION.json). DATE AMBIGUITY RETAINED: the HTML body is dated August 24, 2026 (input line 8), which conflicts with the 2504 identifier; the cause was not determined. Primary class as cited in Zhang v1 [15]. Published version cited as Mach. Learn.: Sci. Technol. 7 (2026) 015002 by Ferko–Mutchler v1 [3], Frank v2 [9], Harvey v2 and Zhang v1 [15]; Universality v1 [9] cites only the 2025 arXiv preprint. Not verified.

[5] Christian Ferko and Aaron Mutchler. Neural Network Field Theory at Finite Width, 2026. URL `https://arxiv.org/abs/2608.21588v1`. arXiv:2608.21588v1 [hep-th], version v1, 21 August 2026.

[6] Christian Ferko, James Halverson, and Aaron Mutchler. Universality of Neural Network Field Theory, 2026. URL `https://arxiv.org/abs/2601.14453v1`. arXiv:2601.14453v1 [hep-th], version v1; arXiv stamp 20 January 2026, HTML body dated 24 August 2026.

[7] Samuel Frank. Constructive Neural Network Field Theory: ϕ_2^4 in Finite Volume, 2026. URL `https://arxiv.org/html/2610.00453v2`. arXiv:2610.00453v2 [hep-th], version v2, 5 October 2026.

[8] J. Fröhlich and B. Simon. Pure states for general $P(\phi)_2$ theories: Construction, regularity and variational equality. *Annals of Mathematics*, 105:493–526, 1977.

[9] J. Fröhlich, R. Israel, E. H. Lieb, and B. Simon. Phase Transitions and Reflection Positivity. I. General Theory and Long Range Lattice Models. *Communications in Mathematical Physics*, 62:1–34, 1978.

[10] Francesco Guerra, Lon Rosen, and Barry Simon. Boundary Conditions for the $P(\phi)_2$ Euclidean Field Theory. *Annales de l'Institut Henri Poincaré, Section A: Physique Théorique*, 25(3):231–334, 1976. URL `http://www.numdam.org/item?id=AIHPA_1976__25_3_231_0`.

[11] James Halverson. Building Quantum Field Theories Out of Neurons, 2021. URL `https://arxiv.org/abs/2112.04527v1`. arXiv:2112.04527v1 as acquired (primary-N-Halverson2021.txt lines 8-15). No journal version is observed in the supplied texts: all citing sources give the arXiv identifier only. Primary class as cited in Ferko–Halverson v1 ref. [2].

[12] James Halverson, Anindita Maiti, and Keegan Stoner. Neural Networks and Quantum Field Theory, 2021. URL https://arxiv.org/abs/2008.08601v2. arXiv:2008.08601v2, latest revision 2021-03-15 per the acquisition record. Supplied only as an excerpt. Primary class as cited in Ferko–Halverson v1 [1] and Zhang v1 [1]. Published version cited as Mach. Learn.: Sci. Technol. 2 (2021) 035002 by Frank v2 [1], Ferko–Mutchler v1 [1], Harvey v2, Universality v1 [15] and Zhang v1 [1]. Ferko–Halverson v1 [15] and Demirtas v2 [21] print an inconsistent volume (104). Ambiguity retained; not verified.

[13] Thomas R. Harvey. What Neural Network Field Theory Can and Cannot Realise on a Computer, 2026. URL https://arxiv.org/abs/2608.21523v2. arXiv:2608.21523v2 [hep-th], version v2, 27 August 2026.

[14] Melrose. 18.155 Lecture 15. MIT course 18.155 lecture notes, 2016. Lecture 15, p. 2, Theorem 1 (kernel theorem for tempered distributions). Bibliographic details not verified.

[15] Werner Nagel and Viola Weiss. Crack STIT tessellations: characterization of stationary random tessellations stable with respect to iteration. *Advances in Applied Probability*, 37(4):859–883, 2005. doi: 10.1239/aap/1134587744.

[16] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green's functions. *Communications in Mathematical Physics*, 31:83–112, 1973. doi: 10.1007/BF01645738.

[17] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green's functions II. *Communications in Mathematical Physics*, 42:281–305, 1975. doi: 10.1007/BF01608978.

[18] Daniel M. Roy and Yee W. Teh. The Mondrian Process. In D. Koller, D. Schuurmans, Y. Bengio, and L. Bottou, editors, *Advances in Neural Information Processing Systems*, volume 21. Curran Associates, Inc., 2008. URL https://proceedings.neurips.cc/paper_files/paper/2008/file/fe8c15fed5f808006ce95eddb7366e35-Paper.pdf.

[19] Barry Simon and Robert B. Griffiths. The $(\phi^4)_2$ Field Theory as a Classical Ising Model. *Communications in Mathematical Physics*, 33:145–164, 1973.

[20] R. F. Streater and A. S. Wightman. *PCT, Spin and Statistics and All That*. Benjamin, New York, 1964.

[21] Zhengkang Zhang. Optimal Architecture and Fundamental Bounds in Neural Network Field Theory, 2026. URL https://arxiv.org/abs/2604.27050v1. arXiv:2604.27050v1, abstract latest revision 2026-04-29 per the acquisition record. Primary class as cited in Frank v2 [12]. Cited only for finite-width cost/error context; nothing imported.

[22] Joey Zou. Math 218 Lecture Notes (Spring 2022). Author-hosted lecture notes, 2022. URL https://sites.math.northwestern.edu/nfw2778/math218_s22_website/lecture_notes.pdf. Printed pages 15–17 consulted; accessed 20 October 2026.